ROYAL CANADIAN NAVY
QUALIFICATION STANDARD AND PLAN (QSP)

Naval Warfare Officer (NWO) (00207)
SONAR Op (00324)

Underwater Warfare Director (UWWD)

NQUAL - ALBV

MITE ID 122164

NTDC(A) Approval – 08/11/2019
NPTG HQ Approval – 13/12/2019
Version No: 1.3
MITE Crse ID : 122164

RECORD OF AMENDMENTS

The training authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Centre Atlantic (NTDC(A)) is the designated development centre responsible for the maintenance of training. Amendments to PO standard are to be requested through NTDC(A) and approved by CNPTG. Amendments to Lesson Guidance are to be approved by CO NTDC(A).

Revision	Approval Date	Authority
1.2	13 Dec 2019	NPTG HQ
1.3	04 Nov 2020	NTDC(A)

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FOREWORD

1. This training is based on the specification found at :

a) A-PD-055-002/PP-001 Naval Warfare Officer (NWO) 00207 Job-Based Occupational Specification dated 07 Aug 2018;

b) A-PD-055-002/PP-002 Canadian Forces Manual of Military Employment Structure Vol. 2 Part 2 NCM Job Based Specifications (JBS) for the Sonar Operator SONAR OP MOS ID 00324 dated 26 Aug 2019;

c) A-PD-055-003/PQ-001 Under Water Warfare Director (UWWD) Occupational Specialty Specification dated 12 Dec 2014;

2. A trainee will achieve the qualification NQUAL **ALBV** when the standards listed in all Performance Objectives have been met.

3. A trainee will achieve MITE Course IDs 122164 when standards listed in all Performance Objectives are met.

4. The designated TEs for this training is indicated below:

TE	Phase / Module	MITE course ID	NQual
NFS(A) / NFS (P)	All	122164	ALBV

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the UWWD (ALBV). This QSP is designed to assist NTDC(A) in developing, Naval Fleet Schools in conducting and evaluating, and NPTG in validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with (IAW) NPTORD 4500-22, Annex B. For this course, a period of instruction is 50 minutes in duration. The recommended schedule is based on **maximum load**. Scheduling cells are to increase/decrease course length based on number of students enrolled on the course.

Course Duration Calculation	
Allocation of ATT	**Periods**
Sum of all EOs/POs	591
Enabling Checks (ECs)	79
Performance Checks (PCs)	61
Actual Training Time (ATT) Sub-total	**91.4 trg days**
Allocation of BAT, TAT, TTT and PT IAW NPTORD 4500-22, Annex B Table 3.1	
Base Administrative Time (BAT)	23
Training Administrative Time (TAT)	23
Training Travel Time (TTT)	0
Physical Fitness (PT)	63
Sub-total	**13.6 trg days**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **400 minutes** **Partial days must be rounded up**	**840** **105 trg days**

REQUIRED COURSE STAFFING

3. Identify key personnel required to conduct the training.

Position	Rank	Qualifications	Qty	PO / EO
Course Director	Lt(N) or above	UWWD	1	All
UWWD Instructor	Lt(N) / PO1	UWWD	1 **(2 if 9+ students)**	All
ORS Instructor	PO2 or above	RQ-PO2 NCIOP	1	PO 001 / EO 001.01.01
SAC Instructor	MS or above	SAC D qualified	1	PO 001 / EO 001.01.01 PO 004 / EO 004.03.04
EW Instructor	PO2 or above	RQ-PO2 NESOP	1	PO 001 / EO 001.01.02

TDL instructor	MS or above	TDL 300	1	PO 001 / EO 001.01.03
IMD Instructor	Lt(N) or PO1	IMDO	1	PO 001 / EO 001.01.04
CSMW Instructor	Lt(N) or PO1	CSMW	1	PO 001 / EO 001.02.06-09
METOC Staff	PO2 or above	METOC staff	1	PO 002 / EO 002.01.09
ADAC Instructor	PO2 or above	ADAC instructor	1	PO 002 / EO 002.01.16
ORO Instructor	Lt(N) or above	Post ORO tour	1	PO 002 / EO 002.01.20 PO 002 / EO 002.02.01 PO 004 / EO 004.03.14
Communications Instructor	MS or above	RQ MS NCI OP	1	PO 004 / EO 004.02.01-02
GUEST LECTURES				
Position	**Rank**	**Qualifications**	**Qty**	**PO / EO**
Sea Training Group	CPO2 or LCdr	STG staff	1	PO 001 / EO 001.01.05
CCFL/P Sonar Chief	CPO2	CCFL/P Sonar Chief	1	PO 001 / EO 001.01.06
TRINITY (ADAC) Staff	PO2 or above	RQ-PO2 SONAR OP	1	PO 001 / EO 001.02.01
TRINITY Mine Warfare Analyst	Lt(N) or PO1	TRINITY Staff	1	PO 001 / EO 001.02.07
CFMWC	PO1 or above	CFMWC staff	1	PO 002 / EO 002.01.21
SUBOPAUTH	Lt(N) or above	SUBOPAUTH staff	1	PO 002 / EO 002.02.02
LRPA Squadron	Capt or above	LRPA TACCO or Pilot	1	PO 004 / EO 004.03.03
Maritime Helicopter Squadron	Capt or above	Maritime Helo TACCO or Pilot	1	PO 004 / EO 004.03.04
Tour Guide Shearwater/Patricia Bay	NA	Shearwater/Patricia Bay Staff	1	PO 004 / EO 004.03.05
CANSUBFOR	Lt(N) or above	CANSUBFOR staff	1	PO 004 / EO 004.01.01
Submarine Command CO/ORO	Lt(N) or above	SMCO or SMORO	1	PO 004 / EO 004.03.08-09

COURSE CAPACITY

4. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40.

   a. Maximum – 8;

   b. Minimum – 4; and

   c. Optimum – 6.

5. For each additional student added above Maximum course load, an addition of **51 training period's total** (41 ATT, 2 BAT, 2 TAT, and 6 PT periods) will need to be added to the overall course length. Naval Fleet Schools Commandant's will need to approve the additional course length based on number of SONAR Op DMILC 2 and NWO BTL manager requirements.

PREREQUISITES

6. Prerequisites. To be selected for this course, the loading authority must ensure that trainees have the following prerequisites:

a. SONAR OP candidates for this training shall:

(1) RQ-SONAR Op-PO2 (ALBD); and

(2) Hold a security clearance of Level 3.

b. NWO candidates for this training shall:

(1) Be of the rank of Sub-Lieutenant or above;

(2) Be recommended by their Commanding Officer; and

(3) Hold a security clearance of Level 3.

COURSE REPORTING

7. Upon completion of the course, students shall receive a paper record of completion by means of form CF 377 Course Report and Form CF 289 (or locally produced equivalent) Certificate of Military Achievement IAW DAOD 31-9 and NPTORD 4500-51.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

8. PLARs shall be conducted IAW NPTORD 4500-27, DAOD 5031-1 and CFITES Vol. 12.

SPECIAL INSTRUCTIONS

9. Nil.

CHAPTER 2: PERFORMANCE OBJECTIVES & LESSON GUIDANCE

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PO 001 – Confirm UWW Team/System Readiness

1. **Performance**: Confirm UWW Team/System Readiness (T0007/T0009)

2. **Conditions**:

   a. Given:

      (1) Subordinates;
      (2) References;
      (3) DWAN and CSNI Computers equipped with:
         (a) Microsoft Office;
         (b) Internet Explorer or web browser;
         (c) Operations Room Layout;
         (d) Flex Schedule Template; and
         (e) Watch and station bill template.

   b. Denied: NIL

   c. Environment: Any (at sea/alongside/shore facility & day/night)

3. **Standard**: The UWWD shall confirm UWW team/system readiness (T0007/T0009) by:

   a. Conducting team training events (G0065) by:

      (1) Identifying team readiness requirements;
      (2) Planning and designing training exercise (G0010) to include:
         (a) Drafting training messages; and
         (b) Contribute to FLEX development
      (3) Assisting in team training events (G0060) by:
         (a) Assessing team training performance (G0120);
         (b) Correcting team training deficiencies (G0070);
         (c) Assessing effectiveness of training (G0130); and
         (d) Recommending improvements to training (L0035).

   b. Optimizing ship's acoustic signature (D0070) by:

      (1) Assisting with tests and trials with outside agencies (B0060);
      (2) Maintaining quiet running bill (D0055) by:
         (a) Developing a noise monitoring plan (D0060);
         (b) Conducting noise rounds (C0130);
         (c) Maintaining ships logs and records (AT0090/BT0380); and
         (d) Drafting deviation request for QRB.
      (3) Supervising mine detection team in mine avoidance environment (C0010)

4. **Training Limitations**: Nil.

EO 001.01 - Conduct Team Training Events

001.01.01 – Operations Team: Crewing and Responsibilities

Conditions		Sequence
Unclassified classroom with DWAN		After course introductions
Time	**Classification**	**Instructor**
150 mins	UNCLASSIFIED	UWWD Instructor ORS Instructor SAC Instructor

Learning Objectives	Method
1) Duties and responsibilities of the UWWD 2) Crewing for operations team 3) Fighting organization 4) Functional organization 5) Actions to address crewing gaps	Distributed learning Video Interactive lecture Jigsaw Collaborative problem solving

Activity Steps	
	1) Students will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on the Duties and Responsibilities of the UWWD and watch the associated video. (45 mins) 2) Students will split into two teams and label all the positions of the Operations room on a provided layout. The instructor will correct any mistakes upon completion. (10 mins) 3) Students are broken into 4 groups to conduct a jigsaw activity. Each group will be assigned one of the following chapters of SSO for which they will become "experts": (20 mins) a. Chapter 2 b. Chapter 4 c. Chapter 5 d. Chapter 6 4) On completion, students will be numbered 1, 2, 3 or 4 forming groups (all 1s to one area, all 2s to another, etc.) and teach each other the content from their readings section. (25 mins) 5) Students will break into pairs and review a provided W/S bill and associated scenario with the aim of identifying crewing shortfalls and providing solutions to include generating a shortage messages using a provided template. Their solution must address the needs of the section to include: (25 mins) a. 1st degree Crewing requirements b. 2nd degree Crewing requirements c. 4th Degree Crewing requirements d. Special Parties 6) On completion, students are to present their scenario along with the impacts of identified crewing shortages and related solutions including shortage messages. (25 mins)

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Resources	References
1) DWAN terminals (1/student) 2) Projector 3) Duties and responsibilities DLN and video 4) Ops Room layout handout 5) Watch and Station Bill scenarios 6) Shortage message template	1) CFCD 120 2) CFCD 129 3) SHOPS 4) SSOs

Notes to Instructor

EO 001.01 - Conduct Team Training Events

001.01.02 – Operations Team: SAC & ORS Liaison

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 001.01.01
Time	**Classification**	**Instructor**
150 mins	UNCLASSIFIED	ORS Instructor SAC Instructor

Learning Objectives	Method
1) Key UWWD support position responsibilities to include: a. SAC; and b. ORS	Guest lecturer

Activity Steps	
	1) Students will receive guest lectures from the SAC instructor (100 mins) to include: a. Aircraft safety and control b. Expected interactions c. TACAIDs/SHOPS d. HVU/ACU duties e. Air Tasking Orders f. Liaising with Air Detachment 2) Students will receive guest lectures from the ORS instructor (50 mins) to include: a. Expected interactions and communications b. Underwater Resolve procedure c. Above Water Liaison and resources d. C4I system management

Resources	References
	1) CFCD 120 2) CFCD 129 3) SHOPS 4) SSOs

Notes to Instructor
Ensure SAC and ORS instructors are identified early to ensure availability

EO 001.01 - Conduct Team Training Events

001.01.03 – Employ Radars and IFF

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 001.01.02
Time	**Classification**	**Instructor**
150 mins	SECRET	UWWD Instructor EW Instructor

Learning Objectives	Method
1) Basic RADAR and EW theory a. Signal generation b. Wave propagation c. Pulse components d. PW/PRI/PRF e. bands/frequencies/ranges/strengths f. environmental effects g. 3D vs 2D h. PD vs Phased Array 2) IFF modes of operations (including mode 5) 3) CPF sensor capabilities and limitations a. SMART-S b. SG-180 c. NAV Radars 4) Integration with CMS including data fusion	Lecture 3-Star Think-Pair-Share

Activity Steps	
	1) The EW instructor will conduct a lecture on basic radar, EW and IFF theory utilizing the provide PowerPoint presentation ensuring the lesson's learning objectives are emphasised. This lecture will also include the current major warship radars as a concrete example. (100 mins) 2) As a review of the content, students will develop a list of their 3 stars from the previous lecture with an emphasis on each point's importance in carrying out the duties of an UWWD. "*Select the top 3 things covered in the lesson that you believe carry the most relevance in your duties as the UWWD*". (10 mins) 3) Students will then pair-up and discuss their individual lists and reasoning. Each pair will then repeat the exercise by generating a new list based on their combined observations. (10 mins) 4) Each pair will then present their lists to the class taking care to explain their reasoning behind each selection and ranking. The other students and the instructor will comment on and discuss their lists. (30 mins)

Resources	References
1) CSNI or secret standalone computer 2) Projector 3) Basic Radar, EW and IFF lecture	1) Command Control System/ Combat Systems computing environment description, operational and maintenance manual 2) HCM operating and technical document for Raytheon navigation radar 3) SEA GIRAFFE 180 operator handbook 4) SIRIUS technical manual 5) SMART-S MK 2 3D radar manual

Notes to Instructor
Ensure EW instructor is identified early to ensure availability.

EO 001.01 - Conduct Team Training Events

001.01.04 – C4ISR

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 001.01.03
Time	**Classification**	**Instructor**
150 mins	UNCLASSIFIED	UWWD Instructor TDL Instructor

Learning Objectives	Method
1) C4ISR definitions a. COP b. CTP c. RMP d. RAP 2) Tactical Data Link definition a. Types b. Theory of operation c. Method of control d. Transmission method capabilities and limitations (UHF, HF, JREAP) e. Integration into CMS f. Command messages g. DLPS definition and location	Lecture Essence

Activity Steps	
	1) The TDL instructor will conduct a lecture on C4ISR and Tactical Data Link utilizing the provide PowerPoint presentation ensuring the lesson's learning objectives are emphasised. (100 mins) 2) In their own words, students will be asked to write a brief (40 words) explanation of how Link contributes to the RMP and why it is so important to ASW. (10 mins) 3) Students will then share their explanation to another student in the class. Students will then be asked to reduce then length of their given explanation to 20 words. (10 mins) 4) Students will again share their explanation with another student (different from previous if possible) in the class before further reducing their explanation to 10 words. (10 mins) 5) Students will then read their explanations to the class. (10 mins) 6) The instructor will provide a Link handout to students at the end of lesson reviewing its content with the class. (10 mins)

Resources	References
1) SECRET computer 2) Projector 3) TDL lecture 4) Link handout	1) ADatP-33

Notes to Instructor

Ensure TDL instructor is identified early to ensure availability.

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EO 001.01 - Conduct Team Training Events

001.01.05 – Identify Team Readiness Requirements

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 001.01.04
Time	**Classification**	**Instructor**
100 mins	UNCLASSIFIED	UWWD Instructor IMD Instructor

Learning Objectives	Method
1) Understanding Combat Readiness Requirements 2) UWW applicable CRRs 3) Training requirements (personnel, equipment, facilities, time, resources) 4) Validity period/maintenance 5) Identifying authority for CRR completion 6) IMD requirements to include: ARMOUR, FLEX and daily SITREP	Lecture Collaborative problem solving

Activity Steps	
	1) Students will be broken into two groups. Each group is to conduct a collaborative brainstorm with CRRs being the theme (on a portion of the whiteboard or on a flip chart). Groups are to write down all the words/items they feel are related to CRRs. Instructor can provide further guidance with probes like, what pub dictates CRRs, how do CRRs impact the ships, management, etc. (7 min) 2) Groups then present their brainstorms to the class, highlighting any differences between the two groups of information (3 min). 3) Students will receive a lecture from the IMD and/or course instructor to include: (40 mins) a. Introduction to CFCD 102 (Release Authority STG) b. Demonstrate how to determine the applicable chapters c. Indicate how to disseminate CRR status d. ARMOUR CRR tracker updates e. FLEX requirements f. Daily SITREP requirements (CRR and expendables) 4) Students are then paired up, assigned a minimum of 2 UWW CRRs and asked to identify the following information: (40 mins) a. CRR title b. Purpose c. Readiness level d. Validity period e. Training requirements (resources, personnel, etc.) f. Standard (assessment criteria) g. Assessment authority 5) In the same groups as above, students will develop and submit a FLEX entry and identify required participants for their respective CRRs. (15 mins) 6) Each group will present their findings (CRR description and FLEX entry) to the class. (45 mins)

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Resources	References
1) DWAN computer (1/student) 2) Projector 3) Example daily sitrep 4) FLEX template 5) Flipcharts and markers	1) CFCD 102 2) CFCD 129

Notes to Instructor

Ensure IMD instructor is identified early to ensure availability.
The course instructor will demonstrate the location of CFCD 102 and ARMOUR with the IMD instructor providing additional details.
Students FLEX submissions to be e-mail to instructor IOT display on projector.

EO 001.01 - Conduct Team Training Events

001.01.06 - Identify Team Readiness Requirements: TRP

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 001.01.05
Time	**Classification**	**Instructor**
200 mins	UNCLASSIFIED	UWWD Instructor Guest Lecturer from STG

Learning Objectives	Method
1) Ships tiered readiness program a. Weapon certification b. TRI 1 & 2 c. OTT 1 & 2 d. SRIs 2) Responsibilities of Sea Training 3) Post readiness training sea training report	Jigsaw Guest lecturer Guided discussion

Activity Steps	
	1) Students are broken into 3 equal groups to conduct a jigsaw activity. Each group will be assigned one of the following readings for which they will become "experts". They may create handouts for their peers or prepare notes as they see fit. Advise groups that they are going to have to teach their peers on their topic at the next rotation. (50 mins) a. CFCD 129 Chapter 4, articles 404, 406, 411, Annex A & B b. CFCD 102 2-51 to 54 c. NAVORD 3137-2 Annex F 2) Once groups have become "experts" in their reading section, students will be numbered 1, 2 or 3. Groups will reform (all 1s to one area, all 2s to another, etc.) and students are to teach each other the content from their readings section covering all key points. (50 mins) 3) Students will receive a guest lecture from STG to include: (50 mins) a. Expected interactions and communications b. Pre SRT preparations and requirements c. Lectures and briefs d. Jazzograms e. Sea training reports f. UWW requirements IAW Readiness Training Guide Annex B 4) Students will review a Post Readiness Training Sea Training report (30 mins) before the instructor leads a guided discussion (20 mins) highlighting the following: a. What is the value of Sea Training reports? i. Expected responses: 1. Reoccurring mistakes 2. Training deficiencies 3. Recommend solutions b. What experiences have you had with sea training reports c. What units/personnel do you think use or should review the Sea Trg Reports?

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	d. How does the information in the Sea Trg report affect you?

Resources	References
1) DWAN computer (1/student) 2) Projector 3) Sea training lecture 4) Sample post readiness training Sea Training report	1) CFCD 102 2) CFCD 129 3) NAVORD 3137-2 4) Readiness Training Guide

Notes to Instructor

Ensure guest lecturer is identified early to ensure availability
Sea Training to supply a brief covering the listed items in the event they cannot attend.

EO 001.01 - Conduct Team Training Events

001.01.07 – Drafting Training Messages: Fleet Training

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 001.01.06
Time	**Classification**	**Instructor**
100 mins	UNCLASSIFIED	UWWD Instructor Guest Lecturer from CCFL/P

Learning Objectives	Method
1) Responsibilities of fleet chief sonar operator 2) Fleet training requirements 3) Drafting fleet training messages	Guest lecturer Student submission

Activity Steps	
	1) Students will receive a guest lecture from the CCFL/P C SONAR Op to include: (50 mins) a. Fleet Chief SONAR Operator responsibilities and expectations b. Fleet training requirements c. Fleet training messages d. Unit responsibilities 2) The instructor will display and explain Fleet Training messages (CREST/IPT). (5 mins) 3) Given a scenario tasking from CCFL/P, students will draft and submit to the instructor a fleet training TABORD as OCS. (35 mins) 4) The instructor will provide feedback on submissions. (10 mins)

Resources	References
1) DWAN computer (1/student) 2) Projector 3) Fleet Chief lecture 4) Sample CREST/IPT message 5) CCFL/P tasking scenario 6) Training message template	1) MARLANTORD 8-1 2) MARPACORD 9-47

Notes to Instructor
Ensure guest lecturer is identified early to ensure availability. CCFL/P to supply a brief covering the listed items in the event they cannot attend. Instructor to detail the submission method as preferred.

EO 001.01 - Conduct Team Training Events

001.01.08 – Liaising with supporting agencies

Conditions		Sequence
Unclassified classroom		Instructed after EO 001.01.07
Time	**Classification**	**Instructor**
50 mins	UNCLASSIFIED	UWWD Instructor

Learning Objectives	Method
1) Identifying available supporting agencies along with their provided resources a. METOC b. PCC(A/P) c. FDU(A/P) route survey office d. DRDC e. CFMWC f. TRINITYADAC g. Campus Atlantic/Pacific	Think/Pair/Share T-Chart

Activity Steps

1) Students will individually create a list of the facilities/units they believe could provide support to the UWW section. (15 mins)
2) Once each student has created a list individually, they are directed to pair up and discuss their list with a peer. (5 mins)
3) Students are then asked to create a class list (share their list). This list will be put up on the white board, for a class T-chart activity. (5 mins)
4) Once the class list is created, the instructor will lead the class in identifying what support these units provide. (25 mins) Anticipated responses may be:

Units	Training Supports Provided
ADAC	Acoustic Analysis Trg/Post analysis feedback
METOC	Acoustic Range prediction trg/bathy feedback
CFMWC	Allied/Canadian Doctrine, lessons learned
DRDC	Future products training
FDU(A/P) (route survey)	Bottom mapping, MCM, UUVs
TRINITY	Training Intelligence
Campus Atlantic/Pacific	Trainers, classrooms, facilities

Resources	References
1) Flip chart/Whiteboard 2) Markers	1) Unit websites

Notes to Instructor

Instructor may add additional agencies/resources based on experience.

EO 001.01 - Conduct Team Training Events

EC 001.01 - Conduct Team Training Events

Conditions		Sequence
Unclassified classroom with DWAN		Conducted after EO 001.01.08
Time	**Classification**	**Instructor**
250 mins	UNCLASSIFIED	UWWD Instructor

Scenario	Your ships has just completed SRT with STG embarked. They have signed off several CRR and noted a few deficiencies to rectify. The ORO wants to know your section's program for the next month including CRRs and CREST/IPT training.
Assessment Steps	Input: Students are provided a CREST or IPT message (coast dependant), section crewing, a Post Readiness Training Sea Training report identifying areas for development within the section and a ship's overall schedule detailing. Performance: Students have 100 mins to create a list of CRRs that must be conducted by the section with substantiation as to why those CRRs must be done. The CRRs selected must aim to correct the deficiencies that have been identified within the section. Once done, they must create a recommended Flex for the UWW section. Deliverables include: • CRR list to be completed with substantiation • Recommended Flex • Crewing considerations and actions On completion, each student will conduct a 15 mins sit down with the instructor to present, explain their plan followed by a 3 mins debrief. (18 mins/student) Output: The trainee must achieve 10/12 SATs to pass this EC. Feedback: Immediate verbal and written feedback provided via EC 001.01 checklist.

Resources	References
1) DWAN computer (1/student) 2) CREST or IPT message 3) Section crewing 4) Post Readiness Training Sea Training report 5) Ship's schedule 6) Flex template 7) CRR list/substantiation template (including manning considerations) 8) CFCD 102	1) CFCD 102 2) Readiness Training Guide 3) SSOs

Notes to Instructor
Students have 100 mins to create their products independently.

NTDC(A) Approval – 08/11/2019
NPTG HQ Approval – 13/12/2019
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Instructor to determine method of submission (email, hard copy, soft copy during sit down)

Limitations	
Impact of Failure	Retrain opportunities and reassess

Item	EC 001.01 - CONDUCT TEAM TRAINING EVENTS	SAT	UNSAT	Remarks
1. CRR substantiation list/plan				
Identified deficient CRRs IAW Sea Training report		☐	☐	
CRR substantiation reflects section needs IAW CFCD 102		☐	☐	
Priority of CRRs identified		☐	☐	
2. FLEX				
Adhered to FLEX format		☐	☐	
CRRs input into FLEX		☐	☐	
Participants required listed in FLEX		☐	☐	
FLEX Timings IAW CFCD 102		☐	☐	
Identified CRRs met by the provided CREST/IPT training serial(s)		☐	☐	
Sequencing of CRR training IAW CFCD 102		☐	☐	
CRR schedule reflects ships sea/alongside periods		☐	☐	
3. Crewing considerations plan				
Identified crewing shortages/impact on ops/considerations		☐	☐	
Identified rectification plan		☐	☐	

Score: **/ 12** The trainee must achieve 10/12 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 001.02 - Optimize Ship's Acoustic Signature

001.02.01 – Maritime Acoustic Signature Policy

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EC 001.01
Time	**Classification**	**Instructor**
100 mins	UNCLASSIFIED	UWWD Instructor Guest lecturer from TRINITY(ADAC)

Learning Objectives	Method
1) Maritime acoustic security authority 2) Maritime acoustic security policy 3) ADAC maritime acoustic security policy / procedures	Reading assignment Guest lecturer

Activity Steps	
	1) Students will read NAVORDs 3470-0. (15 mins) 2) Students will read NAVORDs 3470-1. (35 mins) 3) While conducting the reading assignments, the students will create 2 questions to ask for the Guest Lecture on the MASP. 4) Students will receive a guest lecture from TRINITY(ADAC), or course instructor if not available, emphasising its maritime acoustic security policy. Students will be allowed to ask their questions on MASP to GL or Instructor. (50 mins)

Resources	References
1) DWAN computer (1/student) 2) Projector 3) TRINITY(ADAC) Maritime Acoustic Security Policy lecture	1) NAVORD 3470-0 2) NAVORD 3470-1

Notes to Instructor
Ensure guest lecturer is identified early to ensure availability. Instructor to either ensure TRINITY(ADAC) staff is booked or has reviewed and familiarized themselves with the ADAC MASP brief prior to presenting it.

EO 001.02 - Optimize Ship's Acoustic Signature

001.02.02 – Ranging for Quiet Running Bill

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 001.02.01
Time	**Classification**	**Instructor**
250 mins	UNCLASSIFIED	UWWD Instructor

Learning Objectives	Method
1) Naval signature management policy 2) Types of sound ranging a. Static b. Dynamic c. Quicklook / check ranging 3) Other ranging's a. Degaussing b. Deperming c. SONAR performance figures 4) FMF ranging requirements 5) Liaison and scheduling ranging	5 Star Student presentations Guest lecturer

Activity Steps	
	1) Students will read NAVORD 3395-1. (50 mins) 2) Students are then asked to write down the five most important points contained in the NAVORD. (10 mins) 3) Students are to then share their responses with the class with justification. (20 mins) 4) As a class, students will then develop a list of the top 5 "take aways" from the NAVORD. (20 mins) 5) The instructor will break the students into pairs before assigning each group one type of ranging from the learning objectives list. Students will read and/or review NAVORD 3395-1 and CFCD 109 regarding their ranging IOT prepare a verbal brief. (50 mins) 6) Students will present their briefs to the class. (50 mins) 7) Students will receive a guest lecture from FMF/Range staff (50 mins) to include: a. Trial timelines b. Trial coordination c. Messages d. Communications e. Trial restrictions f. Range reports

Resources	References
1) DWAN computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) FMF range staff lecture	1) CFCD 109 2) NAVORD 3395-1

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Notes to Instructor

Ensure guest lecturer is identified early to ensure availability.

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EO 001.02 - Optimize Ship's Acoustic Signature

001.02.03 – Noise Monitoring Plans

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 001.02.02
Time	**Classification**	**Instructor**
150 mins	UNCLASSIFIED	UWWD Instructor

Learning Objectives	Method
1) Noise monitoring organization 2) Noise rounds requirements 3) Developing noise monitoring plan	Reading assignment T-Chart Think-Pair-Share

Activity Steps

1) Students will read CFCD 109 on their own taking note of the various positions within noise monitoring organization. (50 mins)
2) As a class, the students will develop a T Chart listing the responsibilities of each position along with the rank/trade/position on board ship. Each student/pair will be responsible to write a given role. (40 mins)

Role	Position	Responsibilities
Formation Signature Management Officer (FSMO)	Fleet Staff Officer	• Monitory the implementation of policy • POC for external organizations • Assisting USMO's with trials • Advising fleet readiness
Unit Signature Management Officer (USMO)	CSEO (can also be MSEO, UWWO, AWWO)	• Education of Crew • Establish noise defect identification and rectification procedures • Establish and maintain a noise monitoring cycle • Training of rounds personnel
Unit Noise Management Officer (UNMO)	UWWO	• Activities of the ship's Noise Detection Teams • Noise control indoctrination and awareness training • Brief CO, USMO, UNMT on trials and reports
Assistant Unit Noise Management Officer (A/UNMO)	Senior SONAR OP	• Noise control indoctrination and awareness training • Maintenance and Custody of the ship's Artifacts and Noise Defect Logs

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	Departmental Noise Monitors (DNM)	1x Officer (SLt – Lt(N)) 1 x NCM (S1 – PO2)	• Carry out noise rounds of their departmental spaces reporting to the UNMO • Assist in carrying out noise rounds of departmental spaces • Maintain departmental noise log in consultation with the Senior SONAR OP • Rectify noise defects as detailed by the USMO • Report noise defects which cannot be rectified as they arise to the DNM • Liaise with engineering staff to raise work orders • inform the USMO / Engineering Officer of the rectification of noise defects

3) Using the provided ship's schedule, each student will develop the monthly noise rounds schedule IAW CFCD 109 to include the positions conducting the rounds. (30 mins)
4) In pairs, students will compare and develop a shared schedule (15 mins) before presenting it to the class. (15 mins)

Resources	References
1) DWAN computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) Ship schedule scenario	1) CFCD 109 2) NAVORD 3395-1

Notes to Instructor

EO 001.02 - Optimize Ship's Acoustic Signature

001.02.04 – Logs & Records

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 001.02.03
Time	**Classification**	**Instructor**
100 mins	UNCLASSIFIED	UWWD Instructor

Learning Objectives	Method
1) Purpose of the quiet running bill 2) SONAR quiet states 3) Logs & records 4) Schedule for CO's signature of logs and records 5) Quiet running bill deviation requests	Reading assignment Floating flipchart Lecture Student submission

Activity Steps	
	1) Students will quickly review CFCD 109 regarding the purpose of the Quiet Running Bill and Quiet States. (10 mins) 2) The instructor will set up 3 floating flipcharts/whiteboard stations labelled: a. NORMAL Quiet State b. SONAR Quiet State c. ULTRA Quiet State. Students will be broken into groups (minimum of 3 groups) and go around to each of the stations and provide examples of restrictions IAW CFCD 109 that will occur during each of these states (2 mins per station). Each group will then be asked to explain the findings of one of the stations. (20 mins) 3) The instructor will provide an explanation of what is entered into the various noise logs to include: (20 mins) a. Noise Rounds Log b. Noise Defect Log c. Ship's Artifact Log d. Schedule for CO's review of Noise Organization logs IAW Captain's Standing Orders. 4) The instructor will explain the process for Quiet Running Bill deviation request to include: (10 mins) a. purpose of deviations requests b. when to draft deviation request c. drafting deviation request d. holding of deviation approvals 5) Using a deviation request template along with a scenario, students will draft a QRB Deviation Request and submit it to the instructor. (40 mins)

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Resources	References
1) DWAN computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) Example noise round log 6) Example noise defect log 7) Example Captain Standing Order 8) QRB Deviation Request template 9) QRB deviation scenario	1) CFCD 109

Notes to Instructor

Instructor to use CFCD 109 as answer guide to expected restrictions for each Quiet State.
Instructor to detail the submission method as preferred.

EO 001.02 - Optimize Ship's Acoustic Signature

001.02.05 – Ships UWW Logs & Records

Conditions		Sequence
Ship alongside		Instructed after EO 001.02.04
Time	**Classification**	**Instructor**
150 mins (120 mins instruction) (30 mins travel)	SECRET	UWWD Instructor

Learning Objectives	Method
1) Reviewing ship's quiet running bill 2) Cues to look for during noise monitoring rounds 3) Reviewing ship's logs and records	Ship tour Practical

Activity Steps

1) The instructor will review the following ship's documents to include: (50 mins)
    a. Quiet Running Bill
    b. Sound Ranging Reports (static, dynamic, quicklook)
    c. Noise defect log
    d. Noise artifact log
    e. Noise rounds log

    Students will be given the opportunity to view the publications and ask questions.
2) One at a time, students will be asked to enter a preselected space onboard ship and conduct noise rounds identifying a minimum of 3 potential noise defects and writing them down on a notepad handing it to the instructor before leaving the space. (50 mins)
3) Once all students have done so, the instructor will discuss each identified noise defects indicating which ones were common among the class, and which were unique. (10 mins) The class will then discuss as a group how to rectify some of the more commonly identified noise defects. (10 mins)

Resources	References
1) Sticky pad/note pad 2) Pens/pencils 3) Ship's Quiet Running Bill 4) Ship's sound range reports 5) Ship's noise defect log 6) Ship's noise artifact log 7) Ship's noise rounds log 8) Ship's space(s)	1) CFCD 109 2) NAVORD 3395-1

Notes to Instructor

Instructor to ensure that access to relevant classified documents is confirmed ahead of ship tour. For activity #2, ensure there are sufficient potential noise defects for students or use more than one space. Provided timings include travel to and from ship.

EO 001.02 - Optimize Ship's Acoustic Signature

001.02.06 – NMCM Concepts Overview

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 001.02.05
Time	**Classification**	**Instructor**
200 mins	SECRET	CSMW Instructor

Learning Objectives	Method
1) NMCM concepts overview a. Types of mines b. Actuation/influences of mines 2) Basics of naval mine warfare a. Mine Threat Area (MTA) b. Mine Danger Area (MDA) c. Q-Routes 3) Establishing competencies in NMCM doctrine	Distributed learning Guided discussion PUBEX

Activity Steps	
	1) Students will be directed to complete the animation activities title MCM basics and MCM operational sequence. (50 mins) 2) The instructor will conduct a guided discussion regarding NMCM (50 mins) a. What are the types of mines? i. What do you think the capabilities and limitations are of these generic mine types? (the SO what?) ii. What kinds of signatures can be exploited by mines? b. How do we differentiate mine hunting vs. mine sweeping? i. How would you determine which to use when faced with a mine threat? ii. Name 2 types of mine sweeping. c. Explain the following MCM tasks: i. Precursor ii. Exploratory iii. Clearance d. What is an MTA? e. What is an MDA? f. What is the mine threat in an MTA and an MDA? g. What is a Q route? h. What are some criteria for establishing a Q route from an MCM perspective? i. What do the Q route colour codes mean: i. Green ii. Yellow iii. Red 3) Upon completion of the discussion, the instructor will hand out a list of Mine Warfare publications. Students will use this sheet to track what information (generally) is found in each of the publications by writing down the key concepts covered in each Pub. (50 mins)

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	4) Students will create 2x PUBEX style questions from two different publications and hand the questions along with a separate answer sheet into the instructor. (15 mins) 5) The instructor will dispense the questions to the class. Each student will get 2 questions (not their own) for which they must find the answer. (25 mins) 6) On completion, the instructor will distribute the answers to allow the students to correct any mistakes. (10 mins)

Resources	References
1) CSNI or secret standalone computer with headset (1/student) 2) NMCM Basics animation 3) NCMM Operational sequence animation 4) NMW list of publications	1) ATP-6 2) ATP-24 3) CFCD 106 4) MIDAS mine threat library 5) SSO

Notes to Instructor

Animation can be located on DWAN at Campus Atlantic/NTDCA/Deck Ops
http://halifax.mil.ca/NPTG/CampusAtlantic/NTDCA/DeckOpsDiv/DeckOpsDiv.html

EO 001.02 - Optimize Ship's Acoustic Signature

001.02.07 – Current Mine Threat

Conditions		Sequence
Classified classroom with CSNI		Instructed after EO 001.02.06
Time	**Classification**	**Instructor**
50 mins	SECRET	CSMW Instructor Guest Lecturer from TRINITY Mine Warfare Analyst

Learning Objectives	Method
1) Current mine threats 2) TRINITY resources	Guest lecturer

Activity Steps	
	1) Students will receive a guest lecture from TRINITY mine warfare analyst, or the CSMW instructor if not available, to include: (50 mins) a. Role and responsibility b. Available resources c. Current mine threats worldwide

Resources	References
1) CSNI computer 2) Projector 3) TRINITY lecture	1) ATP-6 2) ATP-24 3) CFCD 106 4) MIDAS mine threat library 5) SSO

Notes to Instructor
Ensure guest lecturer is identified early to ensure availability

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EO 001.02 - Optimize Ship's Acoustic Signature

001.02.08 – Mine Threat Research

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 001.02.07
Time	**Classification**	**Instructor**
150 mins	SECRET	CSMW Instructor

Learning Objectives	Method
1) Creating doctrinal competence in NMW 2) Establishing mine threat awareness 3) Establishing team competencies in NMCM doctrine	Reading assignment Team quiz

Activity Steps	
	1) Students will read the following documents on General NMW, Mine Types, Mine Threat and Mine index. (120 mins) a. General NMW i. ATP 06 Vol 1 (D) V1 (8 pages) : 0103, 0109, 0112, 0113 a. Mine Types (24 pages) i. ATP 06 Vol 1 (D) V1: 0202- 0213, 0215, 0216. b. Mine Threat (12 pages) i. ATP 06 Vol 1 (D) V1: 0217-0220, 0230-0233, 0236-0238, 0240, 0504 c. Mine Index (1page) i. ATP 24 Vol 1(C) Annex A to ch1 2) The instructor will form 4 groups with each creating 3 questions from the assigned readings. (10 mins) 3) Each team will ask their questions to the class with the instructor tracking points earned with each correct answer. (20 mins)

Resources	References
1) CSNI or secret standalone computer (1/student)	1) ATP-6 2) ATP-24

Notes to Instructor
Students may use the publications when creating questions and answering the questions. Be sure to remind them to use and update their publications reference sheet.

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EO 001.02 - Optimize Ship's Acoustic Signature

001.02.09 – Mine Threat Presentation

Conditions		Sequence
Classified Classroom with CSNI		Instructed after EO 001.02.08
Time	**Classification**	**Instructor**
150 mins	SECRET	CSMW Instructor

Learning Objectives	Method
1) Identifying mine types	Student presentation

Activity Steps	
	1) Each student is assigned a mine to brief representing a broad spectrum of existing mines. 2) Utilizing a variety of sources from the references, students will prepare a 10 minutes brief on a chosen mine covering: (70 mins) a. Type (moored/ground/drifting/rising) b. Charge Net Explosive Quantity (NEQ) c. Actuation method d. Special devices e. Effective depth f. Maker/country of origin (if known) g. Interesting Facts h. What countries utilize/holds 3) Students will present their brief to the class. (10 mins/student)

Resources	References
1) DWAN computer 2) CSNI computer (1/student) 3) Projector	1) Internet Open Source 2) Jane's Catalogue 3) MIDAS mine threat library 4) Naval Mines Worldwide 5) Threat Web

Notes to Instructor
At a minimum, students must include subparas a-e from the activity steps para 2 in their briefs as well as 2 items from subparas f-h

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EO 001.02 - Optimize Ship's Acoustic Signature

001.02.10 – Ship Self Protective Measures

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 001.02.09
Time	**Classification**	**Instructor**
250 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Naval mine countermeasures readiness levels a. Mine threat warning level 2) Ship wide responses a. Self-Protective Measures (SPM) b. Ship handling considerations c. Damage control considerations d. Personnel protection requirements	Guided discussion Reading assignment Floating flip chart

Activity Steps	
	1) The instructor will conduct a guided discussion on the principles of Self-Protective Measures with questions to include: (50 mins) a. What are the signatures that may actuate a mine? And what strategies do we use to mitigate the risk: b. TAC SPMs like water depth, HOT, Current, SS… for all plus c. Magnetic – deperming and degaussing plus tactical SPMs d. Acoustic, Material SPMs like shock mounting machinery, placing engines above the WL. NQS, speed and rate of turn management e. Pressure – speed management f. Why do we range ships? g. How does degaussing work? h. What is Deperming? i. What makes Noise hygiene difficult? 2) In order to understand the relevant regulations regarding ship's self-protective measures in a mine threat area, students will read the following: (180 mins) a. SSOs Mine Threat Operations section to include: i. Mine Threat Operations – Halifax class ii. Mine Threat Warning Yellow – Halifax class iii. Mine Threat Warning Red – Halifax class b. CFCD 106 Mine Warfare section to include: i. MARTI 7000 General Naval Mine Countermeasure Principles ii. MARTI 7100 NMCM Readiness Levels and SPMs iii. MARTI 7100 Annex A – Self Protective Measures iv. MARTI 7100 Annex C – Self Protective Measures for Halifax Class Vessels v. MARTI 7105 Surface Ship Mine Detection and Avoidance vi. MARTI 7105 Annex B – Ship Handling Considerations vii. MARTI 7105 Annex C – Damage Control and Special Parties

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	3) After each section of the reading assignment above, students will utilize a Floating Flip chart with the following titles to build a complete picture of ships SPM's with each student providing a minimum of 1 point per chart (to be conducted concurrently with reading assignment): a. Tactical b. Material c. Domestic 4) Students will then compare their Floating Flip charts with the content of MARTI 7105 Annex E – Mine Avoidance Checklist and correct as required. (20 mins)

Resources	References
1) DWAN computer (1/student) 2) CSNI or secret standalone computer (1/student) 3) Flip chart/Whiteboard 4) Markers	1) CFCD 106 2) SSO

Notes to Instructor
Instructor to ensure the Floating Flip chart activity is conducted concurrently with the reading assignment.

EO 001.02 - Optimize Ship's Acoustic Signature

001.02.11 – Total Mine Simulation Software

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 001.02.10
Time	**Classification**	**Instructor**
150 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Applying underwater explosion and damage radius considerations to NMCM operations a. TMSS data; or b. Naval Mine Vulnerability Assessment System (NMVAS) data	Reading assignment Lecture Video Practical

Activity Steps	
	1) Students will read the following documents on the damage radius of underwater explosions. (50 mins) a. ATP 01 Vol 1 para 13013 (1 page) b. ATP 06 Vol 1 paras: 0230- 0237, 0238 subparas 1 and 2 and 0504-0512 (16 pages) 2) The instructor will conduct a lecture on TMSS or NMVAS and play video demonstrating various underwater explosions and damage radius. (50 mins) 3) Students will use the provided UNCLAS TMSS or NMVAS data to conduct a risk assessment for the transit of a MTA. They will role one dice 2 times. The number on the dice for the first role multiplied by 15 will equals the minimum water depth for their ships transit of the MTA. The number on the second role multiplied by 5 will equal the planned transit speed. Students will apply the transit speed and depth to the TMSS/NMVAS data to determine if they are SAFE or UNSAFE. If they are UNSAFE, they will be required to recommend a change in route (minimum safe depth) or transit speed to reduce the risk to a SAFE level. (50 mins)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) TMSS or NMVAS lecture 3) Underwater explosion video 4) UNCLAS TMSS or NMVAS data 5) Dice (2)	1) ATP-1 2) ATP-6

Notes to Instructor

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EO 001.02 - Optimize Ship's Acoustic Signature

001.02.12 – Mine Threat Operations

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 001.02.11
Time	**Classification**	**Instructor**
100 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Mine warfare messages 2) Lead through procedures 3) Force protection for NMCM assets	PUBEX Guided discussion

Activity Steps

1) Utilizing ATP 1 and ATP 24, students will individually answer the following questions and be prepared to cite the source of their information. (50 mins)
    a. How is Naval Mine Warfare coordinated? How are individual units tasked and how do they report their progress?
    b. How can a CPF transit a cleared route within an MTA?
    c. How are NMCM vessels protected from harm? (AAW and small boats)
2) Exploring one question at a time via a guided discussion, the class will discuss the topic by presenting their answer with the instructor ensuring the expected answer is covered as part of the discussion.
    a. Mine Warfare Coordination: The instructor will ensure ATP-1 Vol 1 Table 2-15 and ATP-24 Vol 1 Table 2-2 and 2-3 which highlight the various operational messages employed in NMCM is covered in discussion. (15 mins)
    b. CPF transit of cleared route: While acknowledging other answers such as utilizing Q-Routes, the instructor will focus on Lead-through Operations (ATP-24 Vol 1 art 0111). On conclusion of this discussion, the instructor will present AXP-05 EXTAC 636 which details the lead-through process. (20 mins)
    c. Protection of NMCM asset: While acknowledging self-protective measures on NMCM vessels, CPFs may be tasked to protect NMCM vessels IAW ATP-24 Vol 1 art 0112. The instructor will ensure that support operations IAW ATP-1 Vol 1 art 2181 (specifically situation C) are discussed where the CPF can use its discretion to provide protection without integration into the NMCM task group. (15 mins)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector	1) ATP-1 2) ATP-24 3) AXP-05

Notes to Instructor

While the questions are open to promote the exploration of the source documents, students must be informed that the aim is to identify the relevant articles rather than in depth knowledge.

EO 001.02 - Optimize Ship's Acoustic Signature

001.02.13 – Mine Detection and Avoidance Procedures

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 001.02.12
Time	**Classification**	**Instructor**
150 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Mine threat operations procedures a. Mine Detection Team (MDT) b. Equipment setup for mine detection 2) Use of equipment and assets: a. Hull mounted SONAR b. Aircraft c. Small boats 3) Operations room mine avoidance team procedures	Reading assignment T-Chart 3 Star Role play

Activity Steps

1) Students will read the SSOs sections on the mine detection organization to include: (20 mins)
   a. Mine detection team composition
   b. Roles and responsibilities
   c. Appendix's considerations.
2) Utilizing a T-Chart, each student will be given a number of positions and asked to determine the positions responsibilities, equipment required, communications set up for their positions: (30 mins)

ORO UWWD NAVO SCS HMS Supervisor HMS Operator ASPO SAC (if HELO embarked) OOW/2OOW Port/Starboard Lookout Mine Lookout (MLO) CEROS Operator SIRIUS Operator SEOSS Operator Boat Control Officer (BCO) 2 x Signal Lamp Operators XENON Search Light Operator ORS	Include responsibilities, equipment & communications for each role. See SSO's for correct responses. Students can provide additional information based on experience or potential requirements.

3) Students will read the following sections of CFCD 106: (30 mins)
   a. MARTI 7105 Annex D – AN/SQS-510 Settings and Procedures
   b. MARTI 7105 Annex F – Use of Ship's Boats
   c. MARTI 7105 Annex G – Maritime Helicopter Operations

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	4) Students will identify their “3 stars” for each reading and present them to the class. (20 mins) 5) Students will read the SSOs sections: (20 mins) a. SONAR Detection b. Resolve Procedures c. Conning the ship 6) Utilizing a provided mine detection and avoidance scenario with cue cards for each position, the students will act out (voice only) an assigned position of the mine detection organization IAW SSO’s Appendix 10 Resolve Procedures to include: a. Mine Detection b. Resolve procedure c. Course alteration procedure Students will be given the opportunity to practice multiple scenarios in different positions. (30 mins)

Resources	References
1) DWAN computer (1/student) 2) CSNI or secret standalone computer (1/student) 3) Flip chart/Whiteboard 4) Markers 5) Mine avoidance scenarios with cue cards	1) CFCD 106 2) SSO

Notes to Instructor

During the practical scenario, instructor is to change up positions and randomize the reports to allow for maximum student participation.

EO 001.02 - Optimize Ship's Acoustic Signature

EC 001.02 - Optimize Ship's Acoustic Signature

Conditions		Sequence
Unclassified classroom with DWAN		Conducted after EO 001.02.13
Time	**Classification**	**Instructor**
200 mins (150 mins preparation) (50 mins presentation)	UNCLASSIFIED	UWWD Instructor

Scenario	Your ships is preparing to transit through a mine danger area. Your team is to prepare a lecture (based on CFCD 102 CRR) and present it to command.

Assessment Steps	Input: Students are provided the mine avoidance lecture template and references.
	
	Performance: Students have 150 mins to modify the mine avoidance lecture template including a noise monitoring rounds plan. Students will be assessed in groups of 4-6 to complete and present the brief (25 mins per group).
	Deliverables include:
	• Noise monitoring rounds plan (to be included in the brief)
	• Mine Organization Lecture to include: (conducted in groups 4-6 pers)
	  o Mine Avoidance Organization, to include:
	  ▪ Operations Room crewing;
	  ▪ Duties and responsibilities of organization; and
	  ▪ Individual responsibilities.
	  o Mine avoidance tactics, techniques, and procedures, to include:
	  ▪ Overview of current OPTASK MW and MW SUPP;
	  ▪ How that affects present operations;
	  ▪ Dangers posed by various mine types;
	  ▪ Definition of Mine Danger Area and Q-Routes;
	  ▪ Demonstrates mine avoidance communications and remoting;
	  ▪ Ship manoeuvring;
	  ▪ Self-protective measures;
	  ▪ Acoustic and visual sensor setup;
	  ▪ Helo and small boat tasking's;
	  ▪ Picture compilation reports and procedures;
	  ▪ Mine contact reporting and cueing methods; and
	  ▪ Ops Room mine avoidance team procedures.
	
	Output: The trainees must achieve 16/20 SATs to pass this EC.
	
	Feedback: Immediate verbal and written feedback provided via EC 001.02 checklist.

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Resources	References
1) DWAN computer (1/group) 2) Projector 3) Mine avoidance brief template 4) Standing OPTASK messages	1) APP 11 2) CFCD 102 3) CFCD 109 4) SSO

Notes to Instructor
For ease of instruction/classroom allocation, the content of the brief must remain UNCLASS. Teams are to demonstrate the voice procedures. (acting the positions of the ops room)

Limitations	
Impact of Failure	Feedback, time to make amendments and reassess

Item	EC 001.02 - OPTIMIZE SHIP'S ACOUSTIC SIGNATURE	SAT	UNSAT	Remarks
1. Noise monitoring organization and plan				
Complete the Noise monitoring organization template IAW CFCD 109		☐	☐	
Create rounds schedule IAW CFCD 109		☐	☐	
Responds to question on frequency of monitoring logs (noise rounds log, noise defect log and artifact log) *instructor to ask		☐	☐	
2. Brief Mine Avoidance Organization IAW SSOs				
Identify crewing requirements		☐	☐	
Identify duties/responsibilities of organization		☐	☐	
Identify responsibilities of key personnel IAW SSOs "Mine Detection Team Composition - Halifax Class"		☐	☐	
3. Brief Mine Avoidance tactics, techniques and procedures				
Provides UNCLAS overview of OPTASK MW		☐	☐	
Provides definition of Mine Threat Warnings		☐	☐	
Identifies MW impact on current ops		☐	☐	
Identifies the dangers posed by threat mine types		☐	☐	
Identify SPMs to be utilized in MDA		☐	☐	
Identify ship limitations WRT manoeuvering when in MDA		☐	☐	
Provides definition of Mine Danger Area and Q-Routes		☐	☐	
Explains COMPLAN for mine avoidance transit		☐	☐	
Indicates HMS setup IAW SSOs for mine detection		☐	☐	
Identify helo and small boat employment		☐	☐	
Identify methods of marking mines (smokes, sonobuoys, etc.)		☐	☐	
Explain action upon visual detection of mine		☐	☐	
Team demonstrates the MDT reporting procedures IAW SSOs • Resolve • Alteration recommendation		☐	☐	
Provides response to questions from DS (one question per student)		☐	☐	

Score: / 20

The trainees must achieve 16/20 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidates	Name:	Date:	Signature:
	Name:	Date:	Signature:
	Name:	Date:	Signature:
	Name:	Date:	Signature:
	Name:	Date:	Signature:
	Name:	Date:	Signature:

PROTECTED B (when complete)

PC 001 - Confirm UWW Team/System Readiness

Scope	Conditions	Sequence
PO 001	Unclassified classroom with DWAN	Conducted after EC 001.02
Time	**Classification**	**Instructor**
250 mins	UNCLASSIFIED	UWWD Instructor

Scenario	Your ships has just completed SRT with STG embarked. They have signed off several CRR and noted a few deficiencies to rectify. The ORO wants to know your section's program including CRRs, CREST training and noise rounds requirements for the next month.
Assessment Steps	Input: Students are provided a CREST or IPT message (coast dependant), section crewing for upcoming deployment, a Post Readiness Training Sea Training report identifying areas for development within the section and a ship's overall schedule detailing. The noise monitoring organization is noted as key deficiency within the section. Performance: Students have 100 mins to create a list of CRRs that must be conducted by the section with substantiation as to why those CRRs must be done. The CRRs selected must aim to correct the deficiencies that have been identified within the section. Once done, they must create a recommended Flex for the UWW section along with noise monitoring plan for the section. Deliverables include: • CRR list to be completed with substantiation • Recommended Flex • Noise monitoring rounds plan • Crewing considerations and actions On completion, each student will conduct a 15 mins sit down with the instructor to present, explain their plan followed by a 3 mins debrief. (18 mins/student) Output: The trainee must achieve 13/16 SATs to pass this PC. Feedback: Immediate verbal and written feedback provided via PC 001 checklist.

Resources	References
1) DWAN computer (1/student) 2) CREST or IPT message 3) Section crewing 4) Post Readiness Training Sea Training report 5) Ship's schedule 6) Flex template 7) CRR list/substantiation template (including manning considerations) 8) Noise monitoring plan template 9) Noise monitoring organization chart	1) CFCD 102 2) CFCD 109 3) Readiness Training Guide 4) SSOs

Notes to Instructor
Students have 100 min to create their products independently. Instructor to determine method of submission (email, hard copy, soft copy during sit down)

Limitations	
Impact of Failure	IAW NPTORD 4500-54

Item	PC 001 - CONFIRM UWW TEAM/SYSTEM READINESS	SAT	UNSAT	Remarks
1. CRR substantiation list/plan				
Identified deficient CRRs IAW Sea Training report		☐	☐	
CRR substantiation reflects section needs IAW CFCD 102		☐	☐	
Priority of CRRs identified		☐	☐	
2. FLEX				
Adheres to FLEX format		☐	☐	
CRRs input into FLEX		☐	☐	
Participants required listed in FLEX		☐	☐	
FLEX Timings IAW CFCD 102		☐	☐	
Identifies CRRs met by the provided CREST/IPT training serial(s)		☐	☐	
Sequencing of CRR training IAW CFCD 102		☐	☐	
CRR schedule reflects ships sea/alongside periods		☐	☐	
Noise monitoring CRR included in FLEX		☐	☐	
3. Noise monitoring organization and plan				
Complete the Noise monitoring organization template IAW CFCD 109		☐	☐	
Create rounds schedule IAW CFCD 109		☐	☐	
Responds to question on frequency of monitoring logs (noise rounds log, noise defect log and artifact log) *instructor to ask		☐	☐	
4. Crewing considerations plan				
Identify crewing shortages/impact on ops/considerations		☐	☐	
Identify rectification plan		☐	☐	
Score: **/ 16**	The trainee must achieve 13/16 SATs to pass this PC.			

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PROTECTED B (when complete)

PO 002 – Complete UWW Exercise/Mission Preparations

1. **Performance**: Complete UWW Exercise/Mission Preparations (BT0050)

2. **Conditions**:

   a. Given:

      (1) Mission/Exercise;
      (2) DWAN workstation;
      (3) CSNI workstation;
      (4) References;
      (5) Oceanographic data;
      (6) ASW Sensors;
      (7) Ancillary Equipment;
      (8) Expendables;
      (9) Message drafting software; and
      (10) Operational/Exercise Messages.

   b. Denied: NIL

3. **Environmen**t:

      (1) Any (at sea/alongside/shore facility & day/night)

4. **Standard**: The UWWD shall complete UWW exercise/mission preparations (BT0050) by:

   a. Orienting to the mission by:

      (1) Interpreting operational messages/reports to include:
         (a) COMPLANS, EMCON plans, Frequency Plans (A0035);
         (b) Standing messages (ROE, OPGEN, OPTASK); and
         (c) Evaluation operational reports (BT0140)
      (2) Organizing intelligence (BT0490) by;
         (a) Providing data for RFIs (C0200);
         (b) Establishing caps/lims of platforms (B0045);
         (c) Interpreting oceanographic data (F0045);
            i. Establishing caps/lims of towed array ops in OPAREAs (C0160);
            ii. Calculating array tow speeds, tilt and/or balancing (C0120);
         (d) Determining oceanographic effects on acoustic operations (C0150); and
         (e) Assessing UWW threat information (D0050)

   b. Preparing Operational messages (BT0130) by:
      (1) Preparing UWW exercise support requests (D0045);
      (2) Coordinating water space management (C0165);
      (3) Drafting TABORDs and;
      (4) Drafting OPTASK ASW (Supp)

c. Disseminating intelligence information (B0040) by:
   (1) Displaying intelligence via UWW brief (B0035); and
   (2) Providing advice to command on UWW tactics (T0002)

5. **Training Limitations**:

EO 002.01 - Orient to the Mission

002.01.01 – Mission Briefing

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after PC 001
Time	**Classification**	**Instructor**
50 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Introduction to the course's operational mission	Lecture

Activity Steps	
	1) The course senior instructor will present the mission brief IAW NWTP Vol 4 scenarios to the students providing them with the strategic and tactical overview of the course's overarching mission. The instructor will also details the ASW mission requirement that the students will be expected to provide during their course. (50 min) 2) The Instructor will show the students where to find all the mission operational messages and material. Students will be expected to utilize the information briefed and mission resources to complete and brief the ASW portion during EO 004.03.15 Mission Planning.

Resources	References
1) CSNI or secret standalone computer 2) Projector 3) Mission brief 4) Mission package	1) NWTP Vol 4

Notes to Instructor

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EO 002.01 - Orient to the Mission

002.01.02 – Military Oceanography

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 002.01.01
Time	**Classification**	**Instructor**
150 mins	UNCLASSIFIED	UWWD Instructor

Learning Objectives	Method
1) Introduction to oceanography 2) Military oceanography	Distributed learning Table topics

Activity Steps	
	1) Students will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on UWWD Oceanography. (15 mins) 2) Students will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on Military Oceanography. (85 mins) 3) The instructor will conduct a Table Topics activity where each of the following topic(s) will be put into a hat: a. Why is Oceanography important to UWW operations? b. Oceanography considerations c. Submarine threats d. Passive vs active sound e. Range prediction to include: i. Active detection ii. Active counter detection iii. Passive detection iv. Passive counter detection f. METOC responsibilities 4) Each student will pull a topic from the hat and then verbally explain the topic to the class for 2 min. They will then receive feedback / inputs from the instructor and other students. (5 mins per topic, 50 mins total)

Resources	References
1) DWAN computer (1/student) 2) UWWD oceanography DLN 3) Military oceanography DLN 4) Topics each on a sticky note/piece of paper	1) MILOC manual 2) RP33

Notes to Instructor
Ensure students have access to DWAN with DLN accounts or CD with Distributed Learning available prior to commencement of lecture.

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EO 002.01 - Orient to the Mission

002.01.03 – Ocean Bottom Topography and Composition / Ocean Chemistry

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 002.01.02
Time	**Classification**	**Instructor**
150 mins	UNCLASSIFIED	UWWD Instructor

Learning Objectives	Method
1) Ocean bottom topography and composition 2) Ocean chemistry	Distributed learning Lecture

Activity Steps

1) Students will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on Ocean Bottom Topography and Composition. (50 mins)
2) Students will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on Chemistry of Seawater. (50 mins)
3) The instructor will conduct an interactive lecture on ocean bottom topography. Students will explain the effects different topographies have on sound as it travels through the water and how submarines can exploit them to their advantage. Ensure the following topographies are discussed: (35 mins)
    a. Continental shelf
    b. Continental slope
    c. Continental rise
    d. Abyssal plain
    e. Ridges/Seamounts/Guyot/Pinnacles
    f. Trenches
    g. Bottom compositions and their effects on sound
4) The instructor will conduct a lecture on chemistry of sea water. Students will explain the effects Salinity has on sound as it travels through the water and how submarines can exploit this to their advantage. Ensure the following items are discussed: (15 mins)
    a. States of water
    b. Salinity
    c. Regions of different salinities.

Resources	References
1) DWAN computer (1/student) 2) Ocean bottom topography and composition DLN 3) Chemistry of seawater DLN 4) Ocean bottom topography lecture 5) Chemistry of sea water lecture	1) MILOC manual 2) RP33

Notes to Instructor

Instructor to use the Power Point slides/images for visual representations.

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EO 002.01 - Orient to the Mission

002.01.04 – Physics of Sound / Sound Velocity

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 002.01.03
Time	**Classification**	**Instructor**
200 mins	UNCLASSIFIED	UWWD Instructor

Learning Objectives	Method
1) Physics of sound 2) Sound velocity	Distributed learning Lecture

Activity Steps

1) Students will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on Physics of Sound. (50 mins)
2) Students will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on Sound Velocity. (50 mins)
3) The instructor will conduct an interactive lecture on Physics of Sound ensuring the following items are discussed: (50 mins)
    a. Why use sound for ASW
    b. Compressional waves
    c. Period and frequency
    d. Wavelength (higher vs lower frequencies)
    e. Sound spectrum and military application of sound
4) The instructor will conduct an interactive lecture on Sound Velocity ensuring the following items are discussed: (50 mins)
    a. Temperature
    b. Salinity
    c. Pressure
    d. Changes affecting sound velocity
    e. Sound gradients to include:
        i. Positive
        ii. Negative
        iii. Iso-velocity

Resources	References
1) DWAN computer (1/student) 2) Physics of sound DLN 3) Sound velocity DLN 4) Physics of sound lecture 5) Sound velocity lecture	1) MILOC manual 2) RP33

Notes to Instructor

Instructor to use the Power Point slides/images for visual representations.

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EO 002.01 - Orient to the Mission

002.01.05 – Ocean Thermal Structure / Sound Paths

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 002.01.04
Time	**Classification**	**Instructor**
300 mins	UNCLASSIFIED	UWWD Instructor

Learning Objectives	Method
1) Ocean thermal structure 2) Sound paths	Distributed learning Lecture Practical

Activity Steps

1) Students will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on Ocean Thermal Structure. (65 mins)
2) Students will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on Sound Paths. (85 mins)
3) The instructor will conduct a lecture on Ocean Thermal Structure ensuring the following items are discussed: (30 mins)
    a. Deep water
    b. Permanent thermocline
    c. Surface layer to include
        i. Seasonal thermocline
        ii. Daily thermocline; and
        iii. Effects of weather on surface layer
4) The instructor will conduct an interactive lecture on Sound Paths ensuring the following items are discussed: (60 mins)
    a. Snell's law / HALT rule
    b. Direct path sound propagation to include
        i. Spherical spreading
        ii. Cylindrical spreading
    c. Surface sound channel
    d. Deep sound channel
    e. Bottom bounce
    f. Convergence zones
    g. Depth excess to include
        i. Critical depth
        ii. Conjugate depth
    h. Multipath sound propagation
    i. Cut-off frequency
5) The instructor will use handouts / or draw on whiteboard a Sound Velocity Profile (SVP). Students will be asked to explain/draw/demonstrate what sound paths exists based on the SVP. (5 mins per student, 60 mins total)

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Resources	References
1) DWAN computer (1/student) 2) Ocean thermal structure DLN 3) Sound paths DLN 4) Ocean thermal structure lecture 5) Sound paths lecture	1) MILOC manual 2) RP33

Notes to Instructor
Instructor to use the Power Point slides/images for visual representations.

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EO 002.01 - Orient to the Mission

002.01.06 – Biological Oceanography / Fronts & Eddies

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 002.01.05
Time	**Classification**	**Instructor**
250 mins	UNCLASSIFIED	UWWD Instructor

Learning Objectives	Method
1) Biological oceanography 2) Fronts & eddies 3) Tactical implication of oceanography and weather	Distributed learning Lecture Reading assignment T-Chart

Activity Steps

1) Students will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on Biological Oceanography. (50 mins)
2) The instructor will conduct an interactive lecture on Biological Oceanography ensuring the following items are discussed: (20 mins)
    a. Impact of marine organism on ASW
    b. Deep Scattering Layer (DSL)
    c. Biological false targets
    d. Bioluminescence
3) Students will read SEMS SOP E05A as it pertains to Marine Mammal Mitigation Avoidance Zones (MAZ) for active sonar. (15 mins)
4) As a class, students will develop a T-Chart based on the Recommended MAZ for active sonars used onboard HMC Ships: (10 mins)

Source	Recommended MAZ
SQS 510 (HMS)	4000 yards
SLQ 25A (NIXIE)	1000 yards (not applicable if HMS active)
SSQ-62 DICASS (Sonobuoys)	300 yards
HELRAS (Helo Sonar) < 1000m	4000 yards
HELRAS (Helo Sonar) >1000m	5500 yards

5) Students will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on Fronts and Eddies. (50 mins)
6) The instructor will conduct an interactive lecture on Front, Eddies and Water masses ensuring the following items are discussed: (30 mins)
    a. Cold Water masses
    b. Warm Water masses
    c. Mixing of water masses
    d. Currents
    e. Eddies to include
        i. Cold core
        ii. Warm core
        iii. Eddies effects on sound paths
        iv. Regional eddies (i.e. Mediterranean Sea)

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	7) Students will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on Different Acoustic Domains. (50 mins) 8) The instructor will conduct an interactive lecture on Tactical Implication of oceanography and weather ensuring the following items are discussed: (25 mins) a. Effects of on ASW from i. Wind ii. Precipitation iii. Temperature iv. Ambient Light b. Oceanographic Influences to include i. Sea Swell ii. Ocean Thermal Structure iii. Bottom depth and composition iv. Ice v. Ambient Noise vi. Fronts & Eddies vii. Currents viii. Biologics

Resources	References
1) DWAN computer (1/student) 2) Flip chart/Whiteboard 3) Markers 4) Biological oceanography DLN 5) Front and eddies DLN 6) Different acoustic domains DLN 7) Biological oceanography lecture 8) Front, eddies and water masses lecture 9) Tactical implication of oceanography and weather	1) NAVORD 4003-6 2) MILOC manual 3) RP33 4) SEMS

Notes to Instructor

Instructor to use the Power Point slides/images for visual representations.

EO 002.01 - Orient to the Mission

002.01.07 – Regional Oceanography / Shallow Water Operations

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 002.01.06
Time	**Classification**	**Instructor**
250 mins	UNCLASSIFIED	UWWD Instructor

Learning Objectives	Method
1) Regional Oceanography 2) Shallow Water Operations	Distributed learning Lecture Collaborative problem solving

Activity Steps

1) Students will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on Regional Oceanography. (50 mins)
2) Students will be directed to complete the Distributed Learning lecture / reading assignment found on DWAN / DLN on Shallow Water Operations. (50 mins)
3) The instructor will conduct an interactive lecture on Regional Oceanography ensuring the following items are discussed: (50 mins)
    a. Global water masses
    b. Currents
    c. Shipping noise
    d. Bioluminescence
    e. Regional water masses to include:
        i. North Atlantic regions
        ii. Arctic regions
        iii. North Pacific regions
4) The instructor will conduct an interactive lecture on Shallow Water operations ensuring the following items are discussed: (50 mins)
    a. What defines Shallow water operations?
    b. Bottom topography & composition
    c. Ambient noise to include
        i. Weather and sea state
        ii. Biological noise
        iii. Shipping noise
    d. Sound velocity profile variations to include:
        i. Temperature
        ii. Salinity
        iii. Depth
    e. Existence of sound paths
    f. Considerations of employment of ASW sensors
5) The instructor will split the class into 2-3 teams and conduct a case study on a regional area based on the mission scenarios (OP ARBITER, OP VARANGIAN or other Operational areas) and ask the students to provide their perspective on the oceanography of the area following criteria to the class: (50 mins)
    a. Shallow water areas
    b. Deep water areas
    c. Bottom topography & compositions

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	d. Ambient noise to include: i. Weather and sea state ii. Biological noise iii. Shipping noise e. Currents, fronts & eddies f. Sound velocity profile g. Sound paths h. Considerations of employment of ASW sensors

Resources	References
1) DWAN computer (1/student) 2) Regional oceanography DLN 3) Shallow water operations DLN 4) Regional oceanography lecture 5) Shallow water operations lecture 6) Regional area oceanographic scenarios	1) MILOC manual 2) RP33

Notes to Instructor

Instructor to use the Power Point slides/images for visual representations.

EO 002.01 - Orient to the Mission

002.01.08 – Elements of Acoustic Range Prediction

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 002.01.07
Time	**Classification**	**Instructor**
150 mins	UNCLASSIFIED	UWWD Instructor

Learning Objectives	Method
1) Decibels 2) Low frequency cut-off 3) Introduction to SONAR equations	Reading Assignment Lecture Collaborative Problem solving

Activity Steps

1) Students will read the sections on the MILOC manual pertaining to: (50 mins)
    a. Decibels
    b. Low frequency cut-off
    c. Active sonar equation
    d. Passive sonar equation
2) The instructor will conduct a lecture on frequency cut-off and demonstrate the use of the low frequency cut-off formula with numerous examples IOT allow the students to practice this formula with the instructor as a class. (30 mins)
3) The instructor will split the class into 2-3 teams and conduct a collaborative problem-solving activity. Students will be tasked with using a decibel math equation and the Nomo gram to add multiple ship's output source levels to determine the overall decibel level for all ships combined. Students must answer the question and show their work to the class. The instructor will confirm the accuracy and correct any mistakes. (20 mins)
4) As a class, students will write out the active and passive sonar equations on the boards and explaining what each part of the equation means. (20 mins)
5) The instructor will give each team a different sonar equation scenario and ask them to calculate the figures of merit for both the active and passive equations with each team presenting their solution to the class. (30 mins)

Resources	References
1) DWAN computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) Decibel, frequency cut-off and sonar equations lecture 6) Fleet noise level scenarios 7) Nomo gram for adding Decibels 8) Sonar equation scenarios	1) MILOC manual 2) RP33

Notes to Instructor

Instructor to use the Power Point slides/images for visual representations.

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EO 002.01 - Orient to the Mission

002.01.09 – METOC Products & Services

Conditions		Sequence
METOC Facilities		Instructed after EO 002.01.08
Time	**Classification**	**Instructor**
150 mins (120 mins instruction) (30 mins travel)	SECRET	UWWD Instructor METOC staff

Learning Objectives	Method
1) METOC services a. Deployment Support b. AS-27 / Tactical oceanographic summary (TOS) message	Tour Lecture

Activity Steps	
	1) Students will attend a tour of METOC facilities and receive a brief on METOC products to include: (120 mins) a. Sea Surface Temperature reports b. Ocean Feature Analysis reports c. Pre-deployment packages d. On-deployment support e. AS-27 / Tactical Oceanographic Summary (TOS) message

Resources	References
1) METOC facilities 2) METOC product brief 3) AS-27 TOS message examples	1) APP-11

Notes to Instructor
Ensure tour is booked early to ensure availability

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EO 002.01 - Orient to the Mission

002.01.10 – Tactical Oceanography

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 002.01.09
Time	**Classification**	**Instructor**
200 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Interpret oceanographic data 2) Identify tactical environmental factors that apply to towed array operations 3) Optimize TAS placement 4) Predicted / Tactical Sonar Range	Reading assignment Lecture Collaborative problem solving T-Chart

Activity Steps

1) Students will read ATP-18 sections 0915-16. (20 mins)
2) The instructor will conduct a lecture on tactical considerations of oceanographic data and environmental factors regarding TAS placement to include: (50 mins)
    a. Ocean features (OFA)
    b. Bottom topography
    c. Bottom composition
    d. Sound paths
    e. Tactical implications of cut-off frequencies
    f. Ambient noise sources

    Students are to be provided a copy of the lecture as a training aid.
3) Students are split into two teams and assigned the role of surface TAS ship or submarine unit. Using the cumulative knowledge on oceanography to date, each team is to identify factors or features advantageous to their unit. (20 mins)
4) Each team will present their results IOT to complete a T-Chart. (20 mins)

Oceanographic/Environmental Factors	
Advantages to own ships sensors	Advantages for submarines operating
- Hard sandy bottom (Shallow water) can benefit sonar performance. - Use of Convergence Zone - Use of SLD for sensors placements. - Use of cold Eddies (Decreased range between CZ) - Use of megaphone effect	- Use of shadow zones caused by bottom topography. - Use of the SLD for navigation i.e. surface duct. - Use of high shipping area. - Use of Warm Eddies for navigation (Increased range between CZ)

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	5) In teams of 3 or 4, students will identify the critical tactical considerations of a provided oceanographic/environmental scenario with the aim of recommending an optimal TAS placement. (45 mins) 6) Students will present their scenario to the class and discuss their tactical considerations with feedback and discussion amongst the class. (45 mins)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) Tactical considerations lecture 6) Oceanographic/environmental scenarios	1) ATP-1 2) ATP-18 3) ATP-28 4) CFCD 106 5) MILOC manual 6) RP 33

Notes to Instructor

Team composition should be balanced between NCM and officers to promote sharing of experience

EO 002.01 - Orient to the Mission

002.01.11 – ASW Sensors

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 002.01.10
Time	**Classification**	**Instructor**
250 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) HALIFAX class UWW sensors and equipment a. HMS i. Capabilities and limitations ii. EMCON considerations for active equipment iii. Marine mammal mitigation procedures b. SPS i. Capabilities and limitations c. CANTASS i. Capabilities and limitations d. Underwater telephone i. Capabilities and limitations ii. EMCON considerations iii. Low grade codes iv. Communications 1. Gertrude check 2. Mark Snap procedure 3. Uncle Joe procedure 4. Hooker / Uniform procedure 5. ESUS codes	Lecture Reading assignment 3 Star Jigsaw

Activity Steps	
	1) The instructor will conduct a lecture on the hull mounted sonar to provide an overview including: (50 mins) a. Capabilities and limitations b. EMCON considerations for active equipment 2) Students will read NAVORD 4003-6 and SEMS SOP E05A. (30 mins) 3) Students will write down the 3 most important pieces of information they gained from the readings. (5 mins) 4) Students will share their responses with the class and discuss the relevance and implementation of the procedures at the ship level and in their role as the UWWD. (15 mins) 5) The instructor will conduct a lecture on the sonobuoy processing system to provide an overview including: (50 mins) a. Capabilities and Limitations b. EMCON considerations for sonobuoys c. Sonobuoy employment 6) The instructor will conduct a lecture on the CANTASS to provide an overview of its capabilities and limitations. (50 mins)

	7) The instructor will conduct a lecture on the underwater telephone to provide an overview including: (15 mins) a. Capabilities and limitations b. EMCON considerations 8) The class will be divided into equal groups and assigned one of the following topics to read from AXP-1 and become the "expert": (15 mins) a. Table 5-4: Underwater Signal Code Table b. Table 5-9: Mark Snap procedure c. Chapter 5, para 5-21: Uncle Joe procedure d. Chapter 2, para 2-14 and 2-15: Hooker / Uniform procedure e. Table 5-5: ESUS codes (highlight BELLRINGER) 9) Each group will present their topic to the rest of the class. (20 mins)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) HMS lecture 4) Sonobuoy processing system lecture 5) CANTASS lecture 6) Underwater telephone lecture	1) AN/SQS-510 SONAR System Baseline 6.3 manual 2) AXP-01 3) CANTASS User Manual 4) DUWCS TUUM-6 5) NAVORD 4003-6 6) NCPM 600 7) SEMS SOP E05A 8) SPS User Manual

Notes to Instructor
Instructor can conduct an equipment tour/demonstration onboard ship or in SEACOT if time permits

EO 002.01 - Orient to the Mission

002.01.12 – Towed Array Operations

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 002.01.11
Time	**Classification**	**Instructor**
100 mins	UNCLASSIFIED	UWWD Instructor

Learning Objectives	Method
1) Capabilities and limitations of Canadian Towed Array in OPAREAS to include: a. Locating NAVORD 3137-1 b. Safety considerations 2) Calculating array tow speed, tilt/balancing, time to recover	Think-Pair-Share Reading assignment T-chart Collaborative problem solving

Activity Steps

1) The instructor will pose to class "What safety considerations and restrictions would you have to consider during towed array operations?" with students making an individual list based on their experience. (2 mins)
2) Students are then formed into 3 groups and asked to combined theirs lists (3 mins)
3) Instructor will then provide feedback on the groups lists (5 mins)
4) Students will read NAVORD 3137-1 with each group amending/completing theirs lists as required. (25 mins)
5) The instructor will assign each group a heading from the below T-Chart for them to complete and present to the class. (15 mins)

Anticipated Responses	
Towing Restrictions	Sea State Own Ships Speed Area of Operations Echo Sounder Operations UWON RAM Procedures NIXIE Operations
Operational Restrictions	Array Safety (Weather) Shallow Water Towing Procedures Stopping Procedures Cable Scope Array Depth Rescue Stations Procedures RAS Procedures
Procedures	Strumming Depth Safety Limitations Surface Restrictions (Broaching/Grounding) Use of Cable Cutter ISOPAR M Safety and Precautions

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	6) The instructor will reinforce the topics to ensure students are familiar with the considerations regarding: (5 mins) a. Deployment and retrieval limitations b. Speed considerations c. Depth and surface considerations. 7) In groups of 2, students will utilize the provided the towing chart as well as the manual calculations ($Towing\ Depth = \frac{Cable\ Scope / Ship\ Speed}{2}$) to find the missing information in a variety of scenarios. (5 mins) 8) The instructor will then demonstrate to students how to solve the problem utilizing one of the class scenarios above and explain how to estimate array when the array is "at depth" and stabilization time. (5 mins) 9) Revisiting their scenarios, students will correct their initial calculations, add the time required, and present their results to the class. (10 mins) 10) Students are then formed into new pairs to determine the missing data and provide impact to command for a new scenario. (10 mins) 11) On completion, each pair will present their scenarios along with their results to the class. (15 mins)

Resources	References
1) DWAN computer (1/student) 2) Flip chart/Whiteboard 3) Markers	1) NAVORD 3137-1

Notes to Instructor

EO 002.01 - Orient to the Mission

002.01.13 – Target Motion Analysis

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 002.01.12
Time	**Classification**	**Instructor**
150 mins	UNCLASSIFIED	UWWD Instructor

Learning Objectives	Method
1) Target Motion Analysis a. TMA history b. Sources of information c. TMA plots d. Range calculations 2) TMA Doppler theory	Lecture Think-Pair-Share Video Table top Collaborative problem solving

Activity Steps	
	1) Instructor will conduct a lecture on Target Motion Analysis ensuring the following items are discussed: (50 mins) a. History b. Sources of information c. TMA Plots d. Range calculations 2) Each student will create an explanation of the Doppler Effect based on their experience. (5 mins) 3) Students are formed into pairs to compare their explanation and come up with a shared definition before presenting it to the class. (15 mins) 4) The instructor will show the video on Doppler theory. (5 mins) 5) The instructor will conduct a lecture on the frequency relationships providing examples of Doppler theory and its effect on acoustic frequencies. (15 mins) 6) The instructor will lead a table top using a magnetic fleet in a box and the whiteboard to depict the different possible aspects and situations (opening and closing). (20 mins) 7) The instructor will demonstrate how to interpret a time frequency plot and relationships. (15 mins) 8) Students are broken into pairs and given worksheets where they must interpret time and frequency/relationship problems. (20 mins) 9) The instructor will provide feedback on the worksheets and answer any remaining questions. (5 mins)

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Resources	References
1) DWAN computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) Target motion analysis lecture 6) Doppler effect video 7) Time frequency instructor example 8) Time frequency worksheets	1) NCPM 600 2) NOSTP 200

Notes to Instructor

Instructor will conduct a demonstration of TMA module within CMS 330 as part of EO 003.02.02 Develop ASW Tactical Picture

EO 002.01 - Orient to the Mission

002.01.14 – Ancillary Equipment & Expendables

Conditions		Sequence
Acoustic range prediction trainer (if available) or Unclassified classroom with DWAN		Instructed after EO 002.01.13
Time	**Classification**	**Instructor**
200 mins	UNCLASSIFIED	UWWD Instructor

Learning Objectives	Method
1) HALIFAX class UWW ancillary equipment a. Bathy recorder b. Acoustic range prediction system 2) HALIFAX class UWW expendables a. XSV/XBT b. PTP c. EMATT d. ESUS e. Sonobuoys 3) Familiarize with basic ammunition allocation list and expandable logs/reports 4) DRDC MAPS	Video Instructor demonstration Show and share Practical Lecture

Activity Steps	
	1) The instructor will show the video on Bathythermograph probe launch and demonstrate the MK 21 Bathy program. (25 mins) 2) The instructor will demonstrate the capabilities of the acoustic range prediction system to include: (25 mins) a. The input information required by the ARPS b. Range predictions requirements c. ARPS outputs 3) The instructor will conduct a show and share where students will be provided with a training shape for each of the below expendables along with a demonstration on its operation: (10 mins per item/50 mins total) a. XSV/XBT b. Portable Target Programmer c. EMATT d. ESUS e. Sonobuoys 4) The instructor will present the basic ammunition allocation list for a CPF and explain the expendable requirements based on readiness level of the ship. (10 mins) 5) Each student will be assigned an expendable along with a ship's readiness level IOT determine their unit's allocation, they will then present their answers to the class. (10 mins)

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	6) The instructor will show examples of logs and discuss reporting methods for the below expendables followed by students practicing these logging procedures: (30 mins) a. XSV/XBT b. ESUS c. Sonobuoys d. EMATT 7) Instructor will conduct a lecture on MAPS ensuring the following items are discussed: (50 mins) a. What is MAPS/MAPS Lite? b. Capabilities and limitations c. Deployment considerations

Resources	References
1) DWAN computer (1/student) 2) Projector 3) Bathythermograph probe launch video 4) Laptop with MK 21 Bathy program and sample data 5) Laptop with ARPS software and sample data 6) XSV/XBT training shape 7) Portable Target Programmer 8) EMATT 9) ESUS training shape 10) Sonobuoy training shape 11) Basic ammunition allocation list 12) Sample expendable logs and reports 13) Target motion analysis lecture 14) DRDC MAPS lecture	1) Mk 21 User manual 2) NITES2R user manual

Notes to Instructor

Instructor may have DRDC guest lecture present the MAPS lecture if available.

EO 002.01 - Orient to the Mission

002.01.15 – Acoustic Counter Measures

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 002.01.14
Time	**Classification**	**Instructor**
100 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) HALIFAX class acoustic counter measure systems a. NIXIE b. ADC c. NAEB	Reading assignment Lecture Team quiz Show and share

Activity Steps	
	1) Students will read CFCD 106 MARTI 2120 Annex D and create 3 questions with answers to use in a team quiz activity. (50 mins) 2) The instructor will utilize the spectral breakdown of various NIXIE modes to explain the basic principle behind the system. (10 mins) 3) The class will be divided into two teams with the instructor asking the student generated questions to class and tallying the points for each correct answer. The student that generated a given question may not answer. (20 mins) 4) The instructor will conduct a show and share where students will be provided with a training shape for both ADC and NAEB along with a demonstration on their operation.(10 mins per item/20 mins total)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) NIXIE spectral breakdown 3) ADC training shape 4) NAEB training shape 5) STG CRR 11-05 Expendable ACM Brief	1) ADC technical manual 2) CFCD 106 3) NAEB technical manual 4) NAVORD 3137-0

Notes to Instructor

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EO 002.01 - Orient to the Mission

002.01.16 – Introduction to Passive Acoustics

Conditions		Sequence
ADAC training facility or Classified classroom with CSNI or secret standalone		Instructed after EO 002.01.15
Time	**Classification**	**Instructor**
1200 mins	SECRET	ADAC Instructor

Learning Objectives	Method
1) Introduction to Passive Acoustic Course (ADAC) to include a. Passive aural listening techniques b. Components of a LOFAR Gram c. Sources of radiated noise d. Doppler theory/interference patterns/anomalies e. Steps to analysis	Lecture Practical

Activity Steps	
	1) Students will attend the Introduction to Passive Acoustics training prepared, maintained and delivered by Trinity (ADAC). Training to include: a. Passive Aural Listening techniques and practical exercises (400 mins) b. Identifying Acoustic Sources including: (400 mins) i. Components of a LOFAR gram ii. Sources of radiated noise iii. Doppler theory/Interference patterns and anomalies 2) Using real world data from Trinity (ADAC), students will apply basic acoustic techniques to identify a series non-allied submarines with the assistance of instructors. (400 mins)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) TRINITY(ADAC) Introduction to passive acoustics training package	1) MILOC manual 2) RP33

Notes to Instructor
Students will receive training from ADAC staff that is based on their Introduction to Passive Acoustics Course (IPAC) with the schedule compressed into 3 days from 4 days.

EO 002.01 - Orient to the Mission

002.01.17 – Operational Messages Formatting

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 004.03.14
Time	**Classification**	**Instructor**
100 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Message format 2) Operational hierarchy of messages	Lecture Guided discussion Card Sort

Activity Steps

1) With the students following along, the instructor will utilize APP-11 and the CF message drafter software to demonstrate the basic components of message to include: (70 mins)
    a. action addressees
    b. info addressees
    c. message Classification
    d. Subject line/Message ID
    e. Subject Indicator Code (SIC)
    f. references
    g. Message body
    h. Drafter/Releasing Authority completed
2) The instructor will lead a discussion on the operational hierarchy of messages by asking: (20 mins)
    a. Generally, what you expect to find in the following messages?
        i. Operation Order
        ii. OPGEN
        iii. OPTASKs (ASW, AAW, ASuW, MW, COMMS, etc.)
        iv. OPSTAT Unit
    b. If you had to list these messages IOT of importance, what order would you place them? And Why?
    c. Can you name any additional operation messages?
3) The instructor will then conduct a Card Sort activity, students will be given cards with Operational Message names on them. As a class they will be required to put them in order of hierarchy. Instructor will provide feedback on their results (10 mins)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector	1) AP-11

Notes to Instructor

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EO 002.01 - Orient to the Mission

002.01.18 – Interpret Command Structure & Strategic Guidance

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 002.01.17
Time	**Classification**	**Instructor**
150 mins	SECRET	ORO Instructor

Learning Objectives	Method
1) Interpret command structure a. Task Force b. Task Group c. Task Unit d. Task Element 2) Hierarchy of commands/orders/references 3) TACON/TACOM/OPCON/OPCOM 4) Transfer of command authority (TOCA)	Reading assignment Student presentations Lecture

Activity Steps	
	1) Students will read the following: (110 mins) a. ATP-01 Vol 1 Chapter 2 Articles 2100-2120 b. CFCD 129 Chapter 3 Naval Command and Control 2) Students will form into 3 groups to further develop one of the following topics based on the readings: (20 mins) a. Basic command structure at sea b. Hierarchy of commands/orders c. TACON/TACOM/OPCON/OPCOM 3) Each group will present their results for further discussion with the class with the instructor ensuring that the content of the readings are covered. (10 mins) 4) The instructor will conduct a lecture on the Transfer of Command Authority, CHOP procedures and the significance of TOCA. (10 mins)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers	1) ATP-01 Vol 1 2) CFCD 129

Notes to Instructor

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EO 002.01 - Orient to the Mission

002.01.19 – Interpreting OPGEN & OPTASK Messages

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 002.01.18
Time	**Classification**	**Instructor**
150 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Interpreting OPGEN messages 2) Interpreting OPTASK messages	Reading assignment 3 Star Lecture Practical Guided discussion

Activity Steps	
	1) Students will read an example OPGEN message with specific attention to the general information contained in the message rather than the specifics details. (25 mins) 2) Students will write down the three most important items contained in an OPGEN message. (10 mins) 3) Students will share their responses with the class justifying their selections with any repeated responses skipped. (15 mins) 4) The instructor will utilize APP-11 to lead the class through the format for the OPTASK ASW/SCC message highlighting the components of the message that would be most applicable to the UWWD by building a list on the board. (50 mins) This list should include: a. EXER / OPER identifier b. MSGID identifier c. Period of time d. SUBTHREAT e. CONDUCT f. SCREEN disposition g. FASWMSSN (Force ASW Shipborne Sensor and Equipment) h. FASWTASK i. TDZ/TDA j. ASTD (Anti-Ship Torpedo Defence) instructions k. ASW pre-planned responses l. ASIAP (Active Sonar Interference Avoidance Plan) m. SIAP (Sonobuoy Interference Avoidance Plan) n. Acoustic deception plans 5) In pairs, students will breakdown a provided OPTASK ASW, OPTASK ASW Supp or OPTASK SCC, using the list created as a class during the previous exercise (this should still be on the board). (30 mins) 6) The instructor will call on individual students to provide their answers for each given point with the class discussing any gaps or surplus information. (10 mins) 7) The instructor will display an OPTASK COMMS message and guide the class in identifying the key components related to ASW. (10 mins)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) OPGEN sample message 6) OPTASK ASW/SCC/ASW sup sample message	1) APP-11

Notes to Instructor

EO 002.01 - Orient to the Mission

002.01.20 – Interpreting Miscellaneous Messages

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 002.01.19
Time	**Classification**	**Instructor**
150 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Interpreting OPSTAT UNIT messages 2) Interpreting ROEAUTH message 3) Interpreting weekly OPAREA assignment messages 4) Interpreting water space assignment message 5) Interpreting locator message 6) Interpreting casing message 7) AS 25/27	Practical Lecture Floating flip chart Jig Saw

Activity Steps	
	1) Students will utilize APP-11 to interpret 2 OPSTAT Units, 1 Canadian & 1 Allied nation. (15 mins) 2) The class will be divided in half with each group preparing a presentation of one of the above units on the flip charts/whiteboard. The presentations should contain: (15mins) a. Units name b. Sensors c. Weapons d. Aircrafts e. Equipment limitations/deficiencies f. Caps/lims (speed, endurance, etc.) g. Impact on exercise/operation (“so what”) 3) Each group will present their unit and conclusions to the other (8 mins each/ 16 mins total). 4) The instructor will provide feedback and answer any questions on extracting information from OPSTAT units. (4 mins) 5) The instructor will utilize a sample ROEAUTH message to highlight the key components to include: (5 mins) a. ROE numbering b. Description c. Amplification 6) Using this ROEAUTH message, students will individually identify all ROE relevant to ASW operations. This should include use of force, active emissions and warnings. (10 mins) 7) One a flip chart/whiteboard, each student will take turns writing a ROE # on the board explaining why it is relevant. This will continue until no further ROE are identified. (10 mins) 8) The instructor will present the related ROE Matrix highlighting the identified ROE and who currently holds them. (5 mins)

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	9) In 5 groups, students will utilize APP-11, NAVORDs and operational experience to identify the purpose, format and applicable reference of the following messages (one message per group) and prepare a short presentation to the class: (20 mins) a. Op Area Assignment Message b. Water Space Assignment Message c. Locator Message d. Casing Message e. AS 25/27 Message 10) Groups will present their message to the class with the instructor amplifying by providing a sample message after each presentation. (10 mins per group/50 mins)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) Canadian OPSTAT unit sample message 6) Allied OPSTAT unit sample message 7) ROEAUTH sample message 8) ROE Matrix example 9) OPAREA assignment message sample 10) Water Space Assignment message sample 11) Locator message sample 12) Casing message sample 13) AS 27 message example	1) APP-11 2) ATP-1 3) CANSUBFOR Weekly assignment message 4) MARLANTORD 3255-1 5) MARPACORD 3350-1

Notes to Instructor

Care should be taken in assigning messages to groups during Jig Saw activity to ensure (when possible) that the presenter has some experience with the assigned message.

EO 002.01 - Orient to the Mission

002.01.21 – Evaluation Reports

Conditions		Sequence
DWAN classroom with DWAN		Instructed after EO 002.01.15
Time	**Classification**	**Instructor**
150 mins	UNCLASSIFIED	UWWD Instructor Guest Lecturer from CFMWC

Learning Objectives	Method
1) Types/Sources of lessons learned a. Sea Training Annual Reports b. CFMWC Lessons Learned c. CJOC Lesson Learned	Essence Brainstorm Reading assignment Guided discussion Guest lecturer Lecture Practical

Activity Steps	
	1) As an introduction, students will write a 16-word answer to the question: (3 mins) a. Why is it important for an UWWD to review lessons learned ahead of a deployment/exercise? 2) Student will present their answer to the class. (3 mins) 3) Students will repeat these above steps to shorten their answer to 8 words then 4 words IOT to obtain essence of why the lesson learned process is important. (9 mins) 4) With the instructor writing down the answers, the class will conduct a brainstorming exercise to identify the various sources of lessons learned available to an UWWD. (5 mins) 5) On conclusion of the exercise, the instructor will add any missing resources from the following: (5 mins) a. Peers b. Canadian Forces Maritime Warfare Centre c. Sea Training d. Previous unit (for annual deployment/exercises) e. Fleet chief f. Lesson learned library (KMS) 6) Students will read the latest Sea Training Annual Report with the aim of identifying entries of relevance to the UWWD. (20 mins) 7) The instructor will conduct a guided discussion on these observations with a focus on how the students can help address the identified shortcomings upon completion of the course. (20 mins) 8) Option 1: Students will receive a Guest Lecture from the CFMWC Lesson Learned section presenting the RCN Lesson Learned process to the students to include how to submit observations and review existing information. (50 mins) **OR** Option 2: If unavailable, the instructor will conduct a lecture on the RCN Lesson Learned Process and KMS 3.7. Both lectures are located on CFMWC's DWAN website under the education and training heading. (40 mins)

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	Option 2b: On completion, the instructor will demonstrate the highlighted CFMWC lesson learn resources by locating a document of interest in the KMS library and the RCNLLP share point (it doesn't matter what). (10 mins) 9) The instructor will navigate to the CJOC Lessons learned SharePoint to highlight the existence of this additional resources. A link to this is found on the CFMWC Lesson Learned Section on DWAN. (5 mins) 10) As a consolidation exercise, each student will locate a document of their own choosing or personnel interest using the resources presented with the goal of presenting their findings to the class. The instructor can assign a given resource to each student to ensure variety. (20 mins) 11) Students will present their document to the class including how they went about locating it. (10 mins)

Resources	References
1) DWAN computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) Current Sea Training Annual Report 6) CFMWC LLP lecture 7) KMS 3.7 introduction lecture	1) DAOD 8010-1 2) NAVORD 8010-1

Notes to Instructor

Guest lecturers must be identified early IOT meet schedule. If necessary, this lesson can be moved to accommodate their schedule as its content isn't tested in later lessons.

EO 002.01 - Orient to the Mission

002.01.22 – Organize Intelligence Information

Conditions		Sequence
Classified classroom with CSNI		Instructed after EO 002.01.16
Time	**Classification**	**Instructor**
150 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Identifying various sources of intelligence information a. CSNI capabilities b. Research TRINITY library c. Drafting RFIs	Lecture Student presentation Practical

Activity Steps	
	1) The instructor will navigate CSNI to locate the following applications and provide a short explanation of each: (50 mins) a. Command Net default page b. Search Tools c. Command View websites to include: i. CJOC continental/expeditionary operations ii. CFMWC iii. NTDC iv. TRINITY Intelligence d. Google Earth Tools e. Transverse Chat 2) Students will search throughout CSNI with aim of discovering information pertinent to the role of UWWD of interest to the class with the stipulation that it must not have been mentioned in the class to date. (30 mins) 3) Students will present their findings to the class highlighting its importance and how to locate it on CSNI. (20 mins) 4) The instructor will locate the TRINITY Intelligence RFI Log / Library with the class and direct students to conduct a search for specific items. (15 mins) 5) The instructor will demonstrate how to use the Request for Information (RFI) submission tool (DO NOT SUBMIT AN ACTUAL RFI). (10 mins) 6) Utilizing information from an RFI scenario, students will complete the provided RFI template (simulating the RFI submission tool) and submit it to the instructor. (25 mins)

Resources	References
1) CSNI computer (1/student) 2) Projector	1) NAVORD 2000-2

Notes to Instructor
Instructor and students must have a CSNI account for this lesson. Students may be paired up CSNI access is limited. When using the Transverse Chat tool, login and password is the same as CSNI login / password.

EO 002.01 - Orient to the Mission

002.01.23 - Establishing Capabilities/Limitations of Platforms

Conditions		Sequence
Classified classroom with CSNI		Instructed after EO 002.01.22
Time	**Classification**	**Instructor**
200 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Establishing capabilities and limitations of platforms (friendly and threat) a. Data mining for platform data b. Determining operational impact of platform's gathered information 2) Determining UWW threat 3) Building platform briefs	Student research Practical

Activity Steps	
	1) In pairs, students will conduct research on a submarine taken from the course's overall mission IOT prepare a brief utilizing the provided template to include the following information: (200 min) a. Name/country b. Emitters c. Weapons (torpedoes, missiles & mines) i. Range/speed/run time of torpedoes ii. Method of homing d. Speeds (max and best endurance) e. Engineering configuration f. Operating depth g. National doctrine i. Mission ii. Competency iii. Experience iv. Tactics These briefs will be presented in EO 002.03.01 Platform Intelligence and incorporated in the CASEX and Mission briefs in the later phases of the course.

Resources	References
1) CSNI computer (1/student) 2) DWAN computer (1/student) 3) Projector 4) Platform brief template	1) DIAS Threat Web 2) Jane's catalogue

Notes to Instructor
Pair should be composed of a NWO with a SONAR Op when possible.

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EO 002.01 - Orient to the Mission

EC 002.01 - Orient to the mission

Scope	Conditions	Sequence
EO 002.01	SECRET classroom	Conducted after EO 002.01.23
Time	**Classification**	**Instructor**
250 mins	SECRET	UWWD Instructor

Scenario	Your ship is to participate in an upcoming CASEX. You must sit with your ORO and discuss the oceanographic conditions that will impact the exercise and how you intend on optimizing the ships sensors.
Assessment Steps	Input: Students are provided a CASEX package including relevant message traffic for the CASEX and oceanography "slides" to include: • Area (topography, bottom composition) • SST/OFA • Bathy XSV/XBT • AS 27/Predicted ranges • OPGEN • OPTASK ASW (threat) • OPSTAT unit (own ship with aircraft) • RFI response Performance: Students will have 50 mins to read the provided package, orient to the exercise and be prepared to address items identified in EC 002.01 checklist. On completion, they will then meet with their "ORO" (played by instructor) and discuss the oceanographic conditions and how the ship intends on optimizing its sensors to conduct the CASEX. This is an informal discussion aimed at identifying the checklist criteria and does not require a formal brief. (25 mins per student) Output: The trainee must achieve 26/32 SATs to pass this EC. Feedback: Immediate verbal and written feedback to be provided IAW EC 002.01 checklist.

Resources	References
1) CASEX and oceanography package IAW input	1) APP-11 2) ATP-28 3) AXP-1 4) MILOC Manual 5) NAVORD 2000-2 6) RP 33

Notes to Instructor

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Limitations	
Impact of Failure	Retrain and reassess

Item	EC 002.01 - ORIENT TO THE MISSION	SAT	UNSAT	Remarks
1. Explain the impact of sensor employment based on the Ocean Features Analysis (OFA)				
a. Identify warm water masses		☐	☐	
b. Identify cold water masses		☐	☐	
c. Identify current / fronts		☐	☐	
d. Identify eddies (warm/cold)		☐	☐	
2. Describe the impact of sensor employment based on Bottom Topography and Composition				
a. Bottom composition and effects		☐	☐	
b. Shallow water		☐	☐	
c. Slope water		☐	☐	
d. Deep water		☐	☐	
e. Pinnacle /Seamounts /Ridges /Trenches		☐	☐	
3. Explain the effects of the Chemistry of Water in the region				
a. Temperature		☐	☐	
b. Salinity		☐	☐	
c. Pressure		☐	☐	
d. Biologics		☐	☐	
4. Explain the Sound Velocity Profile				
a. Upper mixed layer		☐	☐	
b. Sonic layer depth		☐	☐	
c. Deep sound channel axis		☐	☐	
d. Conjugate depth		☐	☐	
e. Depth access		☐	☐	

5. Identify the potential Sound Paths present			
a. Direct path	☐	☐	
b. Bottom bounce	☐	☐	
c. Deep sound channel	☐	☐	
d. Convergence zone	☐	☐	
e. Surface duct / Half channel	☐	☐	
6. Based on oceanographic data and threat, provide assessment of submarines COAs			
a. Identify oceanographic features that work in favour of the submarine	☐	☐	
b. Identify topography features that work in favour of the submarine	☐	☐	
c. Identify submarine's likely operating depth to avoid detection	☐	☐	
d. Identify submarines likely operating areas to avoid detection	☐	☐	
7. Provide justifiable recommendations for sensor employment IAW CASEX package			
a. Shipboard active	☐	☐	
b. Helo/Aircraft active	☐	☐	
c. Sonobuoys	☐	☐	
d. Passive towed array	☐	☐	
e. UWT	☐	☐	
Score: /32	The trainee must achieve 26/32 SATs to pass this EC.		

Positive Feedback	**Improvement Feedback**

Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 002.02 - Prepare Operational Messages

002.02.01 – Introduction to the Maritime Tactical Planner (MTP)

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EC 002.01
Time	**Classification**	**Instructor**
150 mins	UNCLASSIFIED	ORO Instructor

Learning Objectives	Method
1) Introduction to the Maritime Tactical Planner (MTP) a. How the MTP improves planning b. How to use the MTP 2) Introduction to Risk Management Process for CAF Operations	Lecture Guided discussion

Activity Steps	
	1) The instructor will conduct a lecture on the Maritime Tactical Planner and demonstrate how it can be used to enable effective planning of UWW training and exercises. (100 mins) 2) Using the Risk Management Document, the instructor will conduct a guided discussion on the Risk Management Process and how inputs at the director level can affect operations. (50 mins)

Resources	References
1) Maritime Tactical Planner lecture 2) Maritime Tactical Planner template	1) Joint Doctrine Manual, Risk Management for CF Operations

Notes to Instructor
As Maritime Technical Planner is in development, LO 2 can be utilized for the entirety of the lesson if MTP not available.

NTDC(A) Approval – 08/11/2019
NPTG HQ Approval – 13/12/2019
Version No: 1.3
MITE Crse ID : 122164

EO 002.02 - Prepare Operational Messages

002.02.02 – Water Space Management Requests

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 002.01.20
Time	**Classification**	**Instructor**
600 mins	SECRET	UWWD Instructor Guest Lecturer from SUBOPAUTH

Learning Objectives	Method
1) Coordination of Water Space Management (SUBOPAUTH) 2) Drafting RSS 3) Drafting UWON	Guest lecture Reading assignment T-Chart Lecture Practical

Activity Steps

1) Students will receive a guest lecture from SUBOPAUTH on water space management. (50 mins)
2) Students will read AHP-6 and complete a T-Chart identifying various water space management areas worldwide and their associated operating authority: (50 mins)

COUNTRY	SUBMARINE OPERATING AUTHORITY (SUBOPAUTH)
Eastern USA (i.e. VACAPES)	
Western USA (i.e. SOCAL, HIOP)	
United Kingdom	
France	
Spain	
Japan	

3) The instructor will conduct a lecture demonstrating the drafting of RSS messages using the CANSUBFOR lecture found on its webpage. (50 mins)
4) Students will develop a RSS template for use in follow on activities and throughout the course using message drafter. (50 mins)
5) The instructor will conduct a lecture demonstrating the drafting of UWON messages using the CANSUBFOR lecture found on its webpage. (50 mins)
6) Students will develop a UWON template for use in follow on activities and throughout the course using message drafter. (50 mins)
7) In pairs, Students will draft an RSS message and a UWON message based on a provided scenario with the instructor providing feedback on completion. (150 mins)
8) Students will draft a second RSS and UWON message individually based on a different scenario with the instructor providing feedback on completion. (150 mins)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) CANSUBFOR RSS lecture 4) CANSUBFOR UWON lecture 5) RSS/UWON Scenarios (2)	1) AHP-6 2) APP-11 3) MARLANTORD 3255-1 4) MARPACORD 3350-1

Notes to Instructor

Ensure guest lecturer is identified early to ensure availability.
SUBOPAUTH maintains the lectures independently from NTDC(A).
Instructor to detail the submission method as preferred.

EO 002.02 - Prepare Operational Messages

002.02.03 – Drafting OPTASK ASW/SCC

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 002.02.02
Time	**Classification**	**Instructor**
500 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Drafting OPTASK ASW/SCC messages	Guided discussion Practical

Activity Steps	
	1) Using message drafter and APP-11, the instructor will lead the class in drafting an OPTASK ASW/SCC template to be used for the rest of the course. (100 mins) 2) In pairs, students will draft an OPTASK ASW/SCC Supplement based on a scenario and the fleet's standing OPTASK ASW/SCC. (200 mins) 3) Each group will present their message to the class with the instructor providing feedback. (50 mins) 4) Using a partially complete message, students will draft a second OPTASK ASW/SCC Supplement individually based on a different scenario and submit it to the instructor. (150 mins) 5) Once corrected, the instructor will provide feedback to each student.

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) Standing OPTASK ASW/SCC 4) OPTASK ASW/SCC Sup scenario (2) 5) Partially complete OPTASK ASW/SCC Supp message	1) APP-11 2) ATP-28

Notes to Instructor
Instructor to detail the submission method as preferred.

EO 002.02 - Prepare Operational Messages

002.02.04 – Interpreting CASEX TABORDs

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 002.02.03
Time	**Classification**	**Instructor**
350 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Types and purpose of CASEX 2) Purpose of relaxations	Reading assignment Student presentation Guided discussion Lecture Practical

Activity Steps	
	1) Using AXP-1, each student will prepare a short brief on an assigned CASEXs identifying the following information: (50 mins) a. What is the purpose of the CASEX? b. What assets will / could be required? c. How much time should be allocated for the training? d. What geographic / area considerations should be considered? e. What sensors should / can be employed? f. What safety considerations are required? 2) Students will present their findings to the class with the instructor ensuring that the key information is discussed. (50 mins) 3) The instructor will demonstrate the use of the AXP-1 Order Table 3-1 by interpreting a sample TABORD with the class. (50 mins) 4) In pairs, students will interpret various TABORDs with each group provided a different message. On completion, they will present their findings to the class using the whiteboard or flip chart. (50 mins) 5) The instructor will conduct a lecture on CASEX relaxation tables to include: a. Authorizing authority b. A short description of each relaxation c. Stared vs normal relaxations based on AXP 1 Order Table 3-2 d. Critical and common relaxations On completion, the instructor will revisit the TABORD in step 2 and identify its relaxations. (50 mins) 6) In the same pairs, students will identify the relaxations found in their respective TABORDS using Order Table 3-2. (25 mins) 7) The instructor will conduct a lecture on exercises instructions to include: a. Use of AXP-1 Order Table 3-3 b. Instructions applicable to submarines c. Instructions applicable to aircraft d. Instructions applicable to ships On completion, the instructor will revisit the TABORD in step 2 and identify its exercise instructions. (50 mins) 8) In the same pairs, students will identify the exercise instructions found in their respective TABORDS using Order Table 3-3. (25 mins)

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Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) Sample TABORDS (5) 6) Order Table 3-1, 3-2 and 3-3 handouts	1) AXP-1

Notes to Instructor

Instructor to fully explain all potential Starred / Non-Starred Relaxations and how they are employed and affect the CASEX.

EO 002.02 - Prepare Operational Messages

002.02.05 – Planning CASEXs

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 002.02.04
Time	**Classification**	**Instructor**
150 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Planning considerations for CASEX 2) Conduct of CASEX	Instructor Lecture / Video Collaborative Problem Solving

Activity Steps	
	1) The instructor will to play an "attention grabber video" on conducting a CASEX. (Japanese ASW video) (accessible through Facebook) (5 mins) https://www.facebook.com/watch/?v=498832447474264 2) The instructor will show the video demonstrating the conduct of a CASEX in full before replaying it in slow time pausing at key events to provide commentary and solicit questions. (45 mins) 3) Students will form 4 groups, each assigned a CASEX scenario to plan and brief to the class using flip charts/whiteboards with details to include: (50 mins) a. Number of Surface units b. Number of Air units c. Initial Screen / Formation d. Location of Submarine(s) e. Distance between surface / air / subsurface units f. Approach method of surface units g. Planned phases / training conduct h. Safety considerations 4) Groups will present their results to the class with the instructor providing feedback and ensuring that key lessons are covered. (50 mins)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) Japanese ASW video (accessible through Facebook) https://www.facebook.com/watch/?v=498832447474264 6) CASEX video 7) CASEX scenarios (A3, C2, C3, C4)	1) AXP-1

Notes to Instructor

NTDC(A) Approval – 08/11/2019
NPTG HQ Approval – 13/12/2019
Version No: 1.3
MITE Crse ID : 122164

EO 002.02 - Prepare Operational Messages

002.02.06 – Drafting CASEXs TABORDs

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 002.02.05
Time	**Classification**	**Instructor**
350 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Drafting TABORDs for CASEX	Practical

Activity Steps	
	1) Utilizing the previous lesson's groups and scenarios, students will draft a CASEX TABORD message. (150 mins) 2) With the messages displayed to the class, the instructor will provide feedback for each TABORD. (50 mins) 3) Students will draft a second CASEX TABORD individually based on a different scenario. (100 mins) 4) Once corrected, the instructor will provide verbal and written feedback to each student. (50 mins)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) Order Table 3-1, 3-2 and 3-3 handouts 4) CASEX scenarios	1) AXP-1

Notes to Instructor
Instructor to detail the submission method as preferred. 3 different scenarios are available for students to use for practice allowing for additional practice if required and/or time permits.

NTDC(A) Approval – 08/11/2019
NPTG HQ Approval – 13/12/2019
Version No: 1.3
MITE Crse ID : 122164

EO 002.02 - Prepare Operational Messages

EC 002.02 - Prepare Operational messages

Scope	Conditions	Sequence
EO 002.02	Classified classroom with CSNI or secret standalone	Conducted after EO 002.02.06
Time	**Classification**	**Instructor**
150 mins	SECRET	UWWD Instructor

Scenario	Your ship is to participate in an upcoming CASEX. Having planned the serial with your ORO already, you must now prepare the TABORD along with any required water space requests by the end of the day.
Assessment Steps	Input: Students are given scenario details in addition to TABORD and UWON message templates. They will each be assigned a different CASEX as these messages will be used for the training exercise in PO 004. Performance: Students have 150 mins to draft a TABORD and UWON. Deliverables to include: • UWON • CASEX TABORD Output: The trainee must achieve 21/26 SATs to pass this EC. Feedback: Immediate verbal feedback provided upon completion of the brief, written feedback via EC 002.02 checklist provided within 72 hrs of receipt of the messages.

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Message drafter software 3) CASEX scenarios (including OPTASK ASW/SCC)	1) APP-11 2) AXP-1 3) MARLANTORD 3255-1 4) MARPACORD 3350-1

Notes to Instructor

UWON will be produced regardless of area of CASEX, this to evaluate students ability to draft this message and is not related to current/future posting of the students.
Instructor to communicate method of submission to the students based on classified content (CSNI email, Classified USB stick, paper copy, or instructor rotating through student workstations).

Limitations	
Impact of Failure	Feedback, reinstruct and reassess

Item	EC 002.02 - PREPARE OPERATIONAL MESSAGES	SAT	UNSAT	Remarks
1. Draft a UWON				
Required action addressees are listed		☐	☐	
Required info addressees are listed		☐	☐	
Classification identified		☐	☐	
Subject line is completed		☐	☐	
Subject Indicator Code (SIC) listed		☐	☐	
All references are included		☐	☐	
Message adheres to UWON format		☐	☐	
Message body info complies with provided exercise direction				
	Exercise participants are listed	☐	☐	
	OPAREA is specified	☐	☐	
	Date/Time of water space requirement is specified	☐	☐	
Drafter/Releasing Authority completed		☐	☐	
2. DRAFT a TABORD				
Required action addressees are listed		☐	☐	
Required info addressees are listed		☐	☐	
Classification identified		☐	☐	
Subject line is completed		☐	☐	
Subject Indicator Code (SIC) listed		☐	☐	
All references are included		☐	☐	
Message adheres to TABORD format		☐	☐	
Message body indicates conduct of exercise		☐	☐	
Relaxations and special instructions are listed		☐	☐	
Safety requirements are listed		☐	☐	
Exercise participants are listed		☐	☐	
Communications and reporting is indicated		☐	☐	
OPAREA is specified		☐	☐	
Date/Time of exercise is specified		☐	☐	
Drafter/Releasing Authority completed		☐	☐	

Score: / 26

The trainee must achieve 21/26 SATs to pass this EC.

Positive Feedback		Improvement Feedback	
Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PROTECTED B (when complete)

EO 002.03 - Disseminate Intelligence Information

002.03.01 – Platform Intelligence

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EC 002.02
Time	**Classification**	**Instructor**
200 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Briefing platform intelligence data 2) Assessing the impact of oceanography on subsurface units 3) Providing advice to command on UWW tactics	Student presentation

Activity Steps

1) Students will present their platform intelligence brief developed in EO 002.01.23 Establishing Capabilities/Limitations of Platforms. The platform intelligence information shall include: (25 mins/student, 200 mins total)
    a. Name/country
    b. Emitters
    c. Weapons (torpedoes, missiles & mines)
        i. Range/speed/run time of torpedoes
        ii. Method of homing
    d. Speeds (max and best endurance)
    e. Engineering configuration
    f. Operating depth
    g. National doctrine
        i. Mission
        ii. Competency
        iii. Experience
        iv. Tactics

    Students will also provide intelligence information on the impact of oceanography on their subsurface platform and provide advice to command on UWW tactics to be utilized against their platform.
2) Students will use this platform, oceanographic and tactical information during EO 002.03.02 CASEX briefing.

Resources	References
1) CSNI or secret standalone computer 2) Projector	1) DIAS Threat Web 2) Jane's catalogue

Notes to Instructor

NTDC(A) Approval – 08/11/2019
NPTG HQ Approval – 13/12/2019
Version No: 1.3
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EO 002.03 - Disseminate Intelligence Information

002.03.02 – CASEX briefing

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 002.03.01
Time	**Classification**	**Instructor**
400 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Building and presenting a CASEX brief	Student presentation

Activity Steps	
	1) Using the information and skills developed in EO 002.02.05 Planning CASEX and a provided template, students will create a CASEX brief covering their group's TABORDs developed in EO 002.02.06 Drafting CASEX TABORDs to include: (200 mins) a. Conduct; b. Execution; c. Submarine Safety. 2) Each group will present their CASEX briefs to the class with the instructor providing feedback. (50 mins/brief, 200 mins total) 3) Students will make corrections to their briefs prior to being presented to the ASW teams in PO 004 Execute Duties of the UWWD for Exercises (CASEX).

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) CASEX brief template	1) AHP-6 2) ATP-28 3) AXP-1 4) CFCD 106 5) MARLANTORD 3255-1 6) MARPACORD 3350-1

Notes to Instructor
Instructor and students will critique the CASEX briefs focussing on Conduct, Execution, and Safety considerations of the CASEX.

EO 002.03 - Disseminate Intelligence Information

EC 002.03 - Disseminate intelligence information

Scope	Conditions	Sequence
EO 002.03	Classified classroom with CSNI or secret standalone	Conducted after EO 002.03.02
Time	**Classification**	**Instructor**
450 mins	SECRET	UWWD Instructor

Scenario	Your ship is to participate in an upcoming CASEX. The water space has already been requested but you must still plan the CASEX, produce the TABORD and brief the rest of the operations team.
Assessment Steps	Input: Students are provided a UWON message from EC 002.02 Prepare Operational Messages, a scenario and a CASEX brief template. Performance: In groups of 2 or 3, students have 350 mins to plan an exercise IAW the provided scenario, draft the required TABORD and build the brief. Deliverables to include: • CASEX TABORD • CASEX brief On completion, each group will have 30 mins to present their brief to the instructor followed by a 3 mins debrief. (33 mins/group) Output: The trainees must achieve 18/22 SATs to pass this EC. Feedback: Immediate verbal and written feedback provided IAW EC 002.03 checklist.

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Message drafter software 3) CASEX scenarios (including OPTASK ASW/SCC) 4) TABORD message template 5) CASEX brief template	1) APP-11 2) AXP-1 3) MARLANTORD 3255-1 4) MARPACORD 3350-1

Notes to Instructor
The message preparation steps are not assessed in this EC and are assigned to students to support the preparation of the brief (which is also used prior to CASEX phase in PO 004).

Limitations	
Impact of Failure	Feedback, reinstruct, and reassess

Item	EC 002.03 - DISSEMINATE INTELLIGENCE INFORMATION	SAT	UNSAT	Remarks
1. Present CASEX brief to include:				
Explains intent of CASEX		☐	☐	
Provides outline of CASEX conduct/timeline		☐	☐	
Identifies friendly force participating units		☐	☐	
Identifies friendly forces capabilities		☐	☐	
Identifies friendly force mission		☐	☐	
Identifies OPFOR units		☐	☐	
Identifies simulated threats capabilities		☐	☐	
Identifies OPFOR mission		☐	☐	
Identifies Start/Stop times		☐	☐	
Identifies Out of Action periods		☐	☐	
Identifies start/stop position for all units		☐	☐	
Identifies OPAREAs/water space		☐	☐	
Identifies special instructions		☐	☐	
Describes execution of conduct		☐	☐	
Describes impact of oceanography		☐	☐	
Actions upon completion of the serial		☐	☐	
2. Provides explanation of Safety requirements for CASEX to include:				
Safety course		☐	☐	
Starred relaxations		☐	☐	
COMPLAN (pyro, UWT, ESUS)		☐	☐	
Red Flare procedures		☐	☐	
HOOKER/UNIFORM procedures		☐	☐	
Answers questions from Directing Staff		☐	☐	

Score: **/ 22** The trainees must achieve 18/22 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
<u>Assessor</u> I have discussed the results of this assessment with the candidate.		**<u>Candidate</u>** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PROTECTED B (when complete)

PC 002 - Complete UWW Exercise/Mission Preparations

Scope	Conditions	Sequence
PO 002	Classified classroom with CSNI or secret standalone	Conducted after EC 002.03
Time	**Classification**	**Instructor**
350 mins	SECRET	UWWD Instructor

Scenario	Your ship is ASWC for an upcoming exercise. As such, the ORO wants you to prepare an ASW Sup in addition to planning a CASEX E1.
Assessment Steps	Input: Students are provided the following: • OPORD • OPGEN • Responses to RFI/intelligence reports • WSM approval messages • OPTASK ASW (Supp) template with blanks • TABORD template • CASEX brief template • Scenario with direction to plan a CASEX E1 Performance: Students have 150 mins to complete the ASW Supp, produce a TABORD and prepare a CASEX brief. Deliverables to include: • OPTASK ASW Supp • CASEX E1 TABORD • CASEX E1 brief On completion, each student will have 20 mins to present their brief to the instructor followed by a 5 mins debrief. (25 mins/student, 200 mins total) Output: The trainee must achieve 26/33 SATs to pass this PC. Feedback: Immediate verbal feedback provided upon completion of the brief and written feedback via PC 002 checklist provided within 72 hrs of receipt of the messages.

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Message drafter software 3) Exercise OPORD and OPGEN 4) RFI responses 5) WSM approval messages 6) OPTASK ASW (Supp) template with blanks 7) TABORD template 8) CASEX brief template 9) Assessment scenario (CASEX E1)	1) APP-11 2) AXP-1 3) MARLANTORD 3255-1 4) MARPACORD 3350-1

Notes to Instructor
Brief focus must be on the conduct and safety aspects of the serial OPTASK ASW Supp is provided with blanks to simplify its development and shorten the evaluation

Limitations	Nil
Impact of Failure	IAW NPTORD 5400-54

Item	PC 002 - COMPLETE UWW EXERCISE/MISSION PREPARATIONS	SAT	UNSAT	Remarks
1. Complete OPTASK ASW Supp				
Adheres to OPTASK ASW format IAW APP 11		☐	☐	
Indicates which exercise/operation the Supp pertains to		☐	☐	
Identifies SUB THREAT		☐	☐	
Dictates SAU Instructions		☐	☐	
Identifies FASW TASK		☐	☐	
Identifies HELO employment		☐	☐	
Identifies HELO alert requirements		☐	☐	
Identifies ASWPN (attack criteria)		☐	☐	
Identifies TDA and TDZ		☐	☐	
Identifies ASTD policy		☐	☐	
Identifies ASIAP/SIAP		☐	☐	
2. Draft CASEX E1 TABORD				
Required action addressees are listed		☐	☐	
Required info addressees are listed		☐	☐	
Classification identified		☐	☐	
Subject line is completed		☐	☐	
Subject Indicator Code (SIC) listed		☐	☐	
All references are included		☐	☐	
Message adheres to TABORD format		☐	☐	
Message body info complies with provided exercise direction				
	Relaxations and special instructions are listed	☐	☐	
	Safety requirements are listed	☐	☐	
	Exercise participants are listed	☐	☐	
	Communications and reporting is indicated	☐	☐	
	OPAREA is specified	☐	☐	
	Date/Time of exercise is specified	☐	☐	
	Drafter/Releasing Authority completed	☐	☐	
3. Present CASEX E1 brief				
Explains intent of CASEX E1		☐	☐	
Provides outline of CASEX conduct/timeline		☐	☐	
Provide description of oceanographic impact on CASEX		☐	☐	
Provides explanation of Safety requirements for CASEX E1 to include:				

Safety course	☐	☐	
Starred relaxations	☐	☐	
COMPLAN	☐	☐	
Red Flare procedures	☐	☐	
HOOKER/UNIFORM procedures	☐	☐	

Score: **/ 33** The trainee must achieve 26/33 SATs to pass this PC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PROTECTED B (when complete)

NTDC(A) Approval – 08/11/2019
NPTG HQ Approval – 13/12/2019
Version No: 1.3
MITE Crse ID : 122164

PO 003 – Operate CS for UWW Weapons Firings

1. **Performance**: Conduct UWW Weapons Firings (D0020)

2. **Conditions**:

   a. Given:

      (1) Operations Room;
      (2) ASW Team;
      (3) References;
      (4) Command System;
      (5) SHINCOM;
      (6) Surface Vessel Torpedo Tubes;
      (7) Weapons firing range procedures; and
      (8) Mission/Exercise.

   b. Denied: NIL

3. **Environmen**t:

      (1) Ship (at sea/alongside & day/night); or
      (2) SEACOT/SEAWOLF trainer.

4. **Standard**: The UWWD shall operate CS for UWW weapons firings (D0020) by:

   a. Preparing for weapons firing by;
      (1) Preparing UWW firing orders (D0015/BT0120); and
      (2) Conducting pre-firing brief

   b. Conducting torpedo firing procedures (all modes) IAW NCPM 1001 by:
      (1) Operating Command System (B0050/T0003) to:
         (a) Compile ASW tactical picture (E0005) to include:
            i. Constructing a Contact Area of Probability (AOP) (C0100);
         (b) Perform UWW weapons systems serviceability checks (airshots) (D0010);
         (c) Performing UWW weapons firing (D0025); and
         (d) Perform weapons system misfire procedures (D0030/D0035);
      (2) Identifying system performance deficiencies, malfunctions or failures (BT0540).

5. **Training Limitations**:

EO 003.01 - Prepare for Weapons Firing

003.01.01 – Weapons capabilities and limitations

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EC 003.02.01
Time	**Classification**	**Instructor**
300 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Capabilities and limitations of UWW weapon system	Reading Assignment Best Summary Floating Flipchart Team Quiz

Activity Steps

1) Students will read CFCD 106 MARTI 2600 & 2610 and NCPM 1001 on the Mk46 Mod5A (SW) Lightweight torpedo. (120 mins)
2) Instructor will assign students with 1 or more of the following topics.
    a. Mk 46 characteristics
    b. Acoustic modes of operation
    c. Pre-Launch programming
    d. Depot settings
    e. Launch phases
    f. System initialization
    g. Search and acquisition
    h. Detect / acquisition range
    i. Attack
    j. Warhead detonation
    k. Countermeasure response
    l. Tactical floor theory
3) Students will create a Best Summary of the assigned topic(s) from the previous readings. (30 mins)
4) Students will then present their summary to the class explaining where they obtained the information. (30 mins)
5) The instructor will conduct a Floating Flipchart activity where students will move between charts adding information for the following categories when able: (30 mins)
    a. Mk 46 Characteristics to include
        i. Dimensions
        ii. Warhead
        iii. Fuel
        iv. Speeds
        v. Endurance
        vi. Depth limitations
        vii. Modes of operation
        viii. Receiver frequencies

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b. Identify Pre-launch programming to include:
   i. Launch modes
   ii. Search ceiling
   iii. Search depth
   iv. Search pattern;
   v. Course heading;
   vi. Search type; and
   vii. Tactical floor.
c. Identify pre-programmed (factory) settings.

6) On completion, the instructor will review the flipcharts and identify any missing information. (10 mins)
7) Students will form 2 teams with each team creating 20 questions on the weapon with their answers. (30 mins)
8) The teams will ask each other questions with the instructor determining if the answer is correct and adding information when required. (50 mins)

Resources

1) CSNI or secret standalone computer (1/student)
2) Projector
3) Flip chart/Whiteboard
4) Markers
5) Pen and paper

References

1) CFCD 106
2) NCPM 1001

Notes to Instructor

Instructor will determine the winning team during the team quiz.

EO 003.01 - Prepare for Weapons Firing

003.01.02 – Range Procedures

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 003.01.01
Time	**Classification**	**Instructor**
350 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Identifying applicable range procedures a. NAVORD 4995-0 b. NAVORD 4995-2 c. NAVORD 3137-2 d. AUTEC/PMFR procedures	Reading assignment Team quiz Best summary Role playing scenario

Activity Steps	
	1) Students will read NAVORD 4995-0 and 4995-2. (80 mins) 2) Students will divide into 2 teams and create 10 questions each that pertain to information within NAVORDs 4995-0 & 4995-2 and their associated answers. These should focus on to the following considerations: (20 mins) a. Electrically vs Mechanically safe b. Range Clear vs Range Foul c. Dangerous Proximity d. Check fire e. Out of Bounds areas f. Location of Signs g. Safety Pipes h. Firing Arcs i. Safety Organization 3) Each team will ask their questions to the other team with the references being available to locate the answer. The instructor will determine if the answers are correct and amplify any points as required. (50 mins) 4) In pairs or individually, students will read and created a Best Summary of an assigned section NAVORD 3137-2 from this list: (30 mins) a. NAVORD 3137-2 Chapters 1 – 5 b. NAVORD 3137-2 Chapters 6 c. NAVORD 3137-2 Annex A (excluding appendices) d. NAVORD 3137-2 Annex B e. NAVORD 3137-2 Annex C & D f. NAVORD 3137-2 Annex E 5) Students will present their Best Summary to the class, using the projector to display the content when required, with instructor ensuring that the relevant content has been discussed. (20 mins) 6) Students will read NAVORD 3137-2 Annex F & Appendix 1 Halifax Class TRI Firing Check-off List. (30 mins)

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	7) The instructor will assign each student one or more of the below positions with each student reviewing their assigned position's requirements IAW NAVORD 3137-2 in preparation for the next activity: (20 mins) a. Commanding Officer (CO) b. Operations Room Officer (ORO) c. Combat Systems Engineering Officer (CSEO) d. UWWD / ASWC e. Operations Room Supervisor (ORS) f. Range Safety Officer (RSO) g. Range Control h. SONAR Control Supervisor (SCS) i. Tubes Operator j. Data Collection Officer (DCO) 8) The instructor will lead the students in conducting the RCN range procedures IAW NAVORD 3137-2 Appendix 1 with each student role playing their position's requirements at each step and the instructor pausing to explain the process as required. (50 mins) 9) Students will read the Range Procedures for AUTEC (east coast) and PMFR (west coast). (50 mins)

Resources	References
1) DWAN computer (1/student) 2) CSNI or secret standalone computer (1/student) 3) Projector	1) AUTEC range manual 2) PMFR range manual 3) NAVORD 3137-2 4) NAVORD 3470-3 5) NAVORD 4995-0 6) NAVORD 4995-2 7) NAVORD 11000-0

Notes to Instructor

Instructor can have students cover more than 1 position during the Role-Playing scenario for range procedures IOT cover all positions.

EO 003.01 - Prepare for Weapons Firing

003.01.03 – Drafting Firing Orders

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 003.01.02
Time	**Classification**	**Instructor**
50 mins	UNCLASSIFIED	UWWD Instructor

Learning Objectives	Method
1) Drafting Firing Orders	EDI

Activity Steps	
	1) The instructor will provide the students with a soft copy of the Firing Orders template, display of copy to the class and explain its format indicating where to find the key information. (10 mins) 2) The instructor will demonstrate how to complete the Firing Orders template using a scenario including a CSE ammunition report. (10 mins) 3) Students will complete their own Firing Orders based on the same scenario and submit them to the instructor for feedback. The instructor will correct and return these before EC 003.01 to allow students to correct any misconceptions. (30 mins)

Resources	References
1) DWAN computer (1/student) 2) Projector 3) Firing Orders template 4) Firing Orders scenario	1) NAVORD 4995-0

Notes to Instructor
Instructor to detail the submission method as preferred.

EO 003.01 - Prepare for Weapons Firing

003.01.04 – Conduct Pre-Firing Brief

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 003.01.03
Time	**Classification**	**Instructor**
150 mins	UNCLASSIFIED	UWWD Instructor

Learning Objectives	Method
1) Building and presenting a pre-firing brief for weapons firing a. Safety considerations for weapons firing	Student presentation

Activity Steps	
	1) The instructor will demonstrate how to locate the Sea Training Group Sample Pre-Firing brief on DWAN. (5 mins) 2) In 3 groups, students will develop a Torpedo Weapon Pre-Firing brief based on the scenario and Firing Orders of the previous lesson (EO 003.01.03 Drafting Firing Orders) to include: (85 mins) a. Key Personnel b. Situation c. General Safety d. Range Clear/Foul procedures e. Check Fire procedures f. Safety Organization g. Communications h. Conduct 3) Each group will present their Pre-Firing Briefs to the class with the instructor and other students providing feedback. (20 mins/group, 60 mins total)

Resources	References
1) DWAN computer (1/student) 2) Projector 3) Sea Training Group Sample Pre-Firing Brief 4) Student drafted Firing Orders	1) NAVORD 3137-2 2) NAVORD 4003-6 3) NAVORD 4995-0 4) NCPM 1001 5) SEMS 6) SVTT Mod 9 manual

Notes to Instructor

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EO 003.01 - Prepare for Weapons Firing

EC 003.01 - Prepare for weapons firing

Scope	Conditions	Sequence
EO 003.01	Unclassified classroom with DWAN	Conducted after EO 003.01.04
Time	**Classification**	**Instructor**
50 mins	UNCLASSIFIED	UWWD Instructor

Scenario	Your ship is conducting a live torpedo firing in the OPAREAs IOT to test a recent upgrade to the SVTT system. You have been asked to provide Firing Orders for this evolution for command approval.
Assessment Steps	Input: Students are provided an OPAREA assignment message, a CSE ammunition report and the Firing Orders template. Performance: Students have 50 mins to draft Firing Orders. Output: The trainee must achieve 5/7 SATs to pass this EC. Feedback: Written feedback provided using EC 003.01 checklist.

Resources	References
1) DWAN computer (1/student) 2) Printer 3) Firing Orders template	1) NAVORD 3137-1 2) NCPM 1001

Notes to Instructor
Instructor to detail the submission method as preferred.

Limitations	
Impact of Failure	Reinstruct/feedback and reassess

Item	EC 003.01 - PREPARE FOR WEAPONS FIRING	SAT	UNSAT	Remarks
1. Draft Firing Orders				
Complies to template format		☐	☐	
References included (NCPM 1001 & NAVORD 4995-0)		☐	☐	
Date/Time of key events indicated IAW OPORD/OPAREA assignment message		☐	☐	
Ammunition information listed (type and serial number) IAW CSE report/scenario		☐	☐	
Safety organization identified IAW NAVORD 4995-0		☐	☐	
Range safety procedures IAW OPORD		☐	☐	
Approving authority identified IAW OPORD		☐	☐	

Score: / 7

The trainee must achieve 5/7 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 003.02 - Conduct Torpedo Firing Procedures

003.02.01 – Command System Setup

Conditions		Sequence
SEACOT (Primary) SEAWOLF (Secondary)		Instructed after EO 004.02.05
Time	**Classification**	**Instructor**
550 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Setting up Combat Management System a. Log In b. TAC Plot c. Picture Compilation d. Configure TEWA e. Tactical Decision Aids f. Stateboards	Demonstration Practical

Activity Steps	
	1) The instructor will explain and demonstrate the process for setting up Combat Management System (CMS 330) by following the provided checklist on the projector to include: (65 min) a. Log in to CMS 330 b. TAC Plot configuration c. Picture Compilation i. Configuration ii. TacPlot d. Configure TEWA i. Threat Score Calculation ii. Threat of the Day iii. UWW Inventory iv. Zone & Reference Points v. UWW Plan vi. UWW Weapons e. Tactical Operations i. Tactical Decision Aids ii. Stateboards 2) As a class, students will imitate the Combat Management System set up with prompting by the instructor for each step. (85 mins) 3) Students will set up the CMS individually without prompting by following the provided job sheet and checklist as guidance while the instructor monitors and assists as required. (400 mins)

Resources	References
1) SEACOT console (1/student) 2) Projector 3) CMS 330 setup, stateboard and inventory checklist 4) CMS 330 setup job sheet	1) CMS user's guide

Notes to Instructor

Instructor to continuously check to ensure students are performing tasks correctly and rectifies deficiencies immediately.

EO 003.02 - Conduct Torpedo Firing Procedures

003.02.02 – Develop ASW Tactical Picture

Conditions		Sequence
SEACOT (Primary) SEAWOLF (Secondary)		Instructed after EO 003.02.01
Time	**Classification**	**Instructor**
1100 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Operating the Command Management System a. ASW tactical picture compilation	Demonstration Practical

Activity Steps	
	1) The instructor will explain and demonstrate how to input various CMS functions contributing to the ASW tactical picture compilation to include: (150 mins) a. Entering the VOSIW (and temperature) b. Entering a Torpedo Danger Zone (TDZ) c. Entering limiting lines (LLSA) d. Entering Torpedo Danger Area (TDA) around subsurface track e. Entering a Datum f. Entering expanding Furthest on Circle (FOC) g. Building an AOP (circular & rectangular) h. Building a water space management area with compensatory allowance i. Inputting a screen / formation j. Entering a Torpedo Run Box (TRB) i. COBRA ii. BLOODHOUND k. Entering CORDON plan l. Entering BEAR plan m. Entering Manual Surface/Subsurface track n. Entering Torpedo Engagement Point o. Entering Reference Point p. Activating water space management q. Deleting contacts r. Repositioning contacts s. Employing the TMA tactical decision aid 2) As a class, students will imitate manipulating and inputting the various functions shown above with prompting by the instructor for each step. (250 mins) 3) Students will practice manipulating the various ASW tactical pictures functions of CMS 330 without prompting by following the provided job sheet while the instructor monitors and assists as required. (400 mins) 4) As a consolidation, students will pair up and enter specific tactical decision aids as prompted by the instructor with the teams verifying each other's work to confirm its accuracy before proceeding. (300 mins)

Resources	References
1) SEACOT console (1/student) 2) Projector 3) CMS 330 setup, stateboard and inventory checklist 4) ASW Tactical picture compilation job sheets	1) CMS user's guide

Notes to Instructor

For the consolidation exercise, the instructor should pair up experienced students with less experienced students and ensure that all functions are covered.

EO 003.02 - Conduct Torpedo Firing Procedures

EC 003.02.01 - Conduct torpedo firing procedures (Operate CS)

Scope	Conditions	Sequence
EO 003.02 CS	SEACOT (Primary) SEAWOLF (Secondary)	Conducted after EO 003.02.02
Time	**Classification**	**Instructor**
150 mins	SECRET	UWWD Instructor

Scenario	You are going on watch for the first time on board your new ship and have decided to review the functions of CMS 330 to ensure you are ready for future operations.
Assessment Steps	Input: Students are provided access to CMS 330. They may use any jobs aids developed during the previous lessons. Performance: Students have 40 mins to setup CMS 330 and manipulate the system IAW EC 003.02 checklist as prompted by the instructor. While the time allocated is 40 mins per student including debrief, there is no assessed time limit for students. Output: The trainee must achieve 17/21 SATs to pass this EC. Feedback: Immediate verbal and written feedback IAW EC 003.02.01 checklist.

Resources	References
1) SEACOT console (1/student) 2) Student developed job aid	1) CMS user's guide

Notes to Instructor
Instructor can assess two students concurrently

Limitations	
Impact of Failure	Retrain and reassess

Item	EC 003.02.01 - CONDUCT TORPEDO FIRING PROCEDURES (OPERATE CS)	SAT	UNSAT	Remarks
1. Setup CS				
Login to CS		☐	☐	
Select RADAR		☐	☐	
Configure sensor setup (HMS, CANTASS, SPS)		☐	☐	
Configure TEWA IAW course job aid to include:				
	Threat score	☐	☐	
	Zones & reference (OSUWW surveillance, threat evaluation, surveillance for HVU)	☐	☐	
	Threat of the day	☐	☐	
	UWW restriction	☐	☐	
	UWW inventory	☐	☐	
2. Setup Tactical Decision Aids to include:				
Plan Bear		☐	☐	
Plan CORDON		☐	☐	
TDA		☐	☐	
TDZ		☐	☐	
Limiting lines		☐	☐	
Torpedo Run Box		☐	☐	
Torpedo Threat Reckon		☐	☐	
FOC		☐	☐	
WSM		☐	☐	
3. Display Tactical Operations Stateboards				
ASW		☐	☐	
Threat		☐	☐	
Call Signs/Radio		☐	☐	
CASEX		☐	☐	

Score: / 21

The trainee must achieve 17/21 SATs to pass this EC.

Positive Feedback		Improvement Feedback	
Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PROTECTED B (when complete)

EO 003.02 - Conduct Torpedo Firing Procedures

003.02.03 – Weapons Firing Procedures

Conditions		Sequence
SEACOT (Primary) SEAWOLF (Secondary)		Instructed after EC 003.01
Time	**Classification**	**Instructor**
500 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Weapons firing procedures a. Primary i. Semi-Automatic ii. Manual iii. Semi-Automatic TCM b. Secondary c. Emergency (misfire procedures)	Demonstration Practical

Activity Steps	
	1) The instructor will explain and demonstrate how to conduct Primary mode firings in CMS for Semi-Automatic firing IAW NCMP 1001. (30 mins) 2) Students will practice the Primary mode Semi-Automatic firing procedures IAW NCPM 1001 while the instructor monitors and assists as required. (70 mins) 3) The instructor will explain and demonstrate how to conduct Primary mode firings in CMS for Manual firing. (30 mins) 4) Students will practice the Primary mode Manual firing procedure while the instructor monitors and assists as required. (70 mins) 5) The instructor will explain and demonstrate how to conduct Primary mode firings in CMS for Semi-Automatic TCM firing. (30 mins) 6) Students will practice the Primary mode Semi-Automatic TCM firing procedure while the instructor monitors and assists as required. (70 mins) 7) The instructor will review the Torpedo Firing Voice reports IAW NCPM 1001 and demonstrate how to conduct the Secondary mode firing procedure. (30 mins) 8) In pairs, students will practice conducting the Secondary firing mode procedure while the instructor monitors and assists as required. (40 mins) 9) The instructor will explain and demonstrate how to conduct the Emergency mode firing procedure. (10 mins) 10) In pairs, students will practice conducting Secondary firing mode procedure including an Emergency while the instructor monitors and assists as required. (20 mins) 11) Students will continue to practice the various firing modes in pairs while the instructor monitors and assists as required. (100 mins)

Resources	References
1) SEACOT console (1/student) 2) Projector 3) Firing Procedures handout from NCPM 1001 (student generated)	1) NAVORD 4995-0 2) NAVORD 4995-2 3) NAVORD 11000-0 4) NCPM 1001

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Notes to Instructor

Instructor to ensure all students successfully complete each firing before presenting another.

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NPTG HQ Approval – 13/12/2019
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EO 003.02 - Conduct Torpedo Firing Procedures

003.02.04 – ID faults

Conditions		Sequence
Unclassified classroom with DWAN SEACOT (Primary) SEAWOLF (Secondary)		Instructed after EO 003.02.03
Time	**Classification**	**Instructor**
150 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Conducting UWW weapons system serviceability check (airshots) 2) Identifying system performance deficiencies, malfunctions and failures	Reading assignment 3 Star Guided discussion

Activity Steps	
	1) In classroom, students will review NCPM 1001, NAVORD 4995-0 and NAVORD 4995-2 on Safety Requirements when conducting: (30 mins) a. Torpedo Air Shots i. At Sea ii. Alongside b. HOTTORP Firings 2) Students will identify their 3 most important safety considerations when conducting these firings. (5 mins) 3) Students will present their 3 Stars to the class justifying their choices with the instructor ensuring that all safety requirements are covered. (15 mins) 4) In SEACOT, the instructor will conduct guided discussion on common weapon system performance deficiencies, malfunctions and failures and demonstrate how to rectify them when able. Deficiencies to include (but not limited to): (100 mins) a. Operator fails to hook contact b. Operator fails to approve firing solution plan c. Tube selected indicates empty / not loaded d. Target is not within reaction zone e. Threat restriction exists f. Torpedo Settings Panel (TSP) is not in remote g. Communication between Combat Management System and TSP is interrupted/broken h. Tubes are electrically Safe “VETO” i. Tubes are mechanically Safe i. Flask not charged ii. Flex hose not connected j. Flask charge is below OPTIMUM pressure k. Actuator Switch not in ON position l. Test Set not communicating with TSP when conducting Airshots m. Students to provide other examples from their experience

Resources	References
1) DWAN computer (1/student) 2) SEACOT console (1/student) 3) Projector	1) NAVORD 4995-0 2) NAVORD 4995-2 3) NCPM 1001

Notes to Instructor

Practical assessment of these skills will be conducted throughout the rest of the course while operating CMS330.

EO 003.02 - Conduct Torpedo Firing Procedures

003.02.05 – Weapons Firing Procedures

Conditions		Sequence
SEACOT (Primary) SEAWOLF (Secondary)		Instructed after EO 003.02.04
Time	**Classification**	**Instructor**
1250 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Conducting weapons firing procedures in a range environment.	Demonstration Fishbowl

Activity Steps

1) Using the NAVORD 3137-2 Annex F & Appendix 1 Halifax Class TRI Firing Check-off List, the instructor will review and demonstrate how to conduct a Weapons firing procedure within a torpedo range environment. (50 mins)
2) Each student will develop muscle memory by practicing all firing modes 3 times on a firing range with other students filling the supporting roles IAW NAVORD 3137-2 Annex F & Appendix 1 Halifax Class TRI Firing Check-off List and NCPM 1001 with the instructor and observers (unused students) debriefing the class after each firing: (150 mins/student, 1200 mins total)
    a. Primary
        i. Semi-Automatic
        ii. Manual
        iii. Semi-Automatic TCM
    b. Secondary
    c. Firing in Emergency (misfire procedures)

Resources	References
1) SEACOT console (1/student) 2) Projector 3) TRI Firing Check-off list from NAVORD 3137-2 4) Firing Procedures handout from NCPM 1001	1) NAVORD 3137-2 2) NAVORD 4995-0 3) NAVORD 4995-2 4) NAVORD 11000-0 5) NCPM 1001

Notes to Instructor

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EO 003.02 - Conduct Torpedo Firing Procedures

EC 003.02.02 - Conduct torpedo firing procedures

Scope	Conditions	Sequence
EO 003.02	SEACOT (Primary) SEAWOLF (Secondary)	Conducted after EO 003.02.05
Time	**Classification**	**Instructor**
350 mins	SECRET	UWWD Instructor

Scenario	Your ship is undergoing the weapons certification process ahead of the deployment. As such, the UWW team is required to demonstrate the proper and safe firing procedures to STG.

Assessment Steps	Input: Students are provided a weapon certification event scenario. They may use any jobs aids developed during the previous lessons in addition to the NCPM 1001 and NAVORD 3137-2 handouts.
	
	Performance: Students have 43 mins including debrief to demonstrate the following deliberate and urgent torpedo firing on a range IAW NCPM 1001 and NAVORD 3137-1 (43 mins/student, 350 mins total):
	• Primary
	  ○ Semi auto
	  ○ Manual
	  ○ Semi auto TCM
	• Secondary
	• Emergency: ONE of the firings will result in a misfire initiating the emergency procedures.
	
	There will be one safety consideration (inject) per student to include:
	• Range foul
	• Dangerous proximity
	• Check fire
	
	Output: Trainee must achieve 35/44 SATs to pass this EC.
	
	Feedback: Immediate verbal and written feedback IAW EC 003.02.02 checklist.

Resources	References
1) SEACOT console (1/student) 2) Weapon certification scenario 3) TRI Firing Check-off list from NAVORD 3137-2 4) Firing Procedures handout from NCPM 1001	1) NAVORD 3137-2 2) NAVORD 4995-0 3) NAVORD 4995-2 4) NAVORD 11000-0 5) NCPM 1001

Notes to Instructor
Support manning IAW NWTP Vol 2

Limitations	
Impact of Failure	Retrain and reassess

Item	EC 003.02.02 - OPERATE CS FOR UWW WEAPONS FIRINGS	SAT	UNSAT	Remarks
1. Conduct Command System Set-up				
	Setup CS with UWWD primary, ORO secondary	☐	☐	
	Confirm setup of stateboards and inventory list IAW preplans and ASW briefs	☐	☐	
	Confirm screens, WSM, Tactical Decision Aids and Special Points input	☐	☐	
	Establish internal communications • Operations team • Bridge & Safety Organization • SVTT operator	☐	☐	
	Confirm safety measures have been implemented IAW NAVORD 4995-0 & 4003-6	☐	☐	
	Conduct closing up report to ORO IAW NCPM 1001	☐	☐	
2. Conduct weapon firing in primary mode IAW NCPM 1001				
Semi-automatic:				
	Set-up weapon firing solution in semi auto mode	☐	☐	
	Confirm weapon settings in UWW torpedo panel (verbally indicate to instructor)	☐	☐	
	Request permission to conduct weapons firing	☐	☐	
	Make tubes ready (upon receipt of permission to conduct weapons firing from ORO)	☐	☐	
	Conduct Firing in primary mode semi auto	☐	☐	
	Confirm weapon away with SVTT operator/UWW weapons panel	☐	☐	
	Direct ORO “make tubes safe”	☐	☐	
	Conduct post firing report and weapons status to ORO	☐	☐	
Manual:				
	Set-up weapon firing solution in manual mode	☐	☐	
	Confirm weapon settings in UWW torpedo panel (verbally indicate to instructor)	☐	☐	
	Request permission to conduct weapons firing	☐	☐	

Make tubes ready (upon receipt of permission to conduct weapons firing from ORO)	☐	☐	
Conduct Firing in primary mode manual	☐	☐	
Confirm weapon away with SVTT operator/UWW weapons panel	☐	☐	
Direct ORO “make tubes safe”	☐	☐	
Conduct post firing report and weapons status to ORO	☐	☐	
Semi-automatic TCM:			
Approve weapon firing solution plan	☐	☐	
Confirm weapon settings in UWW torpedo panel (verbally indicate to instructor)/adjust plan	☐	☐	
Request permission to conduct weapons firing	☐	☐	
Make tubes ready (upon receipt of permission to conduct weapons firing from ORO)	☐	☐	
Conduct Firing in primary mode semi auto TCM	☐	☐	
Confirm weapon away with SVTT operator	☐	☐	
Direct ORO “make tubes safe”	☐	☐	
Conduct post firing report and weapons status to ORO	☐	☐	
Common:			
Conduct post firing procedures (report weapons status to ORO & direct Torpedo Run Box input to CS)	☐	☐	
Manipulate CS IAW NCPM 1001	☐	☐	
Conducts voice reports IAW NCPM 1001	☐	☐	
3. Conduct weapon firing in secondary mode IAW NCPM 1001			
Setup firing in secondary firing mode IAW NCPM 1001	☐	☐	
Request permission to conduct weapons firing	☐	☐	
Make tubes ready (upon receipt of permission to conduct weapons firing from ORO)	☐	☐	
Conduct Firing in secondary mode	☐	☐	
Confirm weapon away with SVTT operator	☐	☐	
Direct ORO “make tubes safe”	☐	☐	

Conduct post firing procedures (report weapons status to ORO & direct Torpedo Run Box input to CS)	☐	☐	
Conducts voice reports IAW NCPM 1001	☐	☐	
4. Conduct weapon firing in emergency mode			
Directs "fire in emergency" to SVTT operator upon indication of weapon misfire	☐	☐	
Conducts voice reports IAW NCPM 1001	☐	☐	
5. Conduct safety procedures IAW NAVORDs & Range Safety Manual			
Adhere to range safety procedures	☐	☐	

Score: /44

Trainee must achieve 35/44 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PROTECTED B (when complete)

PC 003 - Operate CS for UWW Weapons Firings

Scope	Conditions	Sequence
PO 003	Unclassified classroom with DWAN SEACOT (Primary) SEAWOLF (Secondary)	Conducted after EC 003.02.02
Time	**Classification**	**Instructor**
450 mins	SECRET	UWWD Instructor

Scenario	Your ship is undergoing the weapons certification process ahead of the deployment. As such, the UWW team is required to demonstrate the proper and safe firing procedures to STG including the submission of Firing Orders. They do not wish to see you pre firing brief as this was already successfully demonstrated.

Assessment Steps	Input: Students are provided a weapon certification event scenario including an OPAREA assignment message. They may use any jobs aids developed during the previous lessons in addition to the NCPM 1001 and NAVORD 3137-2 handouts.
	
	Performance: Students have 50 mins to draft Firing Orders (50 mins) before demonstrating the following deliberate and urgent torpedo firing on a range IAW NCPM 1001 and NAVORD 3137-1 (50 mins/student, 400 mins total):
	• Primary: Each student will conduct <u>one</u> of the following in Primary mode firings:
	  ○ Semi auto
	  ○ Manual
	  ○ Semi auto TCM
	• Secondary: Each student will conduct <u>one</u> Secondary mode firing.
	• Emergency: one of the above firings will result in a misfire initiating the emergency procedures.
	
	There will be one safety consideration (inject) per student to include:
	• Range foul
	• Dangerous proximity
	• Check fire
	
	Output: Trainee must achieve 31/38 SATs on non-critical factors **AND** all critical factors to pass this PC.
	
	Feedback: Immediate verbal and written feedback IAW PC 003 checklist.

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Resources	References
1) DWAN computer (1/student) 2) Printer 3) SEACOT console (1/student) 4) Weapon certification scenario 5) OPAREA assignment message 6) Firing Orders Template 7) TRI Firing Check-off list from NAVORD 3137-2 8) Firing Procedures handout from NCPM 1001	1) NAVORD 3137-2 2) NAVORD 4003-6 3) NAVORD 4995-0 4) NAVORD 4995-2 5) NAVORD 11000-0 6) NCPM 1001 7) SEMS

Notes to Instructor
Timings calculated for 30 mins run/student followed by a 20 mins debrief and SEACOT reset time. Support manning IAW NWTP Vol 2 and briefed to not to prompt the students.

Limitations	All simulated, no method to determine if torpedo firing was successful. Timelines are compressed in training environment.
Impact of Failure	IAW NPTORD 4500-54

Item	PC 003 - OPERATE CS FOR UWW WEAPONS FIRINGS	SAT	UNSAT	Remarks
1. Draft Firing Orders				
Complies to template format		☐	☐	
References included (NCPM 1001 & NAVORD 4995-0)		☐	☐	
Date/Time of key events indicated IAW OPORD/OPAREA assignment message		☐	☐	
Ammunition information listed (type and serial number) IAW CSE report/scenario		☐	☐	
Safety organization identified IAW NAVORD 4995-0		☐	☐	
Range safety procedures IAW OPORD		☐	☐	
Approving authority identified IAW OPORD		☐	☐	
2. Conduct Command System Set-up				
Setup CS with UWWD primary, ORO secondary		☐	☐	
Confirm setup of stateboards and inventory list IAW preplans and ASW briefs		☐	☐	
Confirm screens, WSM, Tactical Decision Aids and Special Points input is IAW OPORD/scenario		☐	☐	
Establish internal communications • Operations team • Bridge & Safety Organization • SVTT operator		☐	☐	
Confirm external communications with: • Range control • Range craft		☐	☐	
Confirm safety measures have been implemented IAW NAVORD 4995-0 & 4003-6		☐	☐	
Conduct closing up report to ORO IAW NCPM 1001		☐	☐	
3. Conduct weapon firing in primary mode IAW NCPM 1001				
Contact reported to ORO		☐	☐	
Confirm target validity with range control		☐	☐	
Set-up weapon firing solution		☐	☐	
Request conduct of range safety procedures IAW NAVORD 4995-0		☐	☐	
Request permission to conduct weapons firing		☐	☐	

Make tubes ready (upon receipt of permission to conduct weapons firing from ORO)	☐	☐	*** CRITICAL**
Conduct Firing in primary mode	☐	☐	
Confirm weapon away with SVTT operator	☐	☐	
Direct ORO "make tubes safe"	☐	☐	*** CRITICAL**
Conduct post firing report and weapons status to ORO	☐	☐	
Conduct post firing procedures (report weapons status to ORO & direct Torpedo Run Box input to CS)	☐	☐	
Manipulate CS IAW NCPM 1001	☐	☐	
Conducts voice reports IAW NCPM 1001	☐	☐	
Conducts voice reports IAW range safety manual/NAVORD 3137-2	☐	☐	
4. Conduct weapon firing in secondary mode IAW NCPM 1001			
Contact reported to ORO	☐	☐	
Confirm target validity with range control	☐	☐	
Setup firing in secondary firing mode IAW NCPM 1001	☐	☐	
Request conduct of range safety procedures IAW NAVORD 4995-0	☐	☐	
Request permission to conduct weapons firing	☐	☐	
Make tubes ready (upon receipt of permission to conduct weapons firing from ORO)	☐	☐	*** CRITICAL**
Conduct Firing in secondary mode	☐	☐	
Confirm weapon away with SVTT operator	☐	☐	
Direct ORO "make tubes safe"	☐	☐	*** CRITICAL**
Conduct post firing procedures (report weapons status to ORO & direct Torpedo Run Box input to CS)	☐	☐	
Conducts voice reports IAW NCPM 1001	☐	☐	
Conducts voice reports IAW range safety manual/NAVORD 3137-2	☐	☐	
5. Conduct weapon firing in emergency mode			
Directs "fire in emergency" to SVTT operator upon indication of weapon misfire	☐	☐	
Conducts voice reports IAW NCPM 1001	☐	☐	
6. Conduct safety procedures IAW NAVORDs & area range safety manual			
Adhere to range safety procedures	☐	☐	* **CRITICAL**

Score: /38 /5 CRITICAL

Trainee must achieve 31/38 SATs on non-critical factors **AND** all critical factors to pass this PC.

Positive Feedback		Improvement Feedback	
Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PROTECTED B (when complete)

NTDC(A) Approval – 08/11/2019
NPTG HQ Approval – 13/12/2019
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PO 004 – Execute the Duties of an UWWD for Exercises/Operations

1. **Performance**: Execute the Duties of an UWWD for Exercises/Operations (T0001/T0005)

2. **Conditions**:

   a. Given:

      (1) Operations Room;
      (2) ASW Team;
      (3) References;
      (4) Command System;
      (5) SHINCOM;
      (6) Surface Vessel Torpedo Tubes;
      (7) Weapons firing range procedures; and
      (8) Mission/Exercise.

   b. Denied: NIL

3. **Environmen**t:

      (1) Ship (at sea/alongside & day/night); or
      (2) SEACOT/SEAWOLF trainer.

4. **Standard**: The UWWD shall execute the duties of an UWWD for exercises/operations (T0001/T0005)by:

   a. Operating safely with submarines by:

      (1) Advising command on UWW safety (BT0340);
      (2) Applying water space management (D0065);
      (3) Conducting submarine surfacing/diving ops (D0040); and
      (4) Performing SAR/distress/SUBSMASH communication procedures (A0020);

   b. Communicating internally/externally by:

      (1) Authenticating communication transmissions (A0030);
      (2) Encoding/decoding tactical information (A0025);
      (3) Utilizing brevity code words;
      (4) Utilizing voice reports/block reports; and
      (5) Enforcing EMCON restrictions (B0020);

   c. Coordinating UWW operations/exercises (BT0080/E0010/C0145) by:

      (1) Implementing UWW tactics (BT0310) by:
         (a) Coordinating the employment of sensors (BT0210/BT0200

/T0004/C0140);

(b) Coordinating the employment of weapons (BT0190); and

(c) Coordinating the employment of A/C (BT0230/BT0220/T0004/C0140);

(2) Contributing to the RMP (BT0250/BT0280) by:

(a) Amplifying track information (E0020)

(b) Evaluating correlated acoustic information (F0005);

(c) Correlating active/passive/non-acoustic information (C0105);

(d) Identifying changes in contacts readiness and tactical states (F0015); and

(e) Contributing to intelligence gathering operations (B0030);

(3) Adhering to ROE by:

(a) Implementing warnings; and

(b) Coordinating the control of weapons and sensors (BT0180/T0004/C0140);

(4) Assessing the effectiveness of UWW weapons (BT0410); and

(5) Assessing UWW exercises and operations (T0006).

5. **Training Limitations**

a. No onboard ship, training is being simulated in a synthetic training environment;

b. All oceanographic and environmental data is based on a set date and timeframe during the mission scenario and will not change; and

c. Some operational positions are simulated by staff and not individual operators.

EO 004.01 - Operate Safely with Submarines

004.01.01 – Submarine SAR/Distress/SUBSMASH

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after PC 003
Time	**Classification**	**Instructor**
100 mins	UNCLASSIFIED	UWWD Instructor Guest lecturer from CANSUBFOR

Learning Objectives	Method
1) Submarine SAR procedures 2) Types of Submarine distress 3) Actions upon determination of submarine distress 4) Alongside reactions as Ready Duty Ship	Guest Lecturer Reading assignment Class list

Activity Steps	
	1) Students will receive a guest lecture from CANSUBFOR to include: (50 mins) a. Submarine SAR procedures b. Types of Submarine distress c. Actions upon determination of submarine distress 2) Students will read the JTFA/P CONPLAN SUBSAR Annex 13 & 14. (20 mins) 3) In groups of 3 or 4, students will build a list of ships responsibilities based on the above reading. (10 mins) 4) Groups will present their lists to the class to generate a class list with the instructor ensuring all responsibilities are identified and highlight key lessons. (20 mins)

Resources	References
1) DWAN computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) CANSUBFOR lecture 6) CANSUBFOR Website 7) JTFA/P CONPLAN Appendix 13 & 14	1) ATP-10 2) ATP-28 3) ATP-57 4) AXP-1 5) COMSUBLANT/COMSUBPAC OPORD 2137 6) CFCD 103 7) MTP-57

Notes to Instructor
Ensure guest lecturer is identified early to ensure availability JTFA/P COMPLAN SUBSAR can be found on the CANSUBFOR DWAN website.

EO 004.01 - Operate Safely with Submarines

004.01.02 – Sub Safety Procedures

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 004.01.01
Time	**Classification**	**Instructor**
150 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Communications with Submarines a. Pyrotechnics b. Red Flare Procedures c. HOOKER procedures d. UNIFORM procedures e. Surfacing methods	Reading Assignment Demonstration Case Study

Activity Steps	
	1) Students will read the AXP-1 sections pertaining to: (50 mins) a. Underwater Code words b. SUS / ESUS codes c. Pyrotechnics d. Red Flare Procedures e. HOOKER procedures f. UNIFORM procedures g. Surfacing Methods 2) The instructor will explain and demonstrate how to conduct submarine safety procedures to include: (30 mins) a. Red Flare Procedures b. HOOKER procedures c. UNIFORM procedures d. Surfacing Methods ALPHA e. Surfacing Methods BRAVO 3) The instructor will conduct a guided exercise by presenting a series of scenarios on Submarine Safety procedures where the students will identify which safety procedure applies, demonstrate the appropriate communications and describe the required surface ship actions. (70 mins)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) Submarine Safety Procedures scenarios	1) AXP-1

Notes to Instructor
Instructor to ensure each student is involved in a minimum of one Case Study scenario.

NTDC(A) Approval – 08/11/2019
NPTG HQ Approval – 13/12/2019
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EO 004.01 - Operate Safely with Submarines

004.01.03 – Conduct Sub Safety Procedures

Conditions		Sequence
SEACOT (Primary) SEAWOLF (Secondary)		Instructed after EO 004.01.02
Time	**Classification**	**Instructor**
1000 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Conducting CASEX A3 Basic Tracking Exercise 2) Practical exercises on Sub Safety procedures and responses a. Communications with Submarines b. Pyrotechnics c. Red Flare Procedures d. HOOKER procedures e. UNIFORM procedures f. Surfacing methods ALPHA / BRAVO g. SUBSAR procedures	Practical

Activity Steps	
	1) Students will assume the duties of the UWWD for a CASEX A3 Basic Tracking exercise to include: a. Closing up ASW team b. Conduct on watch CASEX serial brief to ASW team c. Conduct detection, localization and tracking of a subsurface contact 2) Students will conduct the appropriate Submarine Safety procedures based on the situation to include: a. Communicating with submarine using underwater code words b. Employment of ESUS codes c. Response / awareness of submarine Pyrotechnics d. HOOKER procedures e. UNIFORM procedures f. Surfacing methods g. Red flare procedures h. SUBSAR 3) Each student will conduct 2 x 60 min runs. (120 mins/student, 1000 mins total) 4) Students not conducting the runs will observe and perform a Fishbowl exercise providing the UWWD student with one positive and one negative constructive observation on completion of each run before they are debriefed by the instructor. This will ensure students who are not participating in the run maintain engagement with the training and improve the UWWD development.

Resources	References
1) SEACOT or SEAWOLF 2) Student generated materials for CASEX A3 3) Submarine safety procedure printouts 4) JTFA/P CONPLAN Annex 13 & 14	1) ATP-10 2) ATP-57 3) AXP- 1 4) COMSUBLANT/COMSUBPAC OPORD 2137

Notes to Instructor

Support manning IAW NWTP Vol 2
The safety scenario can commence once contact is established with the submarine.
The focus is on safety procedures rather than the conduct of the CASEX.

EO 004.01 - Operate Safely with Submarines

EC 004.01 - Operate safely with submarines

Scope	Conditions	Sequence
EO 004.01	SEACOT (Primary) SEAWOLF (Secondary)	Conducted after EO 004.01.03
Time	**Classification**	**Instructor**
500 mins	SECRET	UWWD Instructor

Scenario	Your ship is conducting a CASEX A3 IVO a fishing area.
Assessment Steps	Input: Students are provided the CASEX A3 scenario develop in PO 002. They may use any student generated materials developed during the previous lessons in addition to the submarine safety procedures printouts. Performance: Students have 60 mins in SEACOT/SEAWOLF to complete the following events: (60 mins/student, 500 mins total) • Gain contact with the submarine (minimum 5 mins in contact) • Perform the appropriate HOOKER or UNIFORM procedure based on the situation • The CO will deem that it is unsafe for the submarine to continue and direct that the Sub must surface requiring the student to conduct the appropriate surfacing method ALPHA or BRAVO. • During this surfacing, the submarine will experience an emergency and launches red flares requiring the student to carry out emergency surfacing procedures and follow on SUBSAR considerations. Output: The trainee must achieve 19/23 SATs to pass this EC. Feedback: Immediate verbal and written feedback IAW EC 004.01 checklist.

Resources	References
1) SEACOT or SEAWOLF 2) Student generated materials for CASEX A3 3) Submarine safety procedure printouts 4) JTFA/P CONPLAN Annex 13 & 14	1) ATP-10 2) ATP-57 3) AXP- 1 4) COMSUBLANT/COMSUBPAC OPORD 2137

Notes to Instructor

Support manning IAW NWTP Vol 2
The safety scenario can commence once contact is established with the submarine.
The focus is on the safety procedures rather than the conduct of the CASEX.

Limitations	
Impact of Failure	Retrain and reassess

Item	EC 004.01 - OPERATE SAFELY WITH SUBMARINES	SAT	UNSAT	Remarks
1. Conduct HOOKER/UNIFORM procedures:				
Report fishing vessel to command		☐	☐	
Recommend initiate HOOKER procedures		☐	☐	
Direct HOOKER procedures to ASW team		☐	☐	
Report subsurface contact to command		☐	☐	
Recommend initiate UNIFORM procedures		☐	☐	
Direct UNIFORM procedures to ASW team		☐	☐	
2. Conduct surfacing method ALPHA/BRAVO * CO has directed to surface sub**				
Establish communications with submarine via UWT		☐	☐	
Direct safety course to be steered (or reciprocal)		☐	☐	
Direct cavitation above CIS		☐	☐	
Confirm with bridge Nav lights and echo sounder are on		☐	☐	
Direct ASW team to pass “PPP” via UWT (* not required for BRAVO)		☐	☐	
*** instructor to indicate white smoke or “QQQ”				
Direct ASW team to pass surface SITREP via UWT		☐	☐	
Direct ASW team to pass SAFE/UNSAFE code to submarine IAW AXP 1 (* not required for BRAVO)		☐	☐	
3. Conduct Red Flare procedures and SUBSAR				
Conduct external voice report to all units for emergency surfacing IAW AXP 1		☐	☐	
Ensure ship is driving safety course at CIS		☐	☐	
Direct ASW team to optimize sensor for contact detection		☐	☐	
Direct HMS to cease transmitting or switch to low power if in contact		☐	☐	
Direct/Confirm ASW team to monitor UWT		☐	☐	
Direct ASW team to mark own ship position and datum/bearing to flare		☐	☐	
Recommend to command recovery of towed sensors		☐	☐	
Recommend to command to standby to render assistance		☐	☐	
*** instructor to indicate UWT transmission that SUBSUNK				

Recommend to command to contact SUBOPAUTH to initiate SUBSUNK procedures	☐	☐	
Respond to instructor procedure question IAW JTF CONPLAN or ATP10	☐	☐	

Score: **/ 23** The trainee must achieve 19/23 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PROTECTED B (when complete)

NTDC(A) Approval – 08/11/2019
NPTG HQ Approval – 13/12/2019
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EO 004.02 - Communicate Internally/Externally

004.02.01 – Basic Communications Theory

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after PC 002
Time	**Classification**	**Instructor**
50 mins	UNCLASSIFIED	Communications Instructor & UWWD Instructor

Learning Objectives	Method
1) Basic communication frequencies ranges 2) Power limits and implications to operations 3) Environmental effects	Lecture

Activity Steps	
	1) The instructor will conduct a lecture on basic naval communications ensuring the that the following is covered: (50 mins) a. Frequency ranges (MF, HF, UHF, VHF etc.) b. Their capabilities and limitations c. Power limits and implication to operations d. Environmental effects

Resources	References
1) DWAN computer 2) Projector 3) Basic naval communication lecture	1) ACP-125 2) ACP-176 3) APP-1 4) APP-7 5) CFCD 106 6) NDSOD 7) NOSTP 200 8) Terminal user's instructions for shipboard integrated communications system (HFX CLASS)

Notes to Instructor

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EO 004.02 - Communicate Internally/Externally

004.02.02 – Tactical Communications

Conditions		Sequence
Classified classroom with DWAN		Instructed after EO 004.02.01
Time	**Classification**	**Instructor**
100 mins	UNCLASSIFIED	Communication Instructor

Learning Objectives	Method
1) Tactical communication considerations a. Types of nets b. EMCON c. COMSEC d. Code words and procedures	Lecture Practical Fish bowl

Activity Steps	
	1) The instructor will conduct a lecture on tactical communications ensuring the that the following is covered: (50 mins) a. COMPLANs b. Types of nets c. EMCON and restrictions to communications d. Misfit/charter (delay of changing or early changing of call signs) e. Spoofing f. Jamming g. GUERRILLA report h. NUCO i. Authentication j. COMSEC/EEFI k. Kick/Dance/Waltz l. Tactical codes/low grade codes 2) Using the provided exercise sheets, students will create transmissions incorporating NUCO, challenge and reply and low-grade codes before exchange it with another student for decoding. Students will continue practicing encoding and decoding in pairs using the provided job sheets while the instructor monitors and assists as required. (25 mins) 3) To consolidate, each student will encode an additional exercise transmission. They will then take turns “transmitting” to the class, first to decode the transmission correctly gets a point. Student with most points at the end wins. (25 mins)

Resources	References
1) DWAN computer 2) Projector 3) Tactical communication lecture 4) UNCLASS exercise communications crypto publications 5) Exercise sheets	1) ACP-125 2) ACP-165 3) ACP-176 4) APP-1 5) CFCD 106 6) NDSOD

Notes to Instructor

NTDC(A) Approval – 08/11/2019
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EO 004.02 - Communicate Internally/Externally

004.02.03 – Communication Tools

Conditions		Sequence
SEACOT (Primary) SEAWOLF (Secondary)		Instructed after EO 004.02.02
Time	**Classification**	**Instructor**
100 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) SHINCOM terminal operations a. Internal communications b. External communications c. Secure/Unsecure net 2) Chat tools (Transverse)	Demonstration Practical

Activity Steps	
	1) The instructor will list the communication nets to be monitored by the UWWD during exercises/operations, explain their use and demonstrate how to setup them up on a SHINCOM terminal to include: (15 mins) a. Internal nets b. External nets c. P-Nets 2) Students will set up their SHINCOM terminal as per the above demonstration while the instructor monitors and assists as required. (10 mins) 3) The instructor will explain the differences between operating on secure network vice unsecure network and demonstrate how to change between the two. (5 mins) 4) Students will switch between secure / unsecure nets with the instructor visually confirming it and correcting when required. (5 mins) 5) The instructor will explain and demonstrate how to conduct an internal communication check. (5 mins) 6) In pairs, students will conduct internal communication checks while the instructor monitors and assists as required. (10 mins) 7) The instructor will explain and demonstrate how to conduct an external communication check. (5 mins) 8) In pairs, students will conduct external communication checks while the instructor monitors and assists as required. (10 mins). 9) The instructor will explain and demonstrate the Transverse Chat tool capabilities to include: (15 mins) a. Setting up a room b. Joining an existing room c. Messaging a specific user 10) As a class, all students will join the JIIFC training chat room, while the instructor monitors and assists as required. This will conclude once all students have successfully exchanged a message with the rest of the class. (15 mins)

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Resources	References
1) SEACOT console (1/student) 2) Projector 3) List of unit call signs 4) Communication Aide-Memoire	1) ACP-125 2) ACP-165 3) ACP-176 4) AMSC 608 TC 5) APP-1 6) APP-7 7) ATP-1 8) CFCD 102 9) NOSTP 200

Notes to Instructor

Instructor to have students practice external comms checks in both Secure and Unsecure networks

EO 004.02 - Communicate Internally/Externally

004.02.04 – Brevity Code words

Conditions		Sequence
Unclassified classroom with DWAN		Instructed after EO 004.02.03
Time	**Classification**	**Instructor**
100 mins	UNCLASSIFIED	UWWD Instructor

Learning Objectives	Method
1) Introduction to common ASW brevity code words	Think-Pair-Share Jeopardy

Activity Steps	
	1) Students will read APP-7 and create a list of code words likely to be used during ASW exercises/operations. (25 min) 2) In pairs, students will compare their list to create a joint list. (10 mins) 3) Each pair will present their list justifying their selections IOT create a class list for future use throughout the course. (15 mins) 4) Instructor will conduct the Brevity Code Words Jeopardy game by dividing the class into equal teams (if able) and following standard Jeopardy rules to determine a winner. (50 mins)

Resources	References
1) DWAN computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) Jeopardy Game 6) Bell/Buzzer/Noise maker	1) ACP-125 2) ACP-165 3) ACP-176 4) APP-1 5) APP-7 6) CFCD 106 7) NDSOD

Notes to Instructor
Code words may be periodically quizzed throughout the remainder of the course to ensure students are comfortable with brevity code words

NTDC(A) Approval – 08/11/2019
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EO 004.02 - Communicate Internally/Externally

004.02.05 – Voice Reports

Conditions		Sequence
SEACOT (Primary) SEAWOLF (Secondary)		Instructed after EO 004.02.04
Time	**Classification**	**Instructor**
350 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) SAR format for voice reports 2) Internal ASW block reports 3) External ASW block reports 4) Quality Assurance of voice reports	Demonstration Fishbowl Reading assignment Practical

Activity Steps

1) The instructor will explain the use and demonstrate how to formulate a Situation Assessment Recommendation (SAR) report using a simple example. (10 mins)
2) Using a series of simple scenarios drawn at random, each student will formulate an internal report using the SAR method and present to the class with the instructor and other students provided feedback. (30 mins)
3) Students will read the NCPM 600 chapter on UWW internal voice reports while the instructor hands out the internal voice report aide-memoire. (50 mins)
4) The instructor will explain and demonstrate the following internal voice reports: (50 mins)
    a. Closing up report
    b. Initial Contact reports to include active and passive contacts
    c. ESM detection
    d. SLAM/SLAMEYE
    e. HOT/COLD and amplification reports
    f. Underwater resolve
    g. Torpedo/TCM report
    h. TCM amplification report(s)
    i. ASW plans (CORDON, BEAR, etc.)
    j. Equipment status reports
    k. SAUC/SAC intentions report
5) In pairs, students will practice formulating the above reports based on a provided scenario with one student giving the report while the other student critics it. The instructor will monitor and assist as required. (110 mins)
6) Students will read the APP-1 section on ASW External Voice reports while the instructor hands out the external voice report aide-memoire. (50 mins)
7) The instructor will explain and demonstrate the following external voice reports: (50 mins)
    a. Initial Contact reports to include:
        i. Lance Searcher/Diver
        ii. Helo Ball
        iii. SERPENT
        iv. POINTER

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b. DATUM reports
c. AOP reports
d. COBRA report(s)
e. BLOODHOUND report(s)
f. FLASH reports to include:
    i. Torpedo
    ii. Riser
    iii. Riser gone Sinker
g. Search and Attack Unit Commander report (SAU Cdr)
h. ASW Plans (CORDON, BEAR, etc.)
i. Equipment status reports

8) In pairs, students will practice formulating the above reports based on a provided scenario with one student giving the report while the other student critics it. The instructor will monitor and assist as required. (100 mins)

Resources

1) SEACOT console or Voice Trainer (1/student)
2) Projector signs
3) SAR report scenarios
4) Internal voice report Aide-Mémoire
5) Internal voice report practice scenarios
6) External voice report Aide-Memoire
7) External voice report practice scenarios

References

1) APP-1
2) NCPM 600

Notes to Instructor

Instructor may repeat reports as required. There is an opportunity to develop audio files for each report as an example.

EO 004.03 - Coordinate UWW Operations/Exercises

004.03.01 – ASW Support Operations

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 002.02.01
Time	**Classification**	**Instructor**
100 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Submarine control authorities and responsibilities 2) Command and control procedures of Allied submarines a. Direct support b. Associated support c. Integrated Operations d. BELLRINGER procedures	Reading assignment Cart sort Guided discussion

Activity Steps

1) Students will read ATP-1 chapter 12 article 12110 and 12232 – 12236 on Submarine Support operations/command and control. (30 mins)
2) For the following energizer activity, the instructor will prepare 4 copies of the below table along with its content in strips by column. When ready to start, there should be 4 blank tables and a full set of columns for each of them. In teams of 2 or 3, students will compete to re-create table 6A from ATP-28) as quickly as possible without access to references by taping the appropriate column under the correct heading and raising their hand to request instructor verification. (10 mins)

Example Setup (do one table for each team)

Functions	Area of Operations	Associated Support	Direct Support	Integrated Support
Classified				
classified				
classified				

3) Students will read ATP-28 annex 6A and the definitions of each type of support review the content from the previous exercise. (10 mins)
4) The instructor will conduct a guided discussion covering the following points: (25 mins)
   a. What is the most likely type of submarine support?
   b. Why would a submarine be in each type of support?
5) Students will ATP-28 article 1232 on BELLRINGER procedures. (5 mins)
6) The instructor will conduct a guided discussion covering the following points: (20 mins)
   a. Why would we use this procedure?
   b. When it is likely be ignored?
   c. What are the tactical implications using this procedure?

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Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) ATP-28 ANNEX 6A table cut outs	1) ATP 1 2) ATP 28

Notes to Instructor

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EO 004.03 - Coordinate UWW Operations/Exercises

004.03.02 –Water Space Management vs. Prevention of Mutual Interference

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 004.03.01
Time	**Classification**	**Instructor**
300 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Water Space Management vs Prevention of Mutual interference. 2) Water space management Tools 3) Submarine Safety 4) Types of water space 5) Calculation of Compensatory Allowance (CA) 6) Prevention of mutual interference tools 7) Permanent Submarine Operating Areas	Reading assignment Best summary T-Chart Guided discussion Practical Card sort

Activity Steps

1) The instructor will display the definitions of Water Space Management and Prevention of Mutual Interference where the entire class can review it. They should remain displayed throughout the lesson to emphasize the importance of not confusing the two. Using these, each student will conduct an essence exercise IOT to summarize the definitions into 16 words, then 8, 4 and 2 sharing their solution with the class at each step. (10 mins)
2) Students will read ATP-18 chapter 2 (including Annexes). IOT to focus the reading, the below T-Charts should be displayed on the board as it will be used in the following exercises. (75 mins)
3) Selecting an equal number from the topics below, students will create a Best Summary of their topic, present to the class and add it to the T-Chart. (100 mins)
    a. Principles and restrictions
    b. Safety separation
    c. Sub safe depth
    d. Skylight and stove pipes
    e. Authority to transit/dive to include
        i. Moving haven
        ii. Local Op area
        iii. Sub transit lane assignment
    f. Area partitioning to include
        i. ASWFA
        ii. JTAA
        iii. SAA
        iv. SPA
        v. SSL
        vi. TGSA
        vii. SGSA
        viii. NOTACK
        ix. Compensatory allowance

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g. GRASSHOPPER/FIELD/MEADOW
h. ASW attacking rules

Water Space Management Construct	**Best Summary**	**Weapon Employment**
Name of individual water space management tools such as SGSA.	What it stands for and the best summary for each entry	Who can use an ASW weapon and who is prohibited within each entry when applicable **AND** Is compensatory allowance required around construct

Prevention of Mutual Interference Construct	**Best Summary**
Name of individual prevention of mutual interference tools such as stovepipe.	What it stands for and the best summary for each entry

4) The instructor will review the definition of compensatory allowance from ATP-1 as well as how to calculate it for a Mark 46 torpedo from CFCD 106 before completing the T-chart by filling the last column (weapon employment) with the class. (15 mins)
5) Students will calculate compensatory allowance IAW ATP-28 for a provided scenario while the instructor monitors and assists as required. (50 mins)
6) Students will read AHP-6 to identify which countries have Permanent Submarine Operating Areas. (20 mins)
7) As a review and to emphasise the difference between WSM and PMI, students will take turns drawing the name of various WSM and PMI from a hat and adding them into the appropriate column on a flip chart or whiteboard. (30 mins)
DS Solution:

WSM	PMI
ASWFA JTAA SAA SPA SSL TGSA SGSA NOTACK GRASSHOPPER MEADOW	HORIZONTAL SAFETY SEPARATION VERTICAL SAFETY SEPARATION SKYLIGHT, STOVEPIPE MOVING HAVEN SUBMARINE TRANSIT LANE PERMANENT OPERATION AREA

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Marker 5) Compensatory allowance calculation scenario	1) AHP-6 2) ATP-1 3) ATP-18 4) ATP-28 5) AXP-1 6) CFCD 106

Notes to Instructor

Instructor and students may use maps or illustrations such as the WSM/PMI diagrams in ATP-18 to demonstrate WSM areas.

EO 004.03 - Coordinate UWW Operations/Exercises

004.03.03 – Coordinate the Employment of Fixed Wing AC

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 004.03.02
Time	**Classification**	**Instructor**
300 mins	SECRET	UWWD Instructor Guest Lecturer from LRPA Squadron

Learning Objectives	Method
1) Capabilities and limitations of LRPA assets 2) SAC/ ASAUC duties 3) Employment of Fixed Wing assets	Reading assignment Guest lecture 3 Stars

Activity Steps

1) Students will read the following documents and build a list of questions for the upcoming subject matter expert: (160 mins)
    a. ATP 1 Vol I article 6100-6102, 6107 (including table 6-4), article 12221-12222.
    b. ATP 28 Vol 1 Chapter 4 and annex 4A01-A05, 4A12, 4A14-A19, 4A32-A33.
    c. CFCD 106 MARTI 2000 section 1-25 and MARTI 2010 section 1-3, and MARTI 2580.
    d. ATP-18 Article 0502 section 1-4.
    e. CFCD 120 index.
2) Students will receive a guest lecture on LRPA operations from a TACCO or pilot to include: (120 mins)
    a. Capabilities and limitations
    b. Concept of operations
    c. Crew responsibilities
    d. Communication (chat, radios, link)
    e. Sensors
    f. Command and control (ACU, SAUC, SAC)
    g. What mission(s)/support can they provide?
    h. Implications of re-rolling mid mission
    i. What can/cannot be done concurrently, implications of altitude to mission and ranges?
    j. Weapons
    k. Answer to questions from the readings
3) As a review of the content, students will develop a list of their 3 stars from the readings or guest lecture and share them with the class. (20 mins)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) LRPA operations lecture	1) ATP-1 2) ATP-18 3) ATP-28 4) CFCD 106 5) CFCD 120

Notes to Instructor

Guest lecturer to supply any supporting media such as a PowerPoint and must be identified early.

EO 004.03 - Coordinate UWW Operations/Exercises

004.03.04 – Coordinate the Employment of Rotary Wing A/C

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 004.03.03
Time	**Classification**	**Instructor**
300 mins	SECRET	UWWD Instructor Guest Lecturer from Maritime Helicopter Squadron SAC Instructor

Learning Objectives	Method
1) Capabilities and limitations of rotary wing air assets 2) Allied and Canadian doctrine 3) SAC/ASAUC duties 4) Types of attack	Reading assignment Guest lecture Floating flip chart Guided discussion Case study

Activity Steps

1) Students will read the following documents and build a list of questions for the upcoming subject matter expert: (100 mins)
    a. ATP-28 Vol I article 0512-0525 on ASW Screening
    b. CFCD 106 MARTI 11100 and TACNOTE 11210, 11225, 11240 and 11245
2) Students will receive a guest lecture on Maritime Helicopter operations from a TACCO or pilot to include: (100 mins)
    a. CH 148 overview
    b. ASW sensors
    c. ASW execution
    d. Answer to questions from the readings
3) Students will read the following on the various types of air attacks (INFOTAC/VECTAC/HOVERTAC/SELFTAC): (30 mins)
    a. ATP-28 article 0938
    b. ATP-01 VOL 1 article 12363
4) In 4 groups, students will conduct a floating flip chart activity where each group will have 2 mins to write as many facts as they can remember about a specific type of attacks listed on their starting flip chart. All teams will move to the next flip chart and add as much information as able. This will repeat every 2 minutes until every chart has been visited and not further information can be added. The flip charts will have the following headings: (10 mins)
    a. INFOTAC
    b. VECTAC
    c. HOVERTAC
    d. SELFTAC
5) Groups will present the content from their final chart to the class with the instructor providing feedback and adding any missing information. (10 mins)

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	6) The SAC instructor will conduct a guided discussion covering the following information: (20 mins) a. SAC/ASAU(C) duties, b. Implied and specific tasks and delegation (i.e. TACOM not TACON) c. When does SAC automatically shift? d. What does this imply? 7) Using the provided scenarios, the SAC instructor will conduct a case study of various reports between the SAC and the A/C missing key information such as a specific mission, time, or asset control delegation. The students will highlight the short comings of each situation and its implications to the exercise/mission. The aim is to emphasise to the class the importance of properly compiling and executing SAC/SAU(C) taskings, sitreps and swaps. (30 mins)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) CH148 Cyclone maritime helicopter overview lecture 4) CH148 Cyclone ASW sensors lecture 5) CH148 Cyclone ASW execution lecture 6) SAC case studies	1) ATP 1 2) ATP-28 3) CFCD 106

Notes to Instructor

Guest Lecturer to supply media such as PowerPoint and be identified early.

EO 004.03 - Coordinate UWW Operations/Exercises

004.03.05 – Aircraft Familiarization

Conditions		Sequence
Shearwater/Patricia Bay Squadron (Ops or Trg Sqn)		Instructed after EO 004.03.04
Time	**Classification**	**Instructor**
200 mins	SECRET	Tour Guide

Learning Objectives	Method
1) Capabilities and limitations of CH-148	Tour

Activity Steps	
	1) Students will attend a tour of the Shearwater or Patricia Bay maritime helicopter squadron facilities with a focus on A/C familiarization and real time discussion on operations and training.

Resources	References
1) Transporting to/from location	

Notes to Instructor
Ensure tour is booked early to ensure availability

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004.03.06 – Anti-Ship Torpedo Defense

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 004.03.05
Time	**Classification**	**Instructor**
450 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Generic torpedo countermeasure manoeuvring 2) Task Group Anti-ship Torpedo Defence (TADPOLE's) 3) National TCM doctrine 4) Optimizing sensors for ASTD a. Active sensor use (active and passive modes) b. Passive sensor employment c. Shipborne countermeasure employment and interference d. Use of UWT as a sensor	Lecture Collaborative problem solving Reading assignment T-Chart Video Guided discussion Practical Case study

Activity Steps

1) Using ATP-3, the instructor will explain and demonstrate how to interpret Torpedo Countermeasure Manoeuvring providing students with the Pros and Cons of the one of the plans as an example for the next activity. (20 mins)
2) With the class divided into 2 or 3 groups, each group will select an equal number of manoeuvring plans to interpret identifying Pros and Cons for each. (20 mins)
3) Each group will present their plan along with its Pros and Cons to the class. (15 mins)
4) Students will read the ATP-28 ASTD content covering the TADPOLEs and GRID tables. (40 mins)
5) Students will complete the below T-Chart with the information applicable to each TADPOLE with the instructor making any required corrections and highlighting importing takeaways. (20 mins)

TADPOLE in Effect	MEANING
IAW ATP-28	IAW ATP-28

6) In pairs, students will select an equal number of ASTD grids from ATP-28 to review with the aim of briefing them to the rest of the class. (15 mins)
7) Students will present their GRID to the class indicating its selection criteria with the instructor making any required corrections and highlighting importing takeaways. (50 mins)
8) Students will read CFCD 106 MARTI 2120 on the national TCM doctrine. (50 mins)
9) The instructor will show the video on the correct way to conduct the national TCM doctrine in full before starting over in slow time pausing to explaining each part of the TCM and answer questions. A final viewing in full is recommend solidifying the lesson if time permits. (50 mins)

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10) The instructor will conduct a guided discussion on the national TCM doctrine discussing the following points: (50 mins)
    a. How do you think the TCM works against the various seeker heads?
    b. What happens if you conduct a TCM using a different speed?
    c. What would happen if you place the decoys incorrectly?
    d. How can you maintain contact with the submarine?
    e. Will counter firing have an impact on your overall success?
11) Students will label a provided TCM diagram from memory identifying as much information as able. (25 mins)
12) Students will trade their diagram with another student with each student correcting the other using the completed TCM diagram provided by the instructor. (15 mins)
13) The instructor will show an incorrect TCM videos with students critiquing the conduct of the TCM as a class. (30 mins)
14) The instructor will conduct a guided discussion on optimizing sensors for ASTD discussing the following points: (50 mins)
    a. Based on everything you have learned this lesson and you experience; how can you optimize your sensors to detect torpedo early?
    b. What sensors are most likely to detect a torpedo?
    c. How can active sensors contribute to a task groups overall torpedo defence?
    d. How can active sensors hinder a task groups overall torpedo defence?

Resources

1) CSNI or secret standalone computer (1/student)
2) Projector
3) Flip chart/Whiteboard
4) Markers
5) TCM video (correct)
6) Unlabelled TCM diagram
7) Completed TCM diagram
8) Incorrect TCM case study video(s)

References

1) ATP-1
2) ATP-3
3) ATP-28
4) CFCD 106

Notes to Instructor

EO 004.03 - Coordinate UWW Operations/Exercises

004.03.07 – ASW Screening

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 004.03.06
Time	**Classification**	**Instructor**
350 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) ASW Defence and Screening considerations a. Defence in depth b. Units / Sensors / Weapons c. PSR / TSR d. Fundamental Tasks e. Look zones f. Shadow zones g. Enemy COAs h. Above Water considerations (ASMD, etc.) i. Submarine support 2) ASW Screens j. Kilo k. Expanding screens 3) Sector screen design	Reading assignment 5 Stars Collaborative problem solving Demonstration Homework

Activity Steps	
	1) Students will read ATP-28 Vol 1 Chapter 8 on ASW Defence and Screening and article 0927 on Torpedo Danger Area. (80 mins) 2) On completion, students will identify and order their 5 most important considerations on ASW Defence and Screening. (20 mins) 3) Students will present their 5 Stars to the class justifying their selection with the instructor adding any important missed information. (40 mins) 4) In groups of 2 or 3, students will calculate TDA and TDZ sizes based on different provided scenarios. (30 mins) 5) Each group will present their scenario calculations to the class. (10 mins) 6) Students will read AXP-5 EXTAC 636 ANNEX A02 and ATP-28 Vol II chapter 2 and 3 on ASW Screening. (70 mins) 7) The instructor will demonstrate how to develop a screen Kilo based on a given scenario. (30 mins) 8) In the same group as step 4, students will develop a screen Kilo based on different scenarios. (50 mins) 9) Each group will present their scenario with their solution to the class. (20 mins) 10) Students will develop a screen Kilo based their own individual scenario. This will be assigned as “homework” and must be submitted to the instructor within 3 days with the instructor providing feedback within 72 hours of submission.

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Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) TDA/TDZ calculation scenarios 6) Screen Kilo scenarios (15)	1) ATP-28 2) AXP-5

Notes to Instructor

EO 004.03 - Coordinate UWW Operations/Exercises

004.03.08 – Submarine Search Tactics/Considerations

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 004.03.07
Time	**Classification**	**Instructor**
300 mins	SECRET	UWWD Instructor Guest Lecturer from Submarine CO/ORO

Learning Objectives	Method
1) Submarine Search Tactics for Diesel / SSK / SSC a. Shallow b. Deep 2) Submarine Search Tactics for Nuclear / SSN / SSBN / SSGN c. Shallow d. Deep 3) SM Covert Operations 4) SM Overt Operations 5) Screen considerations to include: e. Look zones f. Shadow zones g. Enemy COAs 6) Submarine Mission	Reading assignment Guest lecturer Tabletop

Activity Steps	
	1. Students will read ATP-18 chapter 5 and CFCD 106 MARTI 2002 on submarine tactics. (100 mins) 2. Students will receive a guest lecture from a submarine CO or ORO on Canadian submarine tactics covering the learning objectives of the lesson. (50 mins) 3. Using a provided scenario including oceanographic information, the instructor will conduct a tabletop exercise utilizing a fleet in a box to discuss the following tactical considerations: (150 mins) a. Why is the sub at PD/Deep? b. Submarine's mission vs ship's mission c. Importance of intelligence d. Shallow vs deep water e. Normal safe operating depth f. Avoidance of weather (diesel boats) g. Torpedo avoidance (sub) h. Evasive steering (i.e. 5-50-5); i. Best method for surface units to avoid detection j. How do subs gain/maintain contact? k. Importance of signature management

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) Submarine tactic lecture 6) Table top scenario 7) Fleet in a box 8) Laminated charts	1) ATP-1 2) ATP-18 3) ATP-28 4) AXP-5 5) CFCD 106

Notes to Instructor

Guest lecturer must be identified early to ensure availability.
Submarine tactic lecture and scenario maintained by NTDC(A) SUBDIV
If available and time permits, students are to watch a submarine firing in the Sub trainer (East coast only)

EO 004.03 - Coordinate UWW Operations/Exercises

004.03.09 – ASW Tactical Implications/Considerations

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 004.03.08
Time	**Classification**	**Instructor**
600 mins	SECRET	UWWD Instructor Guest Lecturer from Submarine CO/ORO

Learning Objectives	Method
1) ASW Fundamental Tasks a. Stop b. Explore c. Hamper d. Mask 2) Implications/considerations a. Geo political b. Go/no go c. Shipping lanes d. Met/ocean conditions e. Static/moving PIM f. Off/defensive ops g. LFA tactics h. Multi-statics	Reading assignment Collaborative problem solving Presentation Demonstration Tabletop Case study

Activity Steps	
	1) Students will read the ATP-28 chapter titled ASW Coordination. (100 mins) 2) In 4 groups, students will fill a blanked version of ATP-28 table 9-1with each group completing a row without the use of references. (15 mins) 3) Each group will present their row's content and attempt to complete it with the assistance of the rest of the class. (20 mins) 4) On completion, the instructor will display the completed table 9-1 and correct any errors in the student generated table. (5 mins) 5) Students to read CFCD 106 TACNOTE 2115. (20 mins) 6) The instructor will present the "Multi Statics and LFA tactics" recorded lectures on Multi Statics and LFA. (50 mins) 7) The instructor will utilize the transit of the Strait of Gibraltar scenario to identify tactical implications and considerations to be considered when developing a plan (all LO2). The instructor may use a fleet in a box and a chart as part of this demonstration and use ATP-18 articles 0915-0916 as a reference. (50 mins) 8) In 2 groups, students will utilize 2 news scenarios and associated charts to plan a mission, highlighting their tactical considerations and identifying the fundamental tasks. Any significant decisions will be justified by listing the pro and cons used to make it. (100 mins) 9) Each group will present their scenarios to the class with the instructor providing feedback as required. (80 mins) 10) In the groups of the previous activity, students will analyse 2 new scenarios with associated charts. Within each group, half will consider the problem from the

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	submarines perspective while the other half will consider it from the ships each with the aim of answering the following questions: (100 mins) a. What areas is the Sub likely to avoid and why? b. What areas will the Sub likely aim to exploit and why? c. Where can the Sub not operate? d. Where will the Sub aim to operate aggressively? e. Where are the surface ships most vulnerable? f. Surface ship's perspective g. What areas will a surface ship likely avoid and why? h. What areas will a surface ship aim to exploit? i. Do the above answers change when using a covert or overt posture? j. What sensor will be optimised to search a given area? k. Where can the ship not operate? l. What other assets are available to support the ship's mission noting that different nations have varying capabilities? m. How will the ship transition from moving to static operations/screening? n. Can the ship front load sanitisation of a static area and if so, what are the implications of this action? o. Where will the ship aim to engage the OPFOR? p. How will weather impact all of the above? 11) Both groups will present their scenario and answers to the above questions starting with the students representing the surface ship's perspective followed by students considering the submarine's perspective. (60 mins)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) Multi Statics Presentation 6) LFA tactics Presentation 7) Straits of Gibraltar transit scenario package a. Laminated chart b. Scenario details c. Oceanographic conditions d. Weather conditions 8) Fleet in a box 9) Student scenario packages (4) a. Laminated charts b. Scenario details c. Oceanographic conditions d. Weather conditions	1) ATP-18 2) ATP-28 3) MILOC manual 4) RP 33

Notes to Instructor

The scenarios are designed to include offensive and defensive postures, good and poor acoustic waters, high and low end threats and differing own force capabilities to ensure a variety of tasks and missions are explored.

If possible, the support from a submarine CO or ORO can be obtained to conduct the final activity and provide the submarine's perspective.

EO 004.03 - Coordinate UWW Operations/Exercises

004.03.10 – Command Classification and DAMASS

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 004.03.09
Time	**Classification**	**Instructor**
350 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Determining command classifications 2) Damage Assessment	Reading assignment T-Chart Jig saw Demonstration Collaborative problem solving Case study Best summary

Activity Steps

1) Students will read AXP-5 EXTAC 198 on command classifications. (60 mins)
2) As a class, the students will complete the below T-Chart covering Command Classification criteria with the instructor correcting any mistakes as required. (30 mins)

Command Classification	Classification Criteria	Example
NONSUB	IAW References	
POSSUB	IAW References	
LOW 1 & 2	IAW References	
HIGH 3 & 4	IAW References	
PROBSUB	IAW References	
CERTSUB	IAW References	

3) Each student will select an equal number of charts from AXP-5 EXTAC 198 for which they will become the “expert”: (10 mins)
    a. Visual
    b. ESM
    c. RADAR
    d. Active SONAR
    e. Sonobuoys
    f. Passive SONAR
    g. MAD
    h. Torpedo/Missile detection or alert means
    i. Tactical coherence criteria
4) Students will present their table to the class with the instructor correcting any mistakes and amplifying significant points. (40 mins)

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	5) The instructor will explain and demonstrate the Command Classification calculations IAW AXP-5 EXTAC 198. (15 mins) 6) In pairs, students will calculate the command classification for an assigned scenario. (15 mins) 7) Each pair will present their scenario and solution to the class with the instructor correcting any mistakes. (20 mins) 8) The instructor will present a series of case studies presenting a variety of command classification problems. As the instructor describes the problem, students will calculate the command classification and provided their answer to the class. In some scenarios, the instructor will continue to progress the problem IOT for students to adjust their command classification by incorporating the new information. (110 mins) 9) Students will read the CFCD 120 TACAID on damage assessment/Category. (20 mins) 10) Each student will select an equal number of damage assessment/categories from CFCD 120 for which they will develop a best summary. (15 mins) 11) Students will present their summary to the class with the instructor emphasising key lessons as required. (30 mins)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) Command classification scenarios (15)	1) ATP-1 2) ATP-28 3) AXP-5 4) CFCD 120

Notes to Instructor

The instructor can provide figure 198-1 Submarine classification flow chart from AXP-5 EXTAC 198 to assist students
CFCD 120 TACAID is available through the vault

EO 004.03 - Coordinate UWW Operations/Exercises

004.03.11 – Initial Responses

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 004.03.10
Time	**Classification**	**Instructor**
300 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Initial contact actions a. Initial contact gain b. Under water resolve c. Initial contact report d. UWWD report i. Manoeuvring requirement ii. Marking/covering with weapon iii. Command classification iv. Plans v. Sensor/AC employment e. SAC/SAU(C) report i. Initial report ii. Amplification reports iii. Lost contact report iv. Plans v. Warnings vi. Weapon priorities/attack intentions f. Primary Warfare Commander actions and reports g. ASW battle rhythm	Lecture Video Matching exercise Reading assignment Card sort Demonstration Practical Guided discussion Lecture

Activity Steps	
	1) The instructor will conduct a lecture introducing the various steps to be carried out upon detection of a sub-surface contact by presenting the flowchart contained in NCPM 600. As this reference may contain dated material, the instructor should note any changes in the process based on the learning objectives. (10 mins) 2) The instructor will show the initial responses demo video, if unavailable the instructor will conduct this demonstration by using a provided scenario script. Either way, the instructor will point out individual steps as they occur. (10 mins) 3) Students will complete the underwater resolve exercise by matching the positions involved with their corresponding responsibility. (5 mins) 4) Students will read the NCPM 600 section detailing the underwater resolve process as well as the provided under water resolve handout before correcting their exercise as required. (10 mins) 5) The instructor will review the underwater resolve procedure with the class highlighting the important lessons. (5 mins) 6) In 3 groups, students will sort the provided set of cards containing various reports into either into SCS or ASPO reports IOT allow the students to review the responsibilities of each role. (20 mins) 7) The instructor will present the solution to the class highlighting the key responsibilities of each position during the initial contact. (10 mins)

8) Students will read ATP-28 articles 0923-0926 and 12333-12338. (60 mins)
9) The following steps will utilize reporting scenario cards containing a series a CMS tactical plot images depicting a contact situation evolving over time with enough situation information to formulate simple reports. Using one of these cards, the instructor will demonstrate the formulation of an UWWD/SAC/SAUC report to include: (25 mins)
    a. Situation
        i. Contact status,
        ii. Sensor(s)/unit(s) in contact
        iii. Tracking information
        iv. Tonal information
    b. Assessment
        i. Command classification
        ii. assessment of targets intentions/situation
    c. Recommendation
        i. Maneuvering intentions
        ii. Issue of warnings
        iii. Marking with weapon
        iv. AC employment
        v. Sensor employment
        vi. Pre-plans
10) Based on additional scenario cards, students will formulate an appropriate their UWWD/SAC/SAUC report. (10 mins)
11) Students will present their scenario and report to the class with the instructor correcting as required. (20 mins)
12) The class will repeat steps 9-11 by formulating initial contact reports for their scenario using the second CMS tactical plot with the instructor going first followed by the students. (20 mins)
13) The class will repeat steps 9-11 by formulating amplification reports for their scenario using the third CMS tactical plot with the instructor going first followed by the students. (20 mins)
14) Student will read ATP-1 Vol 1 article 12107-12109. (10 mins)
15) Based on the class mission's OPGEN, the instructor will conduct a guided discussion IOT for the students to identify the Primary Warfare Commander's considerations and authorities to include: (15 mins)
    a. Launching alert helicopters
    b. Main body manoeuvring
    c. Issuing/Delegating warnings.
16) The instructor will present a sample Primary Warfare Commander report IOT detail the information relevant to a UWWD and discuss the PWC's overall responsibilities. (15 mins)
17) To practice rapidly interpreting a PWC report, the instructor will play a sample PWC report audio file or read for a prepared script while the students use a provided PWC report template to interpret this report and note all relevant information to an UWWD. (25 mins)
18) The instructor will ask students for specific information from the report to ensure that they were able to identify the correct information. (10 mins)
19) The instructor will explain the UWWD's ideal battle rhythm to optimize the management of personnel and data using a provided flowchart. (10 mins)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) Initial submarine contact flowchart handout from NCPM 600 4) Initial responses video or instructor script 5) Underwater resolve matching exercise 6) Underwater resolve procedure handout 7) SCS/ASPO report cards 8) Reporting scenario cards (9) 9) Class mission OPGEN 10) Sample Primary Warfare Commander report 11) Primary Warfare Commander report audio or instructor script 12) Primary Warfare Commander report template 13) UWWD battle rhythm flowchart	1) ATP- 1 2) ATP-28 3) NCPM 600

Notes to Instructor

The underwater water resolve procedure handout includes an explanation of the underwater resolve procedures, the difference between positive and negative resolves and the ships responsibilities in a Task Group IOT to amplify the limited content in NCPM 600.

EO 004.03 - Coordinate UWW Operations/Exercises

004.03.12 – ASW Pre-Plans

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 004.03.11
Time	**Classification**	**Instructor**
150 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Initial Contact and Reactions 2) ASW Pre-Plans considerations 3) Development of ASW Pre-Plan	Guided discussion Reading assignment Tabletop

Activity Steps	
	1) As an energizer, the instructor will conduct a guided discussion on pre-plans where students will develop a set of pre-planned responses as a class based a provided scenario by using a Fleet in a Box, their experience and instructor prompts. (15 mins) 2) The student will read CFCD 106 MARTI 2550 on ASW Pre-Plans. (30 mins) 3) The instructor will conduct a series of tabletop scenarios on the use of ASW Pre-Plans, using the Fleet in a Box or Interactive Table to assist students in identifying the appropriate ASW Pre-Plans to be used in the following situations: (105 mins) a. Inside TDZ b. Outside TDZ c. Outside of Screen / Surveillance Zone by Aircraft d. Observed missile launch e. Unobserved missile launch f. Torpedo detection

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) Interactive table 6) Fleet in a Box 7) ASW pre-plan scenarios	1) ATP-1 2) ATP-28 3) CFCD 106

Notes to Instructor

NTDC(A) Approval – 08/11/2019
NPTG HQ Approval – 13/12/2019
Version No: 1.3
MITE Crse ID : 122164

EO 004.03 - Coordinate UWW Operations/Exercises

004.03.13 – ASW Localization Plans

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 004.03.12
Time	**Classification**	**Instructor**
350 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Search plans (Black) 2) Attack/Avoidance plans (Red)	Reading assignment Lecture Tabletop

Activity Steps

1) Students will read the following on ASW localization plans: (100 mins)
    a. CFCD 106 TACNOTE 11220
    b. ATP-1 Vol I Table 6-1 and article 615.
2) The instructor will conduct a lecture to include how and when the following allied ASW Plans are used: (70 mins)
    a. Plans Red to include:
        i. Cordon
        ii. Bear
    b. Plans Black to include:
        i. Cordon
        ii. Bear
        iii. Oak tree
        iv. Air Plans/Breitler
3) The instructor will conduct a lecture on additional considerations and plans not specifically addressed within Allied doctrine to include: (30 mins)
    a. Escorts forming barriers within their sectors
    b. Maintaining screen integrity
    c. Use of aircraft
4) The instructor will conduct a series of tabletop scenarios on the use of ASW localization plans, using the Fleet in a Box or Interactive Table to assist students in identifying the appropriate ASW localization plans to be used in the following situations. Once identified, students will perform the appropriate voice report to execute the selected plan. The instructor may continue to progress the scenario(s) to allow students to adjust the plans in response to new information. (150 mins)

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) Interactive table 6) Fleet in a Box 7) ASW Table Top scenarios on Power Point	1) ATP-1 2) ATP 28 3) CFCD 106 TACNOTE 11220 and 11245

NTDC(A) Approval – 08/11/2019
NPTG HQ Approval – 13/12/2019
Version No: 1.3
MITE Crse ID : 122164

Notes to Instructor

NTDC(A) Approval – 08/11/2019
NPTG HQ Approval – 13/12/2019
Version No: 1.3
MITE Crse ID : 122164

EO 004.03 - Coordinate UWW Operations/Exercises

004.03.14 – Adhering to ROE

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after EO 004.03.13
Time	**Classification**	**Instructor**
150 mins	SECRET	ORO Instructor

Learning Objectives	Method
1) ROEREQ/ROE AUTH process 2) Coordinating control of weapons and sensors for ROE compliance 3) Hostile Intent vs Hostile Act 4) Defining self-defence 5) Targeting directive and relationship with ROE 6) Warning implementation 7) Exploiting ROE for tactical advantage	Presentation Reading assignment Floating flip chart Guided discussion Tabletop

Activity Steps

1) The instructor will present the "ROE" recorded lecture on Rules of Engagement. (25 mins)
2) Students will read the class mission's OPTASK ROE AUTH identifying all ROEs relevant to the conduct of ASW operations. (25 mins)
3) One a flip chart/whiteboard, each student will take turns writing a ROE # on the board explaining why it is relevant, the relevant criteria for use and the current holding authority. This will continue until no further ROE are identified with the instructor correcting any oversights. (10 mins)
4) The instructor will conduct a guided discussion on the difference between hostile acts and hostile intent addressing the following: (40 mins)
    a. What is considered a hostile act from a threat submarine?
    b. What would constitute a threat submarine having the capability to pose threat to TG or own ship?
    c. What would constitute a threat submarine having the proximity to pose threat to TG or own ship?
    d. How do surface ships determine hostile intent and how can we justify it?
5) The instructor will conduct a series of tabletop scenarios on the application of warnings and ROE, using the Fleet in a Box or Interactive Table to assist students in identifying the applicable ROE for each situation. Once identified, students will state whether use of force is justified by this ROE or self-defence and if not, which warning is appropriate and who holds the required ROE. The instructor may continue to progress the scenario(s) to allow students to adjust their response to new information. (50 mins)

NTDC(A) Approval – 08/11/2019
NPTG HQ Approval – 13/12/2019
Version No: 1.3
MITE Crse ID : 122164

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) Flip chart/Whiteboard 4) Markers 5) Interactive table 6) Fleet in a Box 7) ROE presentation 8) Class mission OPTASK ROE AUTH 9) ROE application tabletop scenarios (8)	1) CFJP 5.1 - Use of Force in CF operation 2) MC362-1 NATO ROE

Notes to Instructor

EO 004.03 - Coordinate UWW Operations/Exercises

004.03.15 – Mission Planning

Conditions		Sequence
Classified classroom with CSNI or secret standalone		Instructed after PC 004.01
Time	**Classification**	**Instructor**
400 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Building and presenting the ASW section of mission brief	Student presentation

Activity Steps	
	1) Using the class mission's information (ARBITER, VARANGIAN, etc.), the skills developed throughout the course and the provided mission brief from EO 002.01.01 Missing Briefing, students will develop the ASW section of the brief as a class to include: (350 mins) a. Expected ASW mission. b. Analysis of AOO (tactical oceanography) c. Allied submarine and relevant WSM/PMI d. Enemy submarines with concept of operations e. Relevant ROEs 2) Student will present this mission brief to the instructor who will provide feedback correcting any significant errors and omissions. (30 mins presentation, 20 mins debrief) 3) Students will apply the briefed information along with any required corrections in the execution EO 004.03.02.

Resources	References
1) CSNI or secret standalone computer (1/student) 2) Projector 3) Mission brief template 4) Class mission package	1) ATP-1 2) ATP-28 3) CFCD 106

Notes to Instructor
Instructor will critique the mission brief with the aim of improving the execution of EO 004.03.02 Coordinate UWW Operations/Exercises (Operations)

EO 004.03 - Coordinate UWW Operations/Exercises

004.03.16 – Consolidation: CASEXs

Conditions		Sequence
SEACOT (Primary) SEAWOLF (Secondary)		Instructed after EC 004.01
Time	**Classification**	**Instructor**
4000 mins (2400 mins CASEX C2) (1600 mins CASEX C4)	SECRET	UWWD Instructor

Learning Objectives	Method
1) Performing the duties of an Underwater Warfare Director for Coordinated Anti-Submarine Exercise	Practical Fishbowl

Activity Steps	
	1) Students will conduct 3 x 90 min CASEX C2 scenario runs IAW AXP-1 where they will set up the Combat Management System, operate comms equipment and coordinate the employment of ships, aircraft, weapons, sensors and decoys in order to execute the duties of an UWWD for ASW exercises. (3x 100 mins/student including 10 mins debrief and trainer reset, 2400 mins total) 2) Students will conduct 2 x 90 min CASEX C4 scenario runs IAW AXP-1 where they will set up the Combat Management System, operate comms equipment and coordinate the employment of ships, aircraft, weapons, sensors and decoys in order to execute the duties of an UWWD for ASW exercises. (2x 100 mins/student including 10 mins debrief and trainer reset, 1600 mins total) 3) Students not conducting the runs will observe and perform a Fishbowl exercise providing the UWWD student with one positive and one negative constructive observation on completion of each run before they are debriefed by the instructor. This will ensure students who are not participating in the run maintain engagement with the training and improve the UWWD development.

Resources	References
1) SEACOT or SEAWOLF 2) Class mission package 3) Student generated materials for CASEX C2 and C4 including a. OPTASK ASW b. CASEX TABORD	1) APP-11 2) APP-7 3) ATP-1 4) ATP-28 5) ATP-57 6) AXP-1 7) AXP-5 8) CFCD 106 9) MILOC manual 10) RP 33 11) NCPM 600 12) NCPM 1001

Notes to Instructor

Students will conduct the CASEX briefing prior to first run of each type.
Support manning IAW NWTP Vol 2

NTDC(A) Approval – 08/11/2019
NPTG HQ Approval – 13/12/2019
Version No: 1.3
MITE Crse ID : 122164

EO 004.03 - Coordinate UWW Operations/Exercises

EC 004.03.01 - Coordinate UWW Operations/Exercises (CASEX)

Scope	Conditions	Sequence
EO 004 CASEX	SEACOT (Primary) SEAWOLF (Secondary)	Conducted after EO 004.03.16
Time	**Classification**	**Instructor**
800 mins	SECRET	UWWD Instructor

Scenario	Your ship has been tasked with conducting a CASEX C4 IAW AXP 1 with task group assets IOT ensure the participating units are ready to support the upcoming of operational mission.

Assessment Steps	Input: Students are provided the CASEX C4 scenario and briefing developed in PO 002. They may use any student generated materials developed during the previous lessons.
	
	Performance: Students have 90 mins in SEACOT/SEAWOLF to conduct a CASEX C3 as OCS following the drafted TABORD IAW EC 004.03.01 checklist.
	(100 mins/student including 10 mins debrief and trainer reset, 800 mins total)
	The scenario must include:
	• Pre-serial briefing to participating units
	• One Submarine safety consideration
	  ○ Red flare
	  ○ HOOKER/UNIFORM procedures
	  ○ Surfacing methods
	  ○ SUBSAR
	• Assumption of SAC/SAUC duties
	• One SWAP
	• Own ship conducts one attack
	• React to attack (green grenade)
	• Exercise hot-wash up
	
	Output: Trainee must achieve 24/30 SATs on non-critical factors **AND** all critical factors to pass this EC.
	
	Feedback: Immediate verbal and written feedback provided IAW EC 004.03.01 checklist.

Resources	References
1) SEACOT or SEAWOLF 2) Class mission package 3) Student generated materials for CASEX C3 including a. OPTASK ASW b. CASEX TABORD	1) APP-11 2) APP-7 3) ATP-1 4) ATP-28 5) ATP-57 6) AXP-1 7) AXP-5 8) CFCD 106 9) MILOC manual 10) RP 33 11) NCPM 600 12) NCPM 1001

Notes to Instructor
Students will conduct the CASEX briefing (unassessed for this EC) prior to first assessed EC run. Support manning IAW NWTP Vol 2 and briefed to not to prompt the students. While the SAR format is encouraged and reinforced throughout the instruction and assessment, it is not a requirement when assessing voice reports.

Limitations	Compressed timeline based on training environment
Impact of Failure	Retrain and reassess

Item	EC 004.03.01 - COORDINATE UWW OPERATIONS (CASEX)	SAT	UNSAT	Remarks
1. Conduct serial preparations				
Conduct CS system setup IAW direction		☐	☐	
Conduct internal/external comms checks		☐	☐	
Conduct closing up report to ORO		☐	☐	
Conduct pre-serial brief to UWW team		☐	☐	
2. Conduct serial as OCS IAW TABORD				
Conduct external serial brief to participating units		☐	☐	
Conduct SAC/SAUC intentions report to include: • Approach method • Course/speed • Asset employment • Search and Attack Plans • Weapons priority • Supplemental instructions		☐	☐	
Report COMEX to ORO upon first attack		☐	☐	
Conduct initial reactions to include:				
Underwater resolve		☐	☐	
Conduct initial contact report to ORO		☐	☐	
Ensure external contact report conducted		☐	☐	
Recommend Command Classification IAW ATP-01 and EXTAC 198		☐	☐	
Assume SAC duties:				
Brief Command on plan/pre-plan		☐	☐	
Conduct SWAP IAW ATP 28 (if required)		☐	☐	
Conduct SAC external report to execute plan		☐	☐	
Direct employment of A/C		☐	☐	
Delegate weapon priority		☐	☐	
Execute pre-plans		☐	☐	
Direct ASW team on picture compilation to include: • WSM / TDA / TDZ / LLSUA • Datums / AOPs / FOCs / Sonobuoys • Contact position / Bearing line / Torpedo Run Box		☐	☐	
Direct the employment of sensors		☐	☐	

Direct the correlation of contacts to include: Datum, active, passive, non-acoustic information	☐	☐	
Update/amend recommended Command Classification IAW ATP-01 and EXTAC 198	☐	☐	
Conduct internal SITREPs to command	☐	☐	
Conduct external SITREPs to participating units	☐	☐	
Provides SITREPs to team (SCS, ASPO, SAC, OOW, ORS, and Command) throughout	☐	☐	
Utilize brevity code words IAW APP 7	☐	☐	
Conduct follow on phase brief to participating units	☐	☐	
Followed TABORD exercise instructions	☐	☐	
3. React to attacks			
Recommend TCM IAW CFCD 106	☐	☐	
Conduct counter-fire	☐	☐	
4. Adhere to CASEX safety procedures			
Adhere to STARRED relaxations	☐	☐	
Direct team's response to submarine safety IAW AXP 1 (one of: red flare, HOOKER/UNIFORM procedures, surfacing methods, SUBSAR)	☐	☐	*** CRITICAL**

Score: / 30 /1 CRITICAL

Trainee must achieve 24/30 SATs on non-critical factors AND all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PROTECTED B (when complete)

PC 004.01 - Execute the Duties of an UWWD for Exercises/Operations (CASEX)

Scope	Conditions	Sequence
PO 004 CASEX	SEACOT (Primary) SEAWOLF (Secondary)	Conducted after EC 004.03.01
Time	**Classification**	**Instructor**
1000 mins	SECRET	UWWD Instructor

Scenario	Your ship has been tasked with conducting a CASEX C3 IAW AXP 1 with task group assets IOT ensure the participating units are ready to support the upcoming of operational mission.

Assessment Steps	Input: Students are provided the CASEX C3 scenario and briefing developed in PO 002. They may use any student generated materials developed during the previous lessons.
	
	Performance: Students have 90 mins in SEACOT/SEAWOLF to conduct a CASEX C3 as OCS following the drafted TABORD IAW PC 004.01 checklist. (100 mins/student including 10 mins debrief and trainer reset, 800 mins total)
	The scenario must include:
	• Pre-serial briefing to participating units
	• One Submarine safety consideration
	  ○ Red flare
	  ○ HOOKER/UNIFORM procedures
	  ○ Surfacing methods
	  ○ SUBSAR
	• Assumption of SAC/SAUC duties
	• One SWAP
	• Own ship conducts one attack
	• React to attack (green grenade)
	• Exercise hot-wash up
	
	Output: Trainee must achieve 24/30 SATs on non-critical factors **AND** all critical factors to pass this EC.
	
	Feedback: Immediate verbal and written feedback provided IAW PC 004.01 checklist.

Resources	References
1) SEACOT or SEAWOLF 2) Class mission package 3) Student generated materials for CASEX C3 including a. OPTASK ASW b. CASEX TABORD	1) APP-11 2) APP-7 3) ATP-1 4) ATP-28 5) ATP-57 6) AXP-1 7) AXP-5 8) CFCD 106 9) MILOC manual 10) RP 33 11) NCPM 600 12) NCPM 1001

Notes to Instructor
Support manning IAW NWTP Vol 2 and briefed to not to prompt the students. While the SAR format is encouraged and reinforced throughout the instruction and assessment, it is not a requirement when assessing voice reports. Time allocated for 2 re-writes in the schedule.

Limitations	Compressed timeline based on training environment
Impact of Failure	IAW NPTORD 4500-54

Item	PC 004.01 - EXECUTE THE DUTIES OF AN UWWD FOR EXERCISES/OPERATIONS	SAT	UNSAT	Remarks
1. Conduct serial preparations				
Conduct CS system setup IAW direction		☐	☐	
Conduct internal/external comms checks		☐	☐	
Conduct closing up report to ORO		☐	☐	
Conduct pre-serial brief to UWW team		☐	☐	
2. Conduct serial as OCS IAW TABORD				
Conduct external serial brief to participating units		☐	☐	
Conduct SAUC intentions report to include: • Approach method • Course/speed • Asset employment • Search and Attack Plans • Weapons priority • Supplemental instructions		☐	☐	
Report COMEX to ORO upon first attack		☐	☐	
Conduct initial reactions to include:				
Underwater resolve		☐	☐	
Conduct initial contact report to ORO		☐	☐	
Ensure external contact report conducted		☐	☐	
Recommend Command Classification IAW ATP-01 and EXTAC 198		☐	☐	
Assume SAC duties:				
Brief Command on plan/pre-plan		☐	☐	
Conduct SWAP IAW ATP 28 (if required)		☐	☐	
Conduct SAC external report to execute plan		☐	☐	
Direct employment of A/C		☐	☐	
Delegate weapon priority		☐	☐	
Execute pre-plans		☐	☐	
Direct ASW team on picture compilation to include: • WSM / TDA / TDZ / LLSUA • Datums / AOPs / FOCs / Sonobuoys • Contact position / Bearing line / Torpedo Run Box		☐	☐	
Direct the employment of sensors		☐	☐	

Direct the correlation of contacts to include: Datum, active, passive, non-acoustic information	☐	☐	
Update/amend recommended Command Classification IAW ATP-01 and EXTAC 198	☐	☐	
Conduct internal SITREPs to command	☐	☐	
Conduct external SITREPs to participating units	☐	☐	
Provides SITREPs to team (SCS, ASPO, SAC, OOW, ORS, and Command) throughout	☐	☐	
Utilize brevity code words IAW APP 7	☐	☐	
Conduct follow on phase brief to participating units	☐	☐	
Followed TABORD exercise instructions	☐	☐	
3. React to attacks			
Recommend TCM IAW CFCD 106	☐	☐	
Conduct counter-fire	☐	☐	
4. Adhere to CASEX safety procedures			
Adhere to STARRED relaxations	☐	☐	
Direct teams response to submarine safety IAW AXP 1 (one of: red flare, HOOKER/UNIFORM procedures, surfacing methods, SUBSAR)	☐	☐	*** CRITICAL**

Score: / 30 /1 CRITICAL

Trainee must achieve 24/30 SATs on non-critical factors AND all critical factors to pass this PC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PROTECTED B (when complete)

EO 004.03 - Coordinate UWW Operations/Exercises

004.03.17 – Consolidation: Operations (Mission)

Conditions		Sequence
SEACOT (Primary) SEAWOLF (Secondary)		Instructed after EO 004.03.15
Time	**Classification**	**Instructor**
4000 mins	SECRET	UWWD Instructor

Learning Objectives	Method
1) Performing the duties of an Underwater Warfare Director in an operational setting	Practical Fishbowl

Activity Steps	
	1) Students will conduct 5 x 90 min multi-threat operation mission scenario runs where they will set up the Combat Management System, operate comms equipment and coordinate the employment of ships, aircraft, weapons, sensors and decoys in order to execute the duties of an UWWD for ASW missions. (5x 100 mins/student including 10 mins debrief and trainer reset, 4000 mins total) 2) Students not conducting the runs will observe and perform a Fishbowl exercise providing the UWWD student with one positive and one negative constructive observation on completion of each run before they are debriefed by the instructor. This will ensure students who are not participating in the run maintain engagement with the training and improve the UWWD development.

Resources	References
1) SEACOT or SEAWOLF 2) Class mission package	1) APP-11 2) APP-7 3) ATP-1 4) ATP-28 5) ATP-57 6) AXP-5 7) CFCD 106 8) MILOC manual 9) RP 33 10) NCPM 600 11) NCPM 1001

Notes to Instructor
Support manning IAW NWTP Vol 2

EO 004.03 - Coordinate UWW Operations/Exercises

EC 004.03.02 - Coordinate UWW Operations/Exercises (Operations)

Scope	Conditions	Sequence
EO 004 Operations	SEACOT (Primary) SEAWOLF (Secondary)	Conducted after EO 004.03.17
Time	**Classification**	**Instructor**
800 mins	SECRET	UWWD Instructor

Scenario	Having completed all your mission preparations and trained your team, your ship is ready to commence its mission IAW the class's overall scenario.
Assessment Steps	Input: Students are provided the class's overall mission package and the brief developed in EO 004.03.15. They may use any student generated materials developed during the previous lessons. Performance: Students have 90 mins in SEACOT/SEAWOLF to conduct a mission run as UWWD IAW EC 004.03.01 checklist. (100 mins/student including 10 mins debrief and trainer reset, 800 mins total) The scenario must include: • Minimum of 1 subsurface contact (gained by a sensor) • Minimum of 2 air assets • Minimum of 2 surface consorts • Minimum of 1 Mission Expeditionary Unit or HVU • Minimum 1 WSM area • Intelligence inject • Upgrading the command classification of subsurface contact • Assumption of SAC/SAUC duties • Execution of plans (to include aircraft) • Opportunity to employ warnings • React to attack (to include counter fire from own ship) Output: Trainee must achieve 33/42 SATs on non-critical factors **AND** all critical factors to pass this EC. Feedback: Immediate verbal and written feedback provided IAW EC 004.03.02 checklist.

Resources	References
1) SEACOT or SEAWOLF 2) Class mission package	1) APP-11 2) APP-7 3) ATP-1 4) ATP-28 5) ATP-57 6) AXP-5 7) CFCD 106 8) MILOC manual 9) RP 33 10) NCPM 600 11) NCPM 1001

Notes to Instructor
Support manning IAW NWTP Vol 2 and briefed to not to prompt the students. While the SAR format is encouraged and reinforced throughout the instruction and assessment, it is not a requirement when assessing voice reports.

Limitations	Compressed timelines due to training environment
Impact of Failure	Retrain and reassess

Item	EC 004.03.02 - COORDINATE UWW OPERATIONS (OPERATIONS)	SAT	UNSAT	Remarks
1. Conduct initial setup/ preparations				
Provide pre planned responses to team		☐	☐	
Conduct CS system setup IAW direction		☐	☐	
Conduct closing up report to ORO		☐	☐	
Disseminate PWC SITREP to team & Command (to include: threats, WSM, assets available, warning levels, and mission specific details)		☐	☐	
Establish WSM in CS (includes brief team & command)		☐	☐	
2. Conduct duties of UWWD in Operations				
Input/Disseminate intel information to team and command		☐	☐	
Direct ASW team on picture compilation to include: • WSM / TDA / TDZ / LLSUA • Datums / AOPs / FOCs / Sonobuoys • Contact position / Bearing line / Torpedo Run Box		☐	☐	
Direct the correlation of contacts to include: Datum, active, passive, non-acoustic information		☐	☐	
Employ UWR procedures upon detection IAW ATP-28		☐	☐	
Recommend Command Classification IAW ATP-01 and EXTAC 198		☐	☐	
Direct the employment of sensors		☐	☐	
Provides SITREP to team (SCS, ASPO, SAC, OOW, ORS)		☐	☐	
Directs/Liaises with ASW team		☐	☐	
Utilize brevity code words IAW APP 7		☐	☐	
Enforce EMCON/ACCON/ROE IAW OPORD		☐	☐	
Provides command with tactical reports/SITREPS		☐	☐	
Assume SAC duties:				
Brief Command on plan/pre-plan		☐	☐	
Conduct SAC external report to execute plan		☐	☐	
Direct employment of A/C		☐	☐	
Direct units to provide barrier for HVU		☐	☐	
Direct warnings to be issued		☐	☐	

Direct weapons priority or attack intentions	☐	☐	
Task Group Recommendations to include:			
• Submarine Warning Level	☐	☐	
• ASTDs (TADPOLES)	☐	☐	
• Alert A/C	☐	☐	
• Main body alterations	☐	☐	
Amend plans based on contact status	☐	☐	
Conduct SAC external update SITREPs based on changes in: • Contact holding • Contact movement • Status of warnings • Plans • Assessment of hostile intent/act • Threat reduction	☐	☐	
3. Use of Force/ ROE			
Report to command/PWC evidence of hostile intent or act	☐	☐	
Direct use/escalation of subsurface warnings	☐	☐	
Recommend to command application, implementation and delegation of ROE	☐	☐	
Adhere to WSM during weapon release	☐	☐	*** CRITICAL**
4. Anti-ship Torpedo Defence			
Recommend/implement ASTD policy IAW ATP 28 to command	☐	☐	
Provide command with initial TCM recommendation IAW CFCD 106			
• Report torpedo bearing from own ship	☐	☐	
• Direct alterations of TCM away from torpedo	☐	☐	*** CRITICAL**
• Provide counter-fire recommendations	☐	☐	
Execute counter fire from SVTT			
• Time IAW ATP 28/CFCD 106	☐	☐	*** CRITICAL**
• Accuracy IAW CFCD 106	☐	☐	
• Report to command status of weapon	☐	☐	
Update command on status of torpedo and TCM	☐	☐	

Provide direction on alteration of TCM MLA to command	☐	☐	
5. Post Firing Actions			
Conduct post firing report to command to include: airborne weapon remaining, weapon train, plans	☐	☐	
Conduct post firing report external to include: airborne weapon remaining, weapon train, plans	☐	☐	
Maintain weapons train	☐	☐	
Direct/Report DAMASS	☐	☐	

Score: / 42 /3 CRITICAL

Trainee must achieve 33/42 SATs on non-critical factors **AND** all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PROTECTED B (when complete)

PC 004.02 - Execute the Duties of an UWWD for Exercises/Operations (Operations)

Scope	Conditions	Sequence
PO 004 Operations	SEACOT (Primary) SEAWOLF (Secondary)	Conducted after EC 004.03.02
Time	**Classification**	**Instructor**
1000 mins	SECRET	UWWD Instructor

Scenario	Having completed all your mission preparations and trained your team, your ship is ready to commence its mission IAW the class's overall scenario.
Assessment Steps	Input: Students are provided the class's overall mission package and the brief developed in EO 004.03.15. They may use any student generated materials developed during the previous lessons. Performance: Students have 90 mins in SEACOT/SEAWOLF to conduct a mission run as UWWD IAW PC 004.02 checklist. (100 mins/student including 10 mins debrief and trainer reset, 800 mins total) The scenario must include: • Minimum of 1 subsurface contact (gained by a sensor) • Minimum of 2 air assets • Minimum of 2 surface consorts • Minimum of 1 Mission Expeditionary Unit or HVU • Minimum 1 WSM area • Intelligence inject • Upgrading the command classification of subsurface contact • Assumption of SAC/SAUC duties • Execution of plans (to include aircraft) • Opportunity to employ warnings • React to attack (to include counter fire from own ship) Output: Trainee must achieve 33/42 SATs on non-critical factors **AND** all critical factors to pass this EC. Feedback: Immediate verbal and written feedback provided IAW PC 004.02 checklist.

Resources	References
1) SEACOT or SEAWOLF 2) Class mission package	1) APP-11 2) APP-7 3) ATP-1 4) ATP-28 5) ATP-57 6) AXP-5 7) CFCD 106 8) MILOC manual 9) RP 33 10) NCPM 600

	11) NCPM 1001

Notes to Instructor
Support manning IAW NWTP Vol 2 and briefed to not to prompt the students. While the SAR format is encouraged and reinforced throughout the instruction and assessment, it is not a requirement when assessing voice reports. Time allocated for 2 re-writes in the schedule.

Limitations	Compressed timelines due to training environment
Impact of Failure	IAW NPTORD 4500-54

Item	PC 004.02 - EXECUTE THE DUTIES OF AN UWWD FOR EXERCISES/OPERATIONS	SAT	UNSAT	Remarks
1. Conduct initial setup/ preparations				
Provide pre planned responses to team		☐	☐	
Conduct CS system setup IAW direction		☐	☐	
Conduct closing up report to ORO		☐	☐	
Disseminate PWC SITREP to team & Command (to include: threats, WSM, assets available, warning levels, and mission specific details)		☐	☐	
Establish WSM in CS (includes brief team & command)		☐	☐	
2. Conduct duties of UWWD in Operations				
Input/Disseminate intel information to team and command		☐	☐	
Direct ASW team on picture compilation to include: • WSM / TDA / TDZ / LLSUA • Datums / AOPs / FOCs / Sonobuoys • Contact position / Bearing line / Torpedo Run Box		☐	☐	
Direct the correlation of contacts to include: Datum, active, passive, non-acoustic information		☐	☐	
Employ UWR procedures upon detection IAW ATP-28		☐	☐	
Recommend Command Classification IAW ATP-01 and EXTAC 198		☐	☐	
Direct the employment of sensors		☐	☐	
Provides SITREP to team (SCS, ASPO, SAC, OOW, ORS)		☐	☐	
Directs/Liaises with ASW team		☐	☐	
Utilize brevity code words IAW APP 7		☐	☐	
Enforce EMCON/ACCON/ROE IAW OPORD		☐	☐	
Provides command with tactical reports/SITREPS		☐	☐	
Assume SAC duties:				
Brief Command on plan/pre-plan		☐	☐	
Conduct SAC external report to execute plan		☐	☐	
Direct employment of A/C		☐	☐	
Direct units to provide barrier for HVU		☐	☐	

Direct warnings to be issued	☐	☐	
Direct weapons priority or attack intentions	☐	☐	
Task Group Recommendations to include:			
• Submarine Warning Level	☐	☐	
• ASTDs (TADPOLES)	☐	☐	
• Alert A/C	☐	☐	
• Main body alterations	☐	☐	
Amend plans based on contact status	☐	☐	
Conduct SAC external update SITREPs based on changes in: • Contact holding • Contact movement • Status of warnings • Plans • Assessment of hostile intent/act • Threat reduction	☐	☐	
3. Use of Force/ ROE			
Report to command/PWC evidence of hostile intent or act	☐	☐	
Direct use/escalation of subsurface warnings	☐	☐	
Recommend to command application, implementation and delegation of ROE	☐	☐	
Adhere to WSM during weapon release	☐	☐	*** CRITICAL**
4. Anti-ship Torpedo Defence			
Recommend/implement ASTD policy IAW ATP 28 to command	☐	☐	
Provide command with initial TCM recommendation IAW CFCD 106			
• Report torpedo bearing from own ship	☐	☐	
• Direct alterations of TCM away from torpedo	☐	☐	*** CRITICAL**
• Provide counter-fire recommendations	☐	☐	
Execute counter fire from SVTT			
• Time IAW ATP 28/CFCD 106	☐	☐	*** CRITICAL**
• Accuracy IAW CFCD 106	☐	☐	
• Report to command status of weapon	☐	☐	

Update command on status of torpedo and TCM	☐	☐	
Provide direction on alteration of TCM MLA to command	☐	☐	
5. Post Firing Actions			
Conduct post firing report to command to include: airborne weapon remaining, weapon train, plans	☐	☐	
Conduct post firing report external to include: airborne weapon remaining, weapon train, plans	☐	☐	
Maintain weapons train	☐	☐	
Direct/Report DAMASS	☐	☐	

Score: / 42 /3 CRITICAL

Trainee must achieve 33/42 SATs on non-critical factors **AND** all critical factors to pass this PC.

Positive Feedback	**Improvement Feedback**

Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
<u>Assessor</u> I have discussed the results of this assessment with the candidate.	**<u>Candidate</u>** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PROTECTED B (when complete)

CHAPTER 3: ASSESSMENT OF TRAINEE

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GENERAL

1. All Performance Checks (PCs) are based on POs in Chapter 2 of this QSP. Trainees must achieve the standard to be awarded the qualification. PCs will be administered in the form of PRACTICAL or WRITTEN assessment and the trainee must pass all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. Enabling Checks (ECs). ECs are formative assessments that help the trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning and increase the retention of material for longer periods in preparation for the PC. Each EO must have at least one EC. EC results do not factor into a trainee's successful completion of the PO.

3. Performance Checks (PCs). PCs are summative assessments used to determine if the trainee has met the standard under the conditions (where possible) of the PO. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

4. Assessment of trainees is to be completed IAW the standards delineated in this QSP and IAW CFITES Volume 7 (Evaluation).

PROGRESS MONITORING

5. Monitoring learning is the responsibility of all instructional staff and supervisors. Monitoring, including unsatisfactory progress, shall adhere to CFITES Volume 6 Conduct of Instruction Programmes.

6. Continuous monitoring of trainee progress is required to provide the following:

   a. Early warning of trainee difficulties;
   b. Feedback on the effectiveness of training;
   c. Information for the Training Review Board (TRB) (if required); and
   d. Information to assist in the completion of the CF 377 Course Report.

7. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW Training Establishment (TE) SOs or SOPs.

8. PC failures and supplemental PC failures are to be handled IAW TE SOs and SOPs. Supplemental ECs and PCs shall not be given no less than 24 hours of the trainee being informed of their EC or PC failure.

TRAINING REVIEW BOARD

9. The aim of the TRB process is to review a trainee's suitability for continued training and education, while ensuring that the trainee is afforded a fair and just assessment of all relevant information before a decision is made. TRBs shall be convened and conducted IAW NPTORD 4500-54 Training Review Board Process.

COURSE REPORTS

10. Course reports (CF 377) shall be prepared and distributed IAW NPTORD 4500-51.

11. Trainees will be graded pass or fail. NPTORD 4500-51 Course Reporting and Certificates, provides direction and guidelines for Letter Grading and Ranking.

12. Narratives will indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager. Training limitations and/or incomplete training must also be included in the course report. A record of trainee counselling is to be maintained by the TE.

Task List

Task #	Task Statement	Trg / No Trg	PO
A0005	Perform communications circuit functional checks (comms check)	N/T	
A0010	Operate communications equipment	N/T	
A0020	Perform Search and Rescue/Distress/SUBSMASH communications procedures	T	004
A0025	Encode/Decode tactical information (with Authenticate communication transmissions)	T	004
A0030	Authenticate communication transmissions	T	004
A0035	Extract information from COMPLANS, EMCON Plans and Frequency Plans	T	002
B0020	Enforce Emission control (EMCON) restrictions	T	004
B0025	Adhere to Emission control (EMCON) restrictions	N/T	
B0030	Contribute to intelligence gathering Ops	T	004
B0035	Display intelligence via UWW brief	T	002
B0040	Disseminate intelligence information	T	002
B0045	Identify Capabilities and limitations of platforms	T	002
B0050	Operate CS	T	003
B0060	Assist with tests and trials with outside agencies	T	001
C0010	Supervise mine detection team in a mine avoidance environment	T	001
C0025	Operate ancillary acoustic equipment	N/T	
C0100	Construct Contact Area of Probability (AOP)	T	003
C0105	Correlate active/passive/non-acoustic information	T	004
C0115	Calculate acoustic sensor range information	N/T	
C0120	Calculate array tow speeds, tilt and/or balancing	T	002
C0130	Conduct noise rounds	T	001
C0140	Coordinate the employment of UWW weapons and sensors	T	004
C0145	Coordinate ASW	T	004
C0150	Determine Oceanographic effects on acoustic operations	T	002
C0155	Direct towed array system operations	N/T	

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C0160	Establish capabilities and limitations of towed array operations in operation areas (OPAREAS)	T	002
C0165	Coordinate water space management	T	002
C0175	Operate Torpedo Setting Panel (TSP)	N/T	
C0200	Provide data for RFIs	T	002
D0005	Conduct UWW weapon systems serviceability checks (i.e. airshot)	N/T	
D0010	Perform UWW weapon systems serviceability checks (i.e. airshot)	T	003
D0015	Prepare UWW firing orders	T	003
D0020	Conduct UWW weapon firings	T	003
D0025	Perform UWW weapon firings	T	003
D0030	Perform weapons system misfire procedures	T	003
D0035	Conduct weapon system misfire procedures	T	003
D0040	Conduct submarine surfacing/diving operations	T	004
D0045	Prepare UWW exercise support requests	T	002
D0050	Assess UWW threat information	T	002
D0055	Maintain quiet running bill	T	001
D0060	Develop noise monitoring plan	T	001
D0065	Apply Water Space Management	T	004
D0070	Optimize ship’s acoustic signature	T	001
E0005	Compile ASW tactical picture	T	003
E0010	Coordinate ASW tactical picture	T	004
E0015	Detect radar contacts	N/T	
E0020	Amplify track information	T	004
F0005	Evaluate correlated acoustic information	T	004
F0015	Identify changes in contact’s readiness and tactical states	T	004
F0045	Interpret oceanographic data	T	002
G0010	Plan/Design training exercises	T	001
G0045	Assist in individual training events	N/T	
G0050	Conduct individual training events	N/T	
G0055	Correct individual training deficiencies	N/T	
G0060	Assist in team training events	T	001
G0065	Conduct team training events	T	001
G0070	Correct team training deficiencies	T	001

G0100	Provide recommendations to PRB/TRB	N/T	
G0115	Assess individual training performance	N/T	
G0120	Assess team training performance	T	001
G0130	Assess effectiveness of training	T	001
L0035	Recommend improvement to training	T	001
AT0090	Maintain ship's logs and records	T	001
BT0050	Plan an UWW operation	T	002
BT0080	Coordinate UWW operation	T	004
BT0120	Draft firing orders	T	003
BT0130	Prepare operational messages	T	002
BT0140	Prepare evaluation operational reports	T	002
BT0170	Supervise UWW weapons firing analysis	N/T	
BT0180	Coordinate the control of weapons	T	004
BT0190	Coordinate the employment of weapons	T	004
BT0200	Coordinate the control of sensors	T	004
BT0210	Coordinate the employment of sensors	T	004
BT0220	Coordinate the control of aircraft	T	004
BT0230	Coordinate the employment of aircraft	T	004
BT0250	Contribute to the Recognized Maritime Picture (RMP)	T	004
BT0280	Supervise the compilation of the UWW picture	T	004
BT0310	Implement UWW tactics	T	004
BT0340	Advise command on UWW safety	T	004
BT0380	Coordinate the maintenance of records for operations analysis	T	001
BT0410	Determine the effectiveness of UWW weapons and sensors	T	004
BT0490	Organize intelligence	T	002
BT0540	Identify systems performance deficiencies, malfunctions or failures	T	003
T0001	Execute duties of an Underwater Warfare Director	T	004
T0002	Provide advice to Command on UWW tactics	T	002
T0003	Operate UWW control systems	T	003
T0004	Direct employment of UWW weapons, sensors and associated equipment	T	004
T0005	Execute UWW operations and exercises	T	004

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T0006	Assess UWW operations and exercises	T	004
T0007	Ensure technical readiness of the UWW system	T	001
T0009	Ensure the operational readiness of UWW teams	T	001

Record of Decisions

Dates: 04 November 2020

QSP Amendment

1. IAW CANFORGEN 112/20 RCN JR Rank Designation Change 041200Z SEP 20, all previous RCN JR Rank designations in this QSP have been amended to reflect the new policy.

2. Addition of previously omitted Chapter 3: Assessment of Trainee.

Dates: 17 – 26 June 2019

Location: CFB Halifax, S15 rm 214

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Lamb	R.J.F.	LCdr (RN)		NTDC(A)
Desjardins	G.R.F.R.	Lt(N)	00207	NTDC(A)
Devine	S.J.	Lt(N)	00207	NFS(A)
Tallman	J.R.	CPO2	00324	NTDC(A)
Coit	R.R.	PO1	00324	NTDC(A)
O'Toole	T.M.	PO1	00324	NFS(A)
Flynn	D.A.	PO1	00324	VDQ
Jawornicki	B.	SLt	00211	NTDC(A)

Working Group Objective:

1. The objective of this working group is to incorporate the lessons learned and address the shortcomings identified by the Combat Training review which took place prior to the working group being convened. The most significant of these shortcomings to be identified were a lack of trainer time and the limited tactical instruction both of which were a focus for the working group.

2. The working group was directly involved in identifying the tasks to be trained from the all three relevant Master Task Lists (Sonar Op, UWWD and NWO) with the justification for no train decisions found in the below table. It also reorganized these tasks into the four POs of this QSP and develop the EC and PCs along with their checklist before disbanding.

3. All remaining content of this QSP including the master lesson guidance, supporting documents and time allocation were completed by NTDC(A) members of the working group over the following months with a final review by the board's TDO to ensure it met modern instruction techniques where able.

Specification Task – No Train Decisions:

Task #	Task Statement	Substantiation
A0005	Perform communications circuit functional checks (comms checks)	This task is instructed on previous courses, no requirement for additional training.

A0010	Operate communications equipment	This task is instructed on previous courses, no requirement for additional training.
B0025	Adhere to Emission control (EMCON) restrictions	This task is instructed on previous courses, no requirement for additional training.
C0025	Operate ancillary acoustic equipment	There is no longer any ancillary acoustic equipment that the UWWD is required to operate.
C0115	Calculate acoustic sensor range information	This is a task performed by RQ-MS and below, no requirement to be performed at the UWWD level.
C0155	Direct towed array system operations	This is a task performed by RQ-MS, no requirement to be performed at the UWWD level.
C0175	Operate Torpedo Setting Panel (TSP)	UWWD TSP panel is now integrated into CMS 330 and is not required to be instructed to UWWD.
D0005	Conduct UWW weapon systems serviceability checks (i.e. airshot)	Task is performed as part of Conducting UWW Weapons firings. Unable to perform serviceability checks task at NFS with only CMS 330. Requires ship with full weapons system availability.
E0015	Detect radar contacts	This task is instructed on previous courses, no requirement for additional training.
G0045	Assist in individual training events	This task is instructed on previous courses, no requirement for additional training. NWO does not require training in this task.
G0050	Conduct individual training events	This task is instructed on previous courses, no requirement for additional training. NWO does not require training in this task.
G0055	Correct individual training deficiencies	This task is instructed on previous courses, no requirement for additional training. NWO does not require training in this task.
G0100	Provide recommendations to PRB/TRB	No requirement for training on this task.
G0115	Assess individual training performance	This task is instructed on previous courses, no requirement for additional training. NWO does not require training in this task.
BT0170	Supervise UWW weapons firing analysis	UWWD do not conduct weapons firing analysis. This is done on completion of an exercise by range staff or embarked CFMWC staff.

Discussions / Decisions:

4. The working group's intent is to create a QSP that reflects the performance of the UWWD. Sequencing of the Performance Objectives is in line with the conducting an exercise/mission from preparations to execution. Starting with individual and ship readiness requirements and progressing into

mission preparations, weapons certification and finally executing the duties of the UWWD while at sea for exercises/operations.

5. Regarding the EO 001.02 on naval mine warfare, it was discussed that UWWD on board ship do not plan NMCM missions as our major warships where they are employed are not mine clearance vessels. These ships are, however, likely to operate or transit in or near a mined area requiring that the director have a thorough understand of the threat and the steps required to limit the risk to the ship. As such, the NMCM content of the course has been modified to align with this focus in mind with EC 001.02 testing this knowledge by way of a mine operations brief deliver to the ship's command team in preparation of a mine transit. It was decided by the board that a PC on this topic is not required as it doesn't represent a critical part of the UWWD's responsibilities as evidenced by the fact the fact that mine warfare as a topic is identified only once within the various master task lists.

6. The group discussed the assessments required for PO 003 (weapons firing). Based on the TDO's recommendation, the group decided to have 3 ECs, including one that exclusively assessed the ability to utilize the Combat Management System. Based on the group's scalaring, the two additional ECs focus on preparations leading up to a weapon firing and executing the weapon firings respectively.

7. The group determined that two separate PCs for PO 004 would be most reflective of the duties of the UWWD as CASEX and Operations have significant differences in conduct, responsibilities and focus. As a consequence, PC 004.01 CASEX and PC 004.02 Operations were created.

8. Regarding the EO 004.02 on communications, while encoding/decoding, challenge and reply are key components for an UWWD to understand the capabilities of the team, they are never done by the Director. It was therefore determined that while they will continue to be trained on this, students will not be assessed on their ability to carry out these procedures. The remaining content of EO 004.02 such as voice reports and brevity code words will be utilized throughout all trainer time and included in various assessments checklists. Consequently, there will not be an EC 004.02.

Guidance to Instructional Staff:

9. As this QSP is significantly longer than its predecessor on per student basis, the maximum load of students was reduced from 12 to 8 in order to limit to overall length to under the six-month mark. Should more students be required, instructional staff are encouraged to submit a CAR to lengthen the course based on chapter 1 recommendation of 51 training periods per additional students along with an additional instructor to support the additional load.

10. In addition to lengthening the course, significant effort was placed in improving the tactical content of the course in EO 004.03. As this includes new content never taught to the previous generations of Sonar Operator now forming the instructor cadre, it is encouraged that additional time be allotted by training establishments for their instructors to adequately prepare during this phase by providing additional resources for early instances of the course.

11. With the many changes made to the QSP, it is anticipated that some supporting resources content may not meet the requirements as envisioned by the master lesson guidance. Frequent exchange of information between the training establishments and NTDC(A) are encouraged to quickly remedy these shortcomings.

Follow Up Action:

Item	Action	OPI	Expected Date of Completion
001	Master Lesson Guidance Supporting documents	PO1 Ron Coit	**2 Dec 19**
002	Tactical Oceanography Review and improvement	CPO2 Jason Tallman	**Fall 2020**
003	Integration of Underwater Sensor Upgrade (UWSU) theoretical content IAW UWSU-IPT	PO1 Ron Coit	**Fall 2020**

Costing Information

Based on 8 students total, 6 local (East or West Coast) and 2 from Out of Area with 105 training days (plus weekends 160 calendar days)

Course: UWWD	Serial:					
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	2	$1,600.00		$3,200	$3,200	$32,000.00
R&Q	2	$78.75	160	$25,200	$25,200	$252,000.00
Incidentals	2	$17.30	160	$5,535	$5,535	$55,360.00
Subtotal				**$33,935.00**	**$33,935.00**	**$339,350.00**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	12	$50.00	1	$600	$600	$6,000.00
Classroom (square meters)	30	$13.00	15	…	…	…
Computer requirements	12	$0.00	1	$0	$0	$0.00
IT Equip (Proj, Smart board etc.)	3	$0.00	1	$0	$0	$0.00
Instructional Materials (Printed / Handouts etc.)	14	$30.00	1	$420	$240	$2,580.00
Subtotal				**$1,020**	**$840**	**$8,580.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1) (incremental)	0	$0.00	0	$0	$0	$0.00
Instructor (Lt(N)) (incremental)	0	$0.00	0	$0	$0	$0.00
Incremental Civilian Instructor	0	0	0	$0	$0	$0.00
		subtotal		**$0.00**	**$0.00**	**$0.00**
	Total Estimated Costs			**$34,955**	**$34,775**	**$347,930**

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Assessment Plan

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria
EC 001.01	EO 001.01	To assess student's competence in planning a section's team training requirement as part of a ship's company	Written/Verbal	Student must obtain 10/12 SATs IAW EC 001.01 Checklist
EC 001.02	EO 001.02	To assess student's competence planning and briefing mine avoidance operations	Practical	Student must obtain 16/20 SATs IAW EC 001.02 Checklist
PC 001	PO 001	To assess student's competence in planning a section's team training and noise monitoring requirement as part of a ship's company	Written/Verbal	Student must obtain 13/16 SATs IAW PC 001 Checklist
EC 002.01	EO 002.01	To assess student's competence in interpreting CASEX messages and oceanographic data to optimize sensors	Verbal	Student must obtain 26/32 SATs IAW EC 002.01 Checklist
EC 002.02	EO 002.02	To assess student's competence in preparing operational messages	Practical	Students must obtain 21/26 SATs IAW EC 002.02 Checklist
EC 002.03	EO 002.03	To assess student's competence in preparing and delivering an exercise brief	Practical	Students must obtain 18/22 SATs IAW EC 002.03 Checklist
PC 002	PO 002	To assess student's competence in completing preparation for an UWW exercise	Practical	Students must obtain 26/33 SATs IAW PC 002 Checklist
EC 003.01	EO 003.01	To assess student's competence in preparing Firing Orders	Written	Students must obtain 5/7 SATs IAW EC 003.01 Checklist
EC 003.02.01	EO 003.02	To assess student's competence in operating the command management system	Practical	Students must obtain 17/21 SATs IAW EC 003.02.01 Checklist
EC 003.02.02	EO 003.02	To assess student's competence in conducting Torpedo firings	Practical	Students must obtain 35/44 SATs IAW EC 003.02.02 Checklist

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PC 003	PO 003	To assess student's competence in preparing and conducting torpedo firing	Practical	Students must obtain 31/38 SATs and ALL CRITICALS IAW PC 003 Checklist
EC 004.01	EO 004.01	To assess student's competence in operating safely with submarines	Practical	Students must obtain 19/23 SATs IAW EC 004.01 Checklist
EC 004.03.01	EO 004.03	To assess student's competence in coordinating an UWW Exercises (CASEX C3)	Practical	Students must obtain 24/30 SATs and ALL CRITICALS IAW EC 004.03.01 Checklist
PC 004.01	EO 004.03	To assess student's competence in Executing the Duties of an UWWD for Exercises (CASEX C3)	Practical	Students must obtain 24/30 SATs and ALL CRITICALS IAW PC 004.01 Checklist
EC 004.03.02	PO 004	To assess student's competence in Coordinating UWW Operations	Practical	Students must achieve 33/42 SATs and ALL CRITICALS IAW EC 004.03.02 Checklist
PC 004.02	PO 004	To assess student's competence in Execute the Duties of an UWWD for Operations	Practical	Students must achieve 33/42 SATs and ALL CRITICALS IAW PC 004.02 Checklist

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TEL Development Opportunities

Title	Type	Description of Requirement	Lesson	SOR #
UWWD video	Video	Develop an updated video of the Duties and Responsibilities of the UWWD	EO 001.01.01	
Sound Ranging video	Video	Develop a video of how Sound Ranging is conducted	EO 001.02.03	
NMVAS	System	Naval Mine Vulnerability Assessment System	EO 001.02.11	
Oceanography Distributed Learning	DL / videos/ Ocean Cube	Update the Oceanography Lectures currently used on the DLN and Distributed Learning Practical implementation of the Ocean Cube or similar oceanographic systems	EO 002.01.02 – 002.01.10	
ASW Equipment Videos	Video	Develop ASW equipment videos that provide demonstration of how the sensor is employed and operated	EO 002.01.11	
CMS 330 TMA demonstration video	Video	Develop a video demonstrating the use of the CMS 330 TMA Tactical Decision Aid	EO 002.01.13	
NIXIE video	Video	Video of what NIXIE does when employed	002.01.15	
ADC/NAEB video	Video	ADC/NAEB's operate in the water and their effectiveness as Acoustic Countermeasures	002.01.15	
CASEX "attention grabber" video	Video	Develop the "attention grabber" video on Japanese/P3 ASW	EO 002.02.05	
Torpedo Firing Video	Video	Develop a video on how to conduct a Surface Vessel Torpedo firing	EO 003.01.01	
Voice Report Audio Files	Audio	Develop a series of short audio files demonstrating each common internal and external voice reports	EO 004.02.05	
TCM Video	Video	Classified video of a ship conducting a Torpedo Counter Measure manoeuvre correctly and incorrectly	EO 004.03.06	
"Multi Statics and LFA tactics" Video	Video	Develop a Multi-static ASW and Low Frequency Active sonar recorded video	EO 004.03.09	
"ROE" Video	Video	Develop a Rules of Engagement recorded video	EO 004.03.14	

Scheduling Information

The recommended schedule is based on **maximum load**. Scheduling cells are to increase/decrease course length based on number of students enrolled on the course.

Scheduling Annex - SONAR OP RQ-PO1 / UWWD													
Lesson		Location	IT Assets					Footprint			Instructor		
#	Time	Location	DWAN	CSNI	SSA	Proj	TEL / Critical Resources	Sec Class	WB	Desks	Type	Qual	Sec Cl
PO 001 - Conduct Team Training Events													
EO 001.01.01	150	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EO 001.01.02	150	Classroom	13	0	0	1	DWAN	UNC	1	13	GS	SAC & RQ-PO2 NCI OP	Sec Cl
EO 001.01.03	150	Classroom	0	1	1	1	CSNI or Secret stand-alone workstations	SEC	1	13	GS	RQ-PO2 NESOP	Sec Cl
EO 001.01.04	150	Classroom	13	0	0	1	DWAN	UNC	1	13	GS	TDL 300 Qualified Instr	Sec Cl
EO 001.01.05	100	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EO 001.01.06	200	Classroom	13	0	0	1	DWAN	UNC	1	13	MIL	UWWD	Sec Cl
EO 001.01.07	100	Classroom	13	0	0	1	DWAN	UNC	1	13	GS	UWWD / CCFL/P GL	Sec Cl
EO 001.01.08	50	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EC 001.01	250	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EO 001.02.01	100	Classroom	13	0	0	1	DWAN	UNC	1	13	GS	UWWD / ADAC GL	Sec Cl
EO 001.02.02	250	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EO 001.02.03	150	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EO 001.02.04	100	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EO 001.02.05	150	Ship	0	0	0	0	Ship Tour	SEC	0	0	Mil	UWWD	Sec Cl
EO 001.02.06	200	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	CSMW Instructor	Sec Cl

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EO 001.02.07	50	Classroom	0	13	0	1	CSNI	SEC	1	13	GS	TRINTY MW GL / UWWD	Sec Cl
EO 001.02.08	150	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	GS	CSMW Instructor	Sec Cl
EO 001.02.09	150	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	GS	CSMW Instructor	Sec Cl
EO 001.02.10	250	Classroom	13	13	13	1	DWAN & CSNI or Secret stand alone	SEC	1	13	Mil	UWWD	Sec Cl
EO 001.02.11	150	Classroom	13	13	13	1	DWAN & CSNI or Secret stand alone	SEC	1	13	Mil	UWWD	Sec Cl
EO 001.02.12	100	Classroom	0	13	0	1	CSNI	CON F	1	13	Mil	UWWD	Sec Cl
EO 001.02.13	150	Classroom	13	13	13	1	DWAN & CSNI or Secret stand alone	SEC	1	13	Mil	UWWD	Sec Cl
EC 001.02	200	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
PC 001	250	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
PO 002 - Complete UWW Exercise/Mission Preparations													
EO 002.01.01	50	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 002.01.02	150	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EO 002.01.03	150	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EO 002.01.04	200	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EO 002.01.05	300	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EO 002.01.06	250	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EO 002.01.07	250	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EO 002.01.08	150	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EO 002.01.09	150	METOC Tour	13	0	0	1	METOC Tour	SEC	1	13	GS	METOC Staff or UWWD	Sec Cl
EO 002.01.10	200	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl

EO 002.01.11	250	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 002.01.12	100	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EO 002.01.13	150	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EO 002.01.14	200	Classroom	13	0	0	1	NITES2R Trainer	UNC	1	13	Mil	UWWD	Sec Cl
EO 002.01.15	100	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 002.01.16	1200	Classroom	0	13	13	1	ADAC or Secret Classroom	SEC	1	13	GS	ADAC Staff or UWWD	Sec Cl
EO 002.01.17	100	Classroom	0	13	13	1	CSNI or Secret standalone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 002.01.18	150	Classroom	0	13	13	1	CSNI or Secret standalone workstations	SEC	1	13	Mil	ORO CO	Sec Cl
EO 002.01.19	150	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 002.01.20	150	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 002.01.21	150	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EO 002.01.22	150	Classroom	0	13	13	1	CSNI	SEC	1	13	Mil	UWWD	Sec Cl
EO 002.01.23	200	Classroom	0	13	13	1	CSNI	SEC	1	13	Mil	UWWD	Sec Cl
EC 002.01	250	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 002.02.01	150	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	Post ORO	Sec Cl
EO 002.02.02	600	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	GS	CANSUBOR Staff or UWWD	Sec Cl
EO 002.02.03	500	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 002.02.04	350	Classroom	0	13	13	1	CSNI or Secret standalone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 002.02.05	150	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl

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EO 002.02.06	350	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EC 002.02	150	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 002.03.01	200	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 002.03.02	400	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EC 002.03	450	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
PC 002	350	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
PO 003 - Operate CS for UWW Weapons Firings													
EO 003.01.01	300	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 003.01.02	350	Classroom	13	13	13	1	DWAN, CSNI or Secret stand	SEC	1	13	Mil	UWWD	Sec Cl
EO 003.01.03	50	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EO 003.01.04	150	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EC 003.01	50	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EO 003.02.01	550	SEACOT	0	0	0	1	SEACOT Trainer	SEC	1	13	Mil	UWWD	Sec Cl
EO 003.02.02	1100	SEACOT	0	0	0	1	SEACOT Trainer	SEC	1	13	Mil	UWWD	Sec Cl
EC 003.02.01	150	SEACOT	0	0	0	1	SEACOT Trainer	SEC	1	13	Mil	UWWD	Sec Cl
EO 003.02.03	500	SEACOT	0	0	0	1	SEACOT Trainer	SEC	1	13	Mil	UWWD	Sec Cl
EO 003.02.04	150	Classroom / SEACOT	13	0	0	1	DWAN & SEACOT Trainer	SEC	1	13	Mil	UWWD	Sec Cl
EO 003.02.05	1250	SEACOT	0	0	0	1	SEACOT Trainer	SEC	1	13	Mil	UWWD	Sec Cl
EC 003.02.02	350	SEACOT	0	0	0	1	SEACOT Trainer	SEC	1	13	Mil	UWWD	Sec Cl
PC 003	450	SEACOT	0	0	0	1	SEACOT Trainer	SEC	1	13	Mil	UWWD	Sec Cl
PO 004 - Execute the Duties of an UWWD for Exercises/Operations													

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EO 004.01.01	100	Classroom	13	0	0	1	DWAN with projector	UNC	1	13	GS	CANSUBFOR Staff or UWWD	Sec Cl
EO 004.01.02	150	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 004.01.03	1000	SEACOT	0	0	0	1	SEACOT or SEAWOLF Trainer	SEC	1	13	Mil	UWWD	Sec Cl
EC 004.01	500	SEACOT	0	0	0	1	SEACOT or SEAWOLF Trainer	SEC	1	13	Mil	UWWD	Sec Cl
EO 004.02.01	50	Classroom	13	0	0	1	DWAN	UNC	1	13	GS	RQ MS NCI OP	Sec Cl
EO 004.02.02	100	Classroom	13	0	0	1	DWAN	UNC	1	13	GS	RQ MS NCI OP	Sec Cl
EO 004.02.03	100	SEACOT	0	0	0	1	SEACOT or Voice Trainer	SEC	1	13	Mil	UWWD	Sec Cl
EO 004.02.04	100	Classroom	13	0	0	1	DWAN	UNC	1	13	Mil	UWWD	Sec Cl
EO 004.02.05	350	SEACOT	0	0	0	1	SEACOT or Voice Trainer	SEC	1	13	Mil	UWWD	Sec Cl
EO 004.03.01	100	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 004.03.02	300	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 004.03.03	300	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	GS	TACCO (ASCO) or UWWD	Sec Cl
EO 004.03.04	300	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	GS	TACCO (ASCO) or UWWD	Sec Cl
EO 004.03.05	200	Tour	0	0	0	0	Tour of Air Wing	SEC	1	13	Mil	UWWD	Sec Cl
EO 004.03.06	450	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 004.03.07	350	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 004.03.08	300	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	GS	SMCO or SMORO	Sec Cl
EO 004.03.09	600	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 004.03.10	350	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl

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EO 004.03.11	300	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 004.03.12	150	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 004.03.13	350	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 004.03.14	150	Classroom	0	13	13	1	CSNI or Secret stand-alone workstations	SEC	1	13	GS	ORO CO or Post ORO	Sec Cl
EO 004.03.15	400	Classroom	0	13	13	1	CSNI or Secret standalone workstations	SEC	1	13	Mil	UWWD	Sec Cl
EO 004.03.16	4000	SEACOT	0	13	13	1	SEACOT with ASW Team	SEC	1	13	Mil	UWWD	Sec Cl
EC 004.03.01	800	SEACOT	0	0	0	1	SEACOT with ASW Team	SEC	1	13	Mil	UWWD	Sec Cl
PC 004.01	1000	SEACOT	0	0	0	1	SEACOT with ASW Team	SEC	1	13	Mil	UWWD	Sec Cl
EO 004.03.17	4000	SEACOT	0	0	0	1	SEACOT with ASW Team	SEC	1	13	Mil	UWWD	Sec Cl
EC 004.03.02	800	SEACOT	0	0	0	1	SEACOT with ASW Team	SEC	1	13	Mil	UWWD	Sec Cl
PC 004.02	1000	SEACOT	0	0	0	1	SEACOT with ASW Team	SEC	1	13	Mil	UWWD	Sec Cl

Total Mins	36550
Periods	731
Days	91.3

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Plan, Train, Do

PO	EO	Maintain Readiness	Plan	Train	Execute
001	01	EO 001.01 Conduct Team Training Events			
	02	EO 001.02 Optimize Ship's Acoustic Signature			
002	01		EO 002.01 Orient to the Mission		
	02		EO 002.02 Prepare Operational Messages		
	03		EO 002.03 Disseminate Intelligence Information		
003	01		EO 003.01 Prepare for Weapons Firing		
	02			EO 003.02 Conduct Torpedo Firing Procedures	
004	01			EO 004.01 Operate Safely with Submarines	
	02			EO 004.02 Communicate Internally/Externally	
	03				EO 004.03 Coordinate UWW Operations/Exercises

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References

All References listed within this QSP:

Ref	Publication
1	ADatP-33
2	ADC technical manual
3	AHP-6
4	AN/SQS-510 SONAR System Baseline 6.3 manual
5	APP-1
6	APP-7
7	APP-11
8	ACP-125
9	ACP-165
10	ACP-176
11	AMSC 608 TC
12	ATP-1
13	ATP-10
14	ATP-6
15	ATP-18
16	ATP-24
17	ATP-28
18	ATP-57
19	AUTEC range manual
20	AXP-1
21	AXP-5
22	COMSUBLANT/COMSUBPAC OPORD 2137
23	CANTASS user manual
24	CFCD 102
25	CFCD 103
26	CFCD 106
27	CFCD 109
28	CFCD 120
29	CFCD 129
30	CFJP 5.1 Use of Force in CF operation
31	Command Control System/ Combat Systems computing environment description, operational and maintenance manual
32	CMS user’s guide

33	DAOD 8010-1
34	DIAS Threat Web
35	DUWCS TUUM-6
36	HCM operating and technical document for Raytheon navigation radar
37	Jane’s Fighting Ships
38	Joint Doctrine Manual, Risk Management for CF Operations
39	MARLANTORD 8-1
40	MARLANTORD 3255-1
41	MARPACORD 9-47
42	MARPACORD 3350-1
43	MC362-1 NATO ROE
44	MIDAS Mine threat library
45	MILOC manual
46	Mk 21 User manual
47	MTP-57
48	NAEB technical manual
49	Naval Mines Worldwide
50	NAVORD 2000-2
51	NAVORD 3137-0
52	NAVORD 3137-1
53	NAVORD 3137-2
54	NAVORD 3395-1
55	NAVORD 3470-0
56	NAVORD 3470-1
57	NAVORD 3470-3
58	NAVORD 4003-6
59	NAVORD 4995-0
60	NAVORD 4995-2
61	NAVORD 8010-1
62	NAVORD 11000-0
63	NCPM 600
64	NCPM 1001
65	NDSOD
66	NITES2R user manual
67	NOSTP 200
68	NWTP Vol 4
69	PMFR range manual

70	Readiness Training Guide (Sea Training)
71	RP 33
72	SEA GIRAFFE 180 operator handbook
73	SEMS
74	Ship Standing Orders (SSO)
75	SHOPS
76	SIRIUS technical manual
77	SMART-S MK 2 3D radar manual
78	SPS user manual
79	SVTT Mod 9 manual
80	Terminal user's instructions for shipboard integrated communications system (HFX CLASS)

Recommended Schedule
Week #1

TIME	DAY 1	DAY 2	DAY 3	DAY 4	DAY 5
Period 1 0800-0850	**Course Introduction & Administration (BAT)**	**PT**	**EO 001.01.04** – C4ISR	**PT**	**EO 001.02.01** – Maritime Acoustic Signature Policy
Period 2 0850-0940	**Course Introduction & Administration (BAT)**	**EO 001.01.02** – SAC Liaison	**EO 001.01.05** – Identify Team Readiness Requirements	**EO 001.01.07** – Drafting Training Messages: Fleet Training	**EO 001.02.01** – Maritime Acoustic Signature Policy
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**Course Introduction & Administration (BAT)**	**EO 001.01.02** – SAC Liaison	**EO 001.01.05** – Identify Team Readiness Requirements	**EO 001.01.08** – Liaising with Supporting Agencies	**EO 001.02.02** – Ranging for Quiet Running Bill
Period 4 1040-1130	**Course Introduction & Administration (BAT)**	**EO 001.01.03** – Employ Radars and IFF	**EO 001.01.06** – Identify Team Readiness Requirements: TRP	**EC 001.01** – Conduct Team Training Events	**EO 001.02.02** – Ranging for Quiet Running Bill
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 001.01.01** – Operations Team: Crewing and Responsibilities	**EO 001.01.03** – Employ Radars and IFF	**EO 001.01.06** – Identify Team Readiness Requirements: TRP	**EC 001.01** – Conduct Team Training Events	**EO 001.02.02** – Ranging for Quiet Running Bill
Period 6 1320-1410	**EO 001.01.01** – Operations Team: Crewing and Responsibilities	**EO 001.01.03** – Employ Radars and IFF	**EO 001.01.06** – Identify Team Readiness Requirements: TRP	**EC 001.01** – Conduct Team Training Events	**EO 001.02.02** – Ranging for Quiet Running Bill
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 001.01.01** – Operations Team: Crewing and Responsibilities	**EO 001.01.04** – C4ISR	**EO 001.01.06** – Identify Team Readiness Requirements: TRP	**EC 001.01** – Conduct Team Training Events	**EO 001.02.02** – Ranging for Quiet Running Bill
Period 8 1510-1600	**EO 001.01.02** – ORS Liaison	**EO 001.01.04** – C4ISR	**EO 001.01.07**– Drafting Training Messages: Fleet Training	**EC 001.01** – Conduct Team Training Events	**PT**

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Week #2

TIME	DAY 6	DAY 7	DAY 8	DAY 9	DAY 10
Period 1 0800-0850	**EO 001.02.03** – Noise Monitoring Plans	**PT**	**PT**	**PT**	**EO 001.02.12** –Mine Threat Operations
Period 2 0850-0940	**EO 001.02.03** – Noise Monitoring Plans	**EO 001.02.06** – NMCM Concepts Overview	**EO 001.02.08** –Mine Threat Research	**EO 001.02.10** –Ship Self Protective Measures	**EO 001.02.13** –Mine Detection and Avoidance Procedures
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 001.02.03** – Noise Monitoring Plans	**EO 001.02.06** – NMCM Concepts Overview	**EO 001.02.09** –Mine Threat Presentation	**EO 001.02.10** –Ship Self Protective Measures	**EO 001.02.13** –Mine Detection and Avoidance Procedures
Period 4 1040-1130	**EO 001.02.04** – Logs & Records	**EO 001.02.06** – NMCM Concepts Overview	**EO 001.02.09** –Mine Threat Presentation	**EO 001.02.10** –Ship Self Protective Measures	**EO 001.02.13** –Mine Detection and Avoidance Procedures
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 001.02.04** – Logs & Records	**EO 001.02.06** – NMCM Concepts Overview	**EO 001.02.09** –Mine Threat Presentation	**EO 001.02.11** –Total Mine Simulation Software	**EC 001.02** – Optimize Ship's Acoustic Signature (Preparation)
Period 6 1320-1410	**EO 001.02.05** – Ships UWW Logs & Records (Ship Tour)	**EO 001.02.07** – Current Mine Threat	**EO 001.02.10** –Ship Self Protective Measures	**EO 001.02.11** –Total Mine Simulation Software	**EC 001.02** – Optimize Ship's Acoustic Signature (Preparation)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 001.02.05** – Ships UWW Logs & Records (Ship Tour)	**EO 001.02.08** –Mine Threat Research	**EO 001.02.10** –Ship Self Protective Measures	**EO 001.02.11** –Total Mine Simulation Software	**EC 001.02** – Optimize Ship's Acoustic Signature (Preparation)
Period 8 1510-1600	**EO 001.02.05** – Ships UWW Logs & Records (Ship Tour)	**EO 001.02.08** –Mine Threat Research	**TAT**	**EO 001.02.12** –Mine Threat Operations	**EC 001.02** – Optimize Ship's Acoustic Signature (Presentation)

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Week #3

TIME	DAY 11	DAY 12	DAY 13	DAY 14	DAY 15
Period 1 0800-0850	**PC 001 (Develop)**	**PT**	**EO 002.01.04** – Physics of Sound	**PT**	**EO 002.01.07** – Regional Oceanography
Period 2 0850-0940	**PC 001 (Develop)**	**EO 002.01.02** –Military Oceanography	**EO 002.01.04** – Physics of Sound	**EO 002.01.05** – Ocean Thermal Structure / Sound Paths	**EO 002.01.07** – Regional Oceanography
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**PC 001 (Present)**	**EO 002.01.02** –Military Oceanography	**EO 002.01.04** – Physics of Sound	**EO 002.01.06** – Biologics, Fronts & Eddies	**EO 002.01.07** – Regional Oceanography
Period 4 1040-1130	**PC 001 (Present)**	**EO 002.01.02** –Military Oceanography	**EO 002.01.05** – Ocean Thermal Structure / Sound Paths	**EO 002.01.06** – Biologics, Fronts & Eddies	**EO 002.01.07** – Regional Oceanography
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**PC 001 (Present)**	**EO 002.01.03** – Ocean Topography/Composition	**EO 002.01.05** – Ocean Thermal Structure / Sound Paths	**EO 002.01.06** – Biologics, Fronts & Eddies	**EO 002.01.08** – Elements of Acoustic Range Prediction
Period 6 1320-1410	**TAT**	**EO 002.01.03** – Ocean Topography/Composition	**EO 002.01.05** – Ocean Thermal Structure / Sound Paths	**EO 002.01.06** – Biologics, Fronts & Eddies	**EO 002.01.08** – Elements of Acoustic Range Prediction
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 002.01.01** –Mission Briefing	**EO 002.01.03** – Ocean Topography/Composition	**EO 002.01.05** – Ocean Thermal Structure / Sound Paths	**EO 002.01.06** – Biologics, Fronts & Eddies	**EO 002.01.08** – Elements of Acoustic Range Prediction
Period 8 1510-1600	**PT**	**EO 002.01.04** – Physics of Sound	**EO 002.01.05** – Ocean Thermal Structure / Sound Paths	**EO 002.01.07** – Regional Oceanography	**TAT**

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Week #4

TIME	DAY 16	DAY 17	DAY 18	DAY 19	DAY 20
Period 1 0800-0850	**EO 002.01.09** – METOC Products & Services	**PT**	**PT**	**PT**	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)
Period 2 0850-0940	**EO 002.01.09** – METOC Products & Services	**EO 002.01.11** – ASW Sensors	**EO 002.01.13** – Target Motion Analysis	**EO 002.01.15** – Acoustic Counter Measures	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 002.01.09** – METOC Products & Services	**EO 002.01.11** – ASW Sensors	**EO 002.01.13** – Target Motion Analysis	**EO 002.01.15** – Acoustic Counter Measures	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)
Period 4 1040-1130	**TAT**	**EO 002.01.11** – ASW Sensors	**EO 002.01.13** – Target Motion Analysis	**EO 002.01.21** – Evaluation Reports	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 002.01.10** – Tactical Oceanography	**EO 002.01.11** – ASW Sensors	**EO 002.01.14** – Ancillary Equipment & Expendables	**EO 002.01.21** – Evaluation Reports	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)
Period 6 1320-1410	**EO 002.01.10** – Tactical Oceanography	**EO 002.01.11** – ASW Sensors	**EO 002.01.14** – Ancillary Equipment & Expendables	**EO 002.01.21** – Evaluation Reports	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 002.01.10** – Tactical Oceanography	**EO 002.01.12** – Towed Array Operations	**EO 002.01.14** – Ancillary Equipment & Expendables	**BAT**	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)
Period 8 1510-1600	**EO 002.01.10** – Tactical Oceanography	**EO 002.01.12** – Towed Array Operations	**EO 002.01.14** – Ancillary Equipment & Expendables	**BAT**	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)

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Week #5

TIME	DAY 21	DAY 22	DAY 23	DAY 24	DAY 25
Period 1 0800-0850	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)	**PT**	**PT**	**PT**
Period 2 0850-0940	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)	**EO 002.01.22** – Organize Intelligence information	**EC 002.01** – Orientate to Mission (Preparation)	**EO 002.02.01** – Introduction to Maritime Tactical Planner
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)	**EO 002.01.22** – Organize Intelligence information	**EC 002.01** – Orientate to Mission (Presentation)	**EO 004.03.01** – ASW Support Operations
Period 4 1040-1130	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)	**EO 002.01.22** – Organize Intelligence information	**EC 002.01** – Orientate to Mission (Presentation)	**EO 004.03.01** – ASW Support Operations
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)	**EO 002.01.23** – Establish Caps & Lims of Platforms	**EC 002.01** – Orientate to Mission (Presentation)	**EO 004.03.02** – WSM vs. PMI
Period 6 1320-1410	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)	**EO 002.01.23** – Establish Caps & Lims of Platforms	**EC 002.01** – Orientate to Mission (Presentation)	**EO 004.03.02** – WSM vs. PMI
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)	**EO 002.01.23** – Establish Caps & Lims of Platforms	**EO 002.02.01** – Introduction to Maritime Tactical Planner	**EO 004.03.02** – WSM vs. PMI
Period 8 1510-1600	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)	**EO 002.01.16** – Introduction to Passive Acoustics (ADAC)	**EO 002.01.23** – Establish Caps & Lims of Platforms	**EO 002.02.01** – Introduction to Maritime Tactical Planner	**TAT**

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Week #6

TIME	DAY 26	DAY 27	DAY 28	DAY 29	DAY 30
Period 1 0800-0850	**PT**	**EO 004.03.03** – Coordinate Fixed Wing Employment	**TAT**	**PT**	**EO 004.03.07** – ASW Screening
Period 2 0850-0940	**EO 004.03.02** – WSM vs. PMI	**EO 004.03.03** – Coordinate Fixed Wing Employment	**EO 004.03.05** – Aircraft Familiarization (Air Wing Tour)	**EO 004.03.06** – Anti-Ship Torpedo Defence	**EO 004.03.07** – ASW Screening
0940-0950	**BREAK**	**BREAK**			**BREAK**
Period 3 0950-1040	**EO 004.03.02** – WSM vs. PMI	**EO 004.03.04** – Coordinate Rotary Wing Employment	**EO 004.03.05** – Aircraft Familiarization (Air Wing Tour)	**EO 004.03.06** – Anti-Ship Torpedo Defence	**EO 004.03.07** – ASW Screening
Period 4 1040-1130	**EO 004.03.02** – WSM vs. PMI	**EO 004.03.04** – Coordinate Rotary Wing Employment	**EO 004.03.05** – Aircraft Familiarization (Air Wing Tour)	**EO 004.03.06** – Anti-Ship Torpedo Defence	**EO 004.03.07** – ASW Screening
1130-1230	**LUNCH**	**LUNCH**			**LUNCH**
Period 5 1230-1320	**EO 004.03.03** – Coordinate Fixed Wing Employment	**EO 004.03.04** – Coordinate Rotary Wing Employment	**EO 004.03.05** – Aircraft Familiarization (Air Wing Tour)	**EO 004.03.06** – Anti-Ship Torpedo Defence	**EO 004.03.07** – ASW Screening
Period 6 1320-1410	**EO 004.03.03** – Coordinate Fixed Wing Employment	**EO 004.03.04** – Coordinate Rotary Wing Employment	**BAT**	**EO 004.03.06** – Anti-Ship Torpedo Defence	**EO 004.03.07** – ASW Screening
1410-1420	**BREAK**	**BREAK**			**BREAK**
Period 7 1420-1510	**EO 004.03.03** – Coordinate Fixed Wing Employment	**EO 004.03.04** – Coordinate Rotary Wing Employment	**EO 004.03.06** – Anti-Ship Torpedo Defence	**EO 004.03.06** – Anti-Ship Torpedo Defence	**EO 004.03.07** – ASW Screening
Period 8 1510-1600	**EO 004.03.03** – Coordinate Fixed Wing Employment	**EO 004.03.04** – Coordinate Rotary Wing Employment	**EO 004.03.06** – Anti-Ship Torpedo Defence	**EO 004.03.06** – Anti-Ship Torpedo Defence	**PT**

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Week #7

TIME	DAY 31	DAY 32	DAY 33	DAY 34	DAY 35
Period 1 0800-0850	**PT**	**EO 004.03.09** – ASW Tactical Implications	**PT**	**PT**	**EO 004.03.11** – Initial Responses
Period 2 0850-0940	**EO 004.03.08** – Submarine Search Tactics	**EO 004.03.09** – ASW Tactical Implications	**EO 004.03.09** – ASW Tactical Implications	**EO 004.03.10** – Command Classification & DAMASS	**EO 004.03.11** – Initial Responses
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 004.03.08** – Submarine Search Tactics	**EO 004.03.09** – ASW Tactical Implications	**EO 004.03.09** – ASW Tactical Implications	**EO 004.03.10** – Command Classification & DAMASS	**EO 004.03.11** – Initial Responses
Period 4 1040-1130	**EO 004.03.08** – Submarine Search Tactics	**EO 004.03.09** – ASW Tactical Implications	**EO 004.03.09** – ASW Tactical Implications	**EO 004.03.10** – Command Classification & DAMASS	**EO 004.03.12** – ASW Pre-Plans
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 004.03.08** – Submarine Search Tactics	**EO 004.03.09** – ASW Tactical Implications	**EO 004.03.10** – Command Classification & DAMASS	**EO 004.03.11** – Initial Responses	**EO 004.03.12** – ASW Pre-Plans
Period 6 1320-1410	**EO 004.03.08** – Submarine Search Tactics	**EO 004.03.09** – ASW Tactical Implications	**EO 004.03.10** – Command Classification & DAMASS	**EO 004.03.11** – Initial Responses	**EO 004.03.12** – ASW Pre-Plans
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 004.03.08** – Submarine Search Tactics	**EO 004.03.09** – ASW Tactical Implications	**EO 004.03.10** – Command Classification & DAMASS	**EO 004.03.11** – Initial Responses	**EO 004.03.13** – ASW Localization Plans
Period 8 1510-1600	**EO 004.03.09** – ASW Tactical Implications	**EO 004.03.09** – ASW Tactical Implications	**EO 004.03.10** – Command Classification & DAMASS	**TAT**	**TAT**

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Week #8

TIME	DAY 36	DAY 37	DAY 38	DAY 39	DAY 40
Period 1 0800-0850	**PT**	**PT**	**EO 002.01.18** – Interpret Command Structure	**PT**	**EO 002.02.02** – Water Space Management Request
Period 2 0850-0940	**EO 004.03.13** – ASW Localization Plans	**EO 004.03.14** – Adhering to ROE	**EO 002.01.19** – Interpreting OPGEN & OPTASK messages	**EO 002.02.02** – Water Space Management Request	**EO 002.02.02** – Water Space Management Request
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 004.03.13** – ASW Localization Plans	**EO 004.03.14** – Adhering to ROE	**EO 002.01.19** – Interpreting OPGEN & OPTASK messages	**EO 002.02.02** – Water Space Management Request	**EO 002.02.02** – Water Space Management Request
Period 4 1040-1130	**EO 004.03.13** – ASW Localization Plans	**EO 004.03.14** – Adhering to ROE	**EO 002.01.19** – Interpreting OPGEN & OPTASK messages	**EO 002.02.02** – Water Space Management Request	**EO 002.02.02** – Water Space Management Request
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 004.03.13** – ASW Localization Plans	**EO 002.01.17** – Operational Messages Formatting	**EO 002.01.20** – Interpreting OPGEN & OPTASK messages	**EO 002.02.02** – Water Space Management Request	**EO 002.02.02** – Water Space Management Request
Period 6 1320-1410	**EO 004.03.13** – ASW Localization Plans	**EO 002.01.17** – Operational Messages Formatting	**EO 002.01.20** – Interpreting OPGEN & OPTASK messages	**EO 002.02.02** – Water Space Management Request	**EO 002.02.03** – Drafting OPTASK ASW/SCC message
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 004.03.13** – ASW Localization Plans	**EO 002.01.18** – Interpret Command Structure	**EO 002.01.20** – Interpreting OPGEN & OPTASK messages	**EO 002.02.02** – Water Space Management Request	**EO 002.02.03** – Drafting OPTASK ASW/SCC message
Period 8 1510-1600	**TAT**	**EO 002.01.18** – Interpret Command Structure	**TAT**	**EO 002.02.02** – Water Space Management Request	**EO 002.02.03** – Drafting OPTASK ASW/SCC message

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Week #9

TIME	DAY 41	DAY 42	DAY 43	DAY 44	DAY 45
Period 1 0800-0850	**PT**	**EO 002.02.04** – Interpreting CASEX TABORDS	**PT**	**PT**	**EO 002.03.01** – Platform Intelligence
Period 2 0850-0940	**EO 002.02.03** – Drafting OPTASK ASW/SCC message	**EO 002.02.04** – Interpreting CASEX TABORDS	**EO 002.02.05** – Planning CASEXs	**EO 002.02.06** – Drafting CASEX TABORDs	**EO 002.03.01** – Platform Intelligence
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 002.02.03** – Drafting OPTASK ASW/SCC message	**EO 002.02.04** – Interpreting CASEX TABORDS	**EO 002.02.05** – Planning CASEXs	**EO 002.02.06** – Drafting CASEX TABORDs	**EO 002.03.01** – Platform Intelligence
Period 4 1040-1130	**EO 002.02.03** – Drafting OPTASK ASW/SCC message	**EO 002.02.04** – Interpreting CASEX TABORDS	**EO 002.02.05** – Planning CASEXs	**EO 002.02.06** – Drafting CASEX TABORDs	**EO 002.03.01** – Platform Intelligence
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 002.02.03** – Drafting OPTASK ASW/SCC message	**EO 002.02.04** – Interpreting CASEX TABORDS	**EO 002.02.06** – Drafting CASEX TABORDs	**TAT**	**EO 002.03.02** – CASEX Briefing
Period 6 1320-1410	**EO 002.02.03** – Drafting OPTASK ASW/SCC message	**EO 002.02.04** – Interpreting CASEX TABORDS	**EO 002.02.06** – Drafting CASEX TABORDs	**EC 002.02** – Prepare Operational Messages	**EO 002.03.02** – CASEX Briefing
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 002.02.03** – Drafting OPTASK ASW/SCC message	**EO 002.02.04** – Interpreting CASEX TABORDS	**EO 002.02.06** – Drafting CASEX TABORDs	**EC 002.02** – Prepare Operational Messages	**EO 002.03.02** – CASEX Briefing
Period 8 1510-1600	**EO 002.02.03** – Drafting OPTASK ASW/SCC message	**TAT**	**EO 002.02.06** – Drafting CASEX TABORDs	**EC 002.02** – Prepare Operational Messages	**EO 002.03.02** – CASEX Briefing

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Week #10

TIME	DAY 46	DAY 47	DAY 48	DAY 49	DAY 50
Period 1 0800-0850	**EO 002.03.02** – CASEX Briefing	**PT**	**PT**	**PT**	**EO 004.02.05** –Voice Reports
Period 2 0850-0940	**EO 002.03.02** – CASEX Briefing	**EC 002.03**– Disseminate intelligence information	**PC 002 (Preparation)**	**EO 004.02.01** – Basic Communications Theory	**EO 004.02.05** –Voice Reports
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 002.03.02** – CASEX Briefing	**EC 002.03**– Disseminate intelligence information	**PC 002 (Preparation)**	**EO 004.02.02** – Tactical Communications	**EO 004.02.05** –Voice Reports
Period 4 1040-1130	**EO 002.03.02** – CASEX Briefing	**EC 002.03**– Disseminate intelligence information	**PC 002 (Preparation)**	**EO 004.02.02** – Tactical Communications	**EO 004.02.05** –Voice Reports
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EC 002.03**– Disseminate intelligence information	**EC 002.03**– Disseminate intelligence information	**PC 002 (Presentation)**	**EO 004.02.03** – Communications types	**EO 004.02.05** –Voice Reports
Period 6 1320-1410	**EC 002.03**– Disseminate intelligence information	**EC 002.03**– Disseminate intelligence information	**PC 002 (Presentation)**	**EO 004.02.03** – Communications types	**EO 004.02.05** –Voice Reports
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EC 002.03**– Disseminate intelligence information	**BAT**	**PC 002 (Presentation)**	**EO 004.02.04** –Brevity Code words	**EO 004.02.05** –Voice Reports
Period 8 1510-1600	**EC 002.03**– Disseminate intelligence information	**TAT**	**PC 002 (Presentation)**	**EO 004.02.04** –Brevity Code words	**TAT**

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Week #11

TIME	DAY 51	DAY 52	DAY 53	DAY 54	DAY 55
Period 1 0800-0850	**PT**	**EO 003.02.01** – Command System Setup	**EO 003.02.02** – Develop ASW Tactical Picture	**PT**	**EO 003.02.02** – Develop ASW Tactical Picture
Period 2 0850-0940	**EO 003.02.01** – Command System Setup	**EO 003.02.01** – Command System Setup	**EO 003.02.02** – Develop ASW Tactical Picture	**EO 003.02.02** – Develop ASW Tactical Picture	**EO 003.02.02** – Develop ASW Tactical Picture
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 003.02.01** – Command System Setup	**EO 003.02.01** – Command System Setup	**EO 003.02.02** – Develop ASW Tactical Picture	**EO 003.02.02** – Develop ASW Tactical Picture	**EO 003.02.02** – Develop ASW Tactical Picture
Period 4 1040-1130	**EO 003.02.01** – Command System Setup	**EO 003.02.01** – Command System Setup	**EO 003.02.02** – Develop ASW Tactical Picture	**EO 003.02.02** – Develop ASW Tactical Picture	**EO 003.02.02** – Develop ASW Tactical Picture
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 003.02.01** – Command System Setup	**EO 003.02.02** – Develop ASW Tactical Picture	**EO 003.02.02** – Develop ASW Tactical Picture	**EO 003.02.02** – Develop ASW Tactical Picture	**EC 003.02.01** – Operate Command System
Period 6 1320-1410	**EO 003.02.01** – Command System Setup	**EO 003.02.02** – Develop ASW Tactical Picture	**EO 003.02.02** – Develop ASW Tactical Picture	**EO 003.02.02** – Develop ASW Tactical Picture	**EC 003.02.01** – Operate Command System
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 003.02.01** – Command System Setup	**EO 003.02.02** – Develop ASW Tactical Picture	**EO 003.02.02** – Develop ASW Tactical Picture	**EO 003.02.02** – Develop ASW Tactical Picture	**EC 003.02.01** – Operate Command System
Period 8 1510-1600	**EO 003.02.01** – Command System Setup	**TAT**	**EO 003.02.02** – Develop ASW Tactical Picture	**EO 003.02.02** – Develop ASW Tactical Picture	**PT**

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Week #12

TIME	DAY 56	DAY 57	DAY 58	DAY 59	DAY 60
Period 1 0800-0850	**PT**	**EO 003.01.02** – Range Procedures	**EO 003.01.03** – Drafting Firing Orders	**PT**	**EO 003.02.04** – ID Faults
Period 2 0850-0940	**EO 003.01.01** – Weapons Capabilities and Limitations	**EO 003.01.02** – Range Procedures	**EO 003.01.04** – Conduct Pre-Firing Brief	**EO 003.02.03** – Weapons Firing Procedures	**EO 003.02.04** – ID Faults
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 003.01.01** – Weapons Capabilities and Limitations	**EO 003.01.02** – Range Procedures	**EO 003.01.04** – Conduct Pre-Firing Brief	**EO 003.02.03** – Weapons Firing Procedures	**EO 003.02.04** – ID Faults
Period 4 1040-1130	**EO 003.01.01** – Weapons Capabilities and Limitations	**EO 003.01.02** – Range Procedures	**EO 003.01.04** – Conduct Pre-Firing Brief	**EO 003.02.03** – Weapons Firing Procedures	**EO 003.02.05** – Weapons Firing Procedures (Practical)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 003.01.01** – Weapons Capabilities and Limitations	**EO 003.01.02** – Range Procedures	**EC 003.01**– Prepare for Weapons Firing	**EO 003.02.03** – Weapons Firing Procedures	**EO 003.02.05** – Weapons Firing Procedures (Practical)
Period 6 1320-1410	**EO 003.01.01** – Weapons Capabilities and Limitations	**EO 003.01.02** – Range Procedures	**EO 003.02.03** – Weapons Firing Procedures	**EO 003.02.03** – Weapons Firing Procedures	**EO 003.02.05** – Weapons Firing Procedures (Practical)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 003.01.01** – Weapons Capabilities and Limitations	**EO 003.01.02** – Range Procedures	**EO 003.02.03** – Weapons Firing Procedures	**EO 003.02.03** – Weapons Firing Procedures	**EO 003.02.05** – Weapons Firing Procedures (Practical)
Period 8 1510-1600	**TAT**	**PT**	**EO 003.02.03** – Weapons Firing Procedures	**EO 003.02.03** – Weapons Firing Procedures	**BAT**

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Week #13

TIME	DAY 61	DAY 62	DAY 63	DAY 64	DAY 65
Period 1 0800-0850	**PT**	**PT**	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**PT**	**PC 003 (Firing Practical)**
Period 2 0850-0940	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**EC 003.02.01** – Operate Command System	**PC 003 (Firing Practical)**
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**EC 003.02.01** – Operate Command System	**PC 003 (Firing Practical)**
Period 4 1040-1130	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**EC 003.02.02** – Conduct Weapons	**PC 003 (Firing Practical)**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**EC 003.02.01** – Operate Command System	**PC 003 (Firing Practical)**
Period 6 1320-1410	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**EC 003.02.01** – Operate Command System	**PC 003 (Firing Practical)**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**EC 003.02.01** – Operate Command System	**PC 003 (Firing Practical)**
Period 8 1510-1600	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**EO 003.02.05** – Weapons Firing Procedures (Practical)	**TAT**	**EC 003.02.01** – Operate Command System	**PC 003 (Firing Practical)**

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Week #14

TIME	DAY 66	DAY 67	DAY 68	DAY 69	DAY 70
Period 1 0800-0850	**PC 003 (Draft Firing Orders)**	**EO 004.01.03** – Conduct Submarine Safety Procedures	**PT**	**PT**	**EC 004.01** – Operate Safely with Submarines
Period 2 0850-0940	**BAT**	**EO 004.01.03** – Conduct Submarine Safety Procedures	**EO 004.01.03** – Conduct Submarine Safety Procedures	**EO 004.01.03** – Conduct Submarine Safety Procedures	**EC 004.01** – Operate Safely with Submarines
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 004.01.01** – Submarine SAR / SMASH	**EO 004.01.03** – Conduct Submarine Safety Procedures	**EO 004.01.03** – Conduct Submarine Safety Procedures	**EO 004.01.03** – Conduct Submarine Safety Procedures	**EC 004.01** – Operate Safely with Submarines
Period 4 1040-1130	**EO 004.01.01** – Submarine SAR / SMASH	**EO 004.01.03** – Conduct Submarine Safety Procedures	**EO 004.01.03** – Conduct Submarine Safety Procedures	**EO 004.01.03** – Conduct Submarine Safety Procedures	**EC 004.01** – Operate Safely with Submarines
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 004.01.02** – Submarine Safety Procedures	**EO 004.01.03** – Conduct Submarine Safety Procedures	**EO 004.01.03** – Conduct Submarine Safety Procedures	**EO 004.01.03** – Conduct Submarine Safety Procedures	**EC 004.01** – Operate Safely with Submarines
Period 6 1320-1410	**EO 004.01.02** – Submarine Safety Procedures	**EO 004.01.03** – Conduct Submarine Safety Procedures	**EO 004.01.03** – Conduct Submarine Safety Procedures	**EO 004.01.03** – Conduct Submarine Safety Procedures	**EC 004.01** – Operate Safely with Submarines
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 004.01.02** – Submarine Safety Procedures	**EO 004.01.03** – Conduct Submarine Safety Procedures	**EO 004.01.03** – Conduct Submarine Safety Procedures	**EC 004.01** – Operate Safely with Submarines	**EC 004.01** – Operate Safely with Submarines
Period 8 1510-1600	**PT**	**EO 004.01.03** – Conduct Submarine Safety Procedures	**EO 004.01.03** – Conduct Submarine Safety Procedures	**EC 004.01** – Operate Safely with Submarines	**EC 004.01** – Operate Safely with Submarines

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Week #15

TIME	DAY 71	DAY 72	DAY 73	DAY 74	DAY 75
Period 1 0800-0850	**EO 004.03.16** – Conduct CASEXs (C2)	**PT**	**EO 004.03.16** – Conduct CASEXs (C2)	**PT**	**EO 004.03.16** – Conduct CASEXs (C2)
Period 2 0850-0940	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	
Period 3 0950-1040	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)
Period 4 1040-1130	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	
Period 5 1230-1320	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)
Period 6 1320-1410	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	
Period 7 1420-1510	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)
Period 8 1510-1600	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**PT**

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Week #16

TIME	DAY 76	DAY 77	DAY 78	DAY 79	DAY 80
Period 1 0800-0850	**PT**	**EO 004.03.16** – Conduct CASEXs (C2)	**PT**	**PT**	**EO 004.03.16** – Conduct CASEXs (C4)
Period 2 0850-0940	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C4)	**EO 004.03.16** – Conduct CASEXs (C4)	**EO 004.03.16** – Conduct CASEXs (C4)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C4)	**EO 004.03.16** – Conduct CASEXs (C4)	**EO 004.03.16** – Conduct CASEXs (C4)
Period 4 1040-1130	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C4)	**EO 004.03.16** – Conduct CASEXs (C4)	**EO 004.03.16** – Conduct CASEXs (C4)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C4)	**EO 004.03.16** – Conduct CASEXs (C4)	**EO 004.03.16** – Conduct CASEXs (C4)	**EO 004.03.16** – Conduct CASEXs (C4)
Period 6 1320-1410	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C4)	**EO 004.03.16** – Conduct CASEXs (C4)	**EO 004.03.16** – Conduct CASEXs (C4)	**EO 004.03.16** – Conduct CASEXs (C4)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C4)	**EO 004.03.16** – Conduct CASEXs (C4)	**EO 004.03.16** – Conduct CASEXs (C4)	**EO 004.03.16** – Conduct CASEXs (C4)
Period 8 1510-1600	**EO 004.03.16** – Conduct CASEXs (C2)	**EO 004.03.16** – Conduct CASEXs (C4)	**EO 004.03.16** – Conduct CASEXs (C4)	**EO 004.03.16** – Conduct CASEXs (C4)	**TAT**

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Week #17

TIME	DAY 81	DAY 82	DAY 83	DAY 84	DAY 85
Period 1 0800-0850	**EO 004.03.16** – Conduct CASEXs (C4)	**EC 004.03.01** – Coordinate UWW CASEXs (C4)	**EC 004.03.01** – Coordinate UWW CASEXs (C4)	**PT**	**PC 004.01 (CASEX)**
Period 2 0850-0940	**EO 004.03.16** – Conduct CASEXs (C4)	**EC 004.03.01** – Coordinate UWW CASEXs (C4)	**EC 004.03.01** – Coordinate UWW CASEXs (C4)	**PC 004.01 (CASEX)**	**PC 004.01 (CASEX)**
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 004.03.16** – Conduct CASEXs (C4)	**EC 004.03.01** – Coordinate UWW CASEXs (C4)	**EC 004.03.01** – Coordinate UWW CASEXs (C4)	**PC 004.01 (CASEX)**	**PC 004.01 (CASEX)**
Period 4 1040-1130	**EO 004.03.16** – Conduct CASEXs (C4)	**EC 004.03.01** – Coordinate UWW CASEXs (C4)	**EC 004.03.01** – Coordinate UWW CASEXs (C4)	**PC 004.01 (CASEX)**	**PC 004.01 (CASEX)**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 004.03.16** – Conduct CASEXs (C4)	**EC 004.03.01** – Coordinate UWW CASEXs (C4)	**EC 004.03.01** – Coordinate UWW CASEXs (C4)	**PC 004.01 (CASEX)**	**PC 004.01 (CASEX)**
Period 6 1320-1410	**EO 004.03.16** – Conduct CASEXs (C4)	**EC 004.03.01** – Coordinate UWW CASEXs (C4)	**EC 004.03.01** – Coordinate UWW CASEXs (C4)	**PC 004.01 (CASEX)**	**PC 004.01 (CASEX)**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 004.03.16** – Conduct CASEXs (C4)	**EC 004.03.01** – Coordinate UWW CASEXs (C4)	**EC 004.03.01** – Coordinate UWW CASEXs (C4)	**PC 004.01 (CASEX)**	**BAT**
Period 8 1510-1600	**PT**	**EC 004.03.01** – Coordinate UWW CASEXs (C4)	**EC 004.03.01** – Coordinate UWW CASEXs (C4)	**PC 004.01 (CASEX)**	**PT**

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Week #18

TIME	DAY 86	DAY 87	DAY 88	DAY 89	DAY 90
Period 1 0800-0850	**PC 004.01 (CASEX)**	**EO 004.03.15** – Mission Planning	**EO 004.03.15** – Mission Planning (Brief)	**PT**	**EO 004.03.17** – Conduct ASW Operations
Period 2 0850-0940	**PC 004.01 (CASEX)**	**EO 004.03.15** – Mission Planning	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**PC 004.01 (CASEX)**	**EO 004.03.15** – Mission Planning	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations
Period 4 1040-1130	**PC 004.01 (CASEX)**	**EO 004.03.15** – Mission Planning	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**PC 004.01 (CASEX)**	**EO 004.03.15** – Mission Planning	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations
Period 6 1320-1410	**PC 004.01 (CASEX)**	**EO 004.03.15** – Mission Planning	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**PC 004.01 (CASEX)**	**EO 004.03.15** – Mission Planning	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations
Period 8 1510-1600	**PT**	**TAT**	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**PT**

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Week #19

TIME	DAY 91	DAY 92	DAY 93	DAY 94	DAY 95
Period 1 0800-0850	**PT**	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**PT**	**EO 004.03.17** – Conduct ASW Operations
Period 2 0850-0940	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations
Period 4 1040-1130	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations
Period 6 1320-1410	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations
Period 8 1510-1600	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**TAT**	**EO 004.03.17** – Conduct ASW Operations	**PT**

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Week #20

TIME	DAY 96	DAY 97	DAY 98	DAY 99	DAY 100
Period 1 0800-0850	**EO 004.03.17** – Conduct ASW Operations	**PT**	**EO 004.03.17** – Conduct ASW Operations	**PT**	**EC 004.03.02** – Coordinate UWW Operations
Period 2 0850-0940	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EC 004.03.02** – Coordinate UWW Operations	**EC 004.03.02** – Coordinate UWW Operations
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EC 004.03.02** – Coordinate UWW Operations	**EC 004.03.02** – Coordinate UWW Operations
Period 4 1040-1130	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EC 004.03.02** – Coordinate UWW Operations	**EC 004.03.02** – Coordinate UWW Operations
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EC 004.03.02** – Coordinate UWW Operations	**EC 004.03.02** – Coordinate UWW Operations
Period 6 1320-1410	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EC 004.03.02** – Coordinate UWW Operations	**EC 004.03.02** – Coordinate UWW Operations
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EC 004.03.02** – Coordinate UWW Operations	**EC 004.03.02** – Coordinate UWW Operations
Period 8 1510-1600	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EO 004.03.17** – Conduct ASW Operations	**EC 004.03.02** – Coordinate UWW Operations	**PT**

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Week #21

TIME	DAY 101	DAY 102	DAY 103	DAY 104	DAY 105
Period 1 0800-0850	**PT**	**PC 004.02 (Operations)**	**PC 004.02 (Operations)**	**PT**	**PT**
Period 2 0850-0940	**EC 004.03.02** – Coordinate UWW Operations	**PC 004.02 (Operations)**	**PC 004.02 (Operations)**	**TAT**	**BAT (Graduation)**
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EC 004.03.02** – Coordinate UWW Operations	**PC 004.02 (Operations)**	**PC 004.02 (Operations)**	**TAT**	**BAT (Graduation)**
Period 4 1040-1130	**TAT**	**PC 004.02 (Operations)**	**PC 004.02 (Operations)**	**BAT**	**BAT (Course Reports)**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**PC 004.02 (Operations)**	**PC 004.02 (Operations)**	**PC 004.02 (Operations)**	**BAT**	**BAT (Out Routine)**
Period 6 1320-1410	**PC 004.02 (Operations)**	**PC 004.02 (Operations)**	**PC 004.02 (Operations)**	**BAT (PCD)**	**BAT (Out Routine)**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**PC 004.02 (Operations)**	**PC 004.02 (Operations)**	**PC 004.02 (Operations)**	**BAT (PCD)**	**BAT (Out Routine)**
Period 8 1510-1600	**PC 004.02 (Operations)**	**PC 004.02 (Operations)**	**PC 004.02 (Operations)**	**BAT**	**BAT (Out Routine)**

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IPAC (EO 002.01.16) Schedule

TIME	DAY 1	DAY 2	DAY 3
Period 1 0800-0850	**EO** – PAL Techniques (Interactive Lecture)	**EO -** Identify Acoustic Sources (Components of a LOFAR Gram), (Signature Characteristics)	**EO** – Identify Passive Acoustic Contacts (Non-Allied Diesel Lecture)
Period 2 0850-0940	**EO** – PAL Techniques (Interactive Lecture)	**EO -** Identify Acoustic Sources (Components of a LOFAR Gram), (Signature Characteristics)	**EO** – Identify Passive Acoustic Contacts (Non-Allied Diesel Lecture)
0940-0950	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO** – PAL Techniques (Interactive Lecture)	**EO -** Identify Acoustic Sources (Sources of Radiated Noise)	**EO** – Classify Passive Acoustic Contacts (Instructor Guided Lab)
Period 4 1040-1130	**EO** – PAL Techniques (Interactive Lecture)	**EO -** Identify Acoustic Sources (Sources of Radiated Noise)	**EO** – Classify Passive Acoustic Contacts (Instructor Guided Lab)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO** – PAL (LAB #1)	**EO -** Identify Acoustic Sources (Sources of Radiated Noise)	**EO** – Classify Passive Acoustic Contacts (Instructor Guided Lab)
Period 6 1320-1410	**EO** – PAL (LAB #2)	**EO -** Identify Acoustic Sources (Sources of Radiated Noise)	**EO** – Classify Passive Acoustic Contacts (Instructor Guided Lab)
1410-1420	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 007.03** – PAL (LAB #3)	**EO -** Identify Acoustic Sources (Doppler Theory / Interference Patterns / Anomalies)	**EO** – Classify Passive Acoustic Contacts (Instructor Guided Lab)
Period 8 1510-1600	**EO** – PAL Miscellaneous Sources (Interactive Lecture)	**EO -** Identify Acoustic Sources (Doppler Theory / Interference Patterns / Anomalies)	**EO** – Classify Passive Acoustic Contacts (Instructor Guided Lab)

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